TIRust benchmark

Batched one-loop reduction: TIRust, IREGI, COLLIER and MadLoop

2026-10-30 · Archival comparisons: AMD EPYC 9754 · CPUs 0 / 254 · identified kernel snapshots
Direct engine: same-session TIRust/IREGI/COLLIER · AMD EPYC 9754 · CPU 153 · binary64
Historical opening examples (26 October): CPU 18 · previous TIRust engine

Reading guide. Every displayed timing ratio has TIRust in the numerator: *TIRust / IREGI*, *TIRust / COLLIER*, or *TIRust(candidate) / TIRust(baseline)*. Physical graphs use *TIRust / MadLoop*. In comparison cells, green means TIRust is faster, white parity and red slower. The archival reference-only table shows absolute times and outcomes. Gray is unmeasured; amber marks pending or inconclusive evidence. The opening example tables use blue/green to identify reduction/master components, not relative speed. Source inspection does not establish numerical agreement. **Archival comparison note:** Measured TIRust6b24 passes eight numerical cases but retains seven timing regressions/one noisy row. It predates later APIs/preparation. Original-IREGI/COLLIER and physical sessions remain separate; private improvements, including the new unadopted lazy/eager cohort, do not close performance acceptance.

Current priorities

Feature completion	The documented finite N≤6/rank≤6 numerical checkpoint, including dots, is validated. Higher-N qualification remains separate.	**Validated**
Performance optimization	Direct numerical engine is the binary64 default; same-session TIRust/IREGI/COLLIER tables follow. Further optimization continues.	**Active**
Physical validation	Current fixed-graph checks appear below; historical timing failures remain visible	**Scoped**

1 Direct numerical engine: same-session comparison

What changed. The public binary64 default (`TensorReductionPlan::new` + `TensorWorkspace`) now evaluates the validated PV recurrences numerically per lane with direct oneloopmaster scalar kernels, instead of interpreting compiled symbolic expressions. Affinely dependent sub-topologies use their constant-one denominator identity; a dotted propagator is a Taylor jet in its squared mass with master derivatives from the IBP relation; D=6 uses `C(λ+h,κ)·(−2)^h`. Lanes with ill-conditioned systems, and for N≥4 (equal powers) lanes failing a loop-rotation test, are rerun by the same direct engine at 106/256/512 bits (see the stability section), then by the symbolic engine.

Protocol. All codes ran in one session on CPU 153 (AMD EPYC 9754 128-Core Processor), one pinned core, load average 238.30 / 237.94 / 237.83 at start. For every row and batch size each candidate got a fresh process; a pilot fixed the calls per block (≈20 ms); then 11 (batch 1) or 7 (batch 32) rounds ran one block per candidate in seeded random order. Ratios are medians of same-round block ratios with 99% bootstrap intervals; every ratio has TIRust in the numerator (green: TIRust faster). E/A is the dimensionally rescaled whole-array error of lane 0 versus the archived reference tensors (admitted against quadruple-precision original IREGI). IREGI uses its fastest mode with a complete finite array within E/A≤10^{-3}. COLLIER uses the archived selected public configuration. Rejected or nonfinite outputs are shown amber with diagnostic times and never enter a ratio. Builds: TIRust release (thin LTO), Fortran `-O3`, all baseline x86-64; binary SHA-256 in `validation/benchmark/direct-engine-20261030/summary-029.json`. TIRust source: commit `e76dd38`. This capture (029, 4 November, load 225 to 238) measures TIRust e76dd38. With the precision rescue on, lanes the a priori method choice does not keep go to the rescue directly, without the binary64 retries below the gate, by the expansions, by the alternative PV and with every box and triangle off PV (on the hexagon amplitude these kept 2 of 8 box lanes at about 2 ms per lane, where the double-double tier takes about 0.6 ms per box lane and gives 106 bits); lanes the choice keeps without a clear agreement go to the double-double tier first, and the off-PV retry runs only for the lanes that tier refuses. The binary64-only modes keep their retries. An amplitude call also allocates less, bit-identically: the native masters' store keys use one reused buffer and a hit reads the stored value in place, and the lightlike coefficients and the expansions reuse their scratch vectors. Binary64 is unchanged bit for bit (TIRust's binary64-only arrays of all 27500 stability points are byte-identical to those of TIRust 7022322). These rows rarely need the retries: every TIRust error and status is that of capture 028. TIRust's rows match capture 028 at the median (+0.1%; COLLIER's +1.2%, IREGI's +0.7%) and are within −7 and +5% of it except batch-1 N1R0 binary64-only contracted (+8%, 0.05 µs); the dotted N4R6 row with a cubed propagator, 5 to 9% slower in capture 028, is 2 to 7% faster in every mode, back at capture 027′s level. 19 of the 38 batch-1 and batch-32 measurements started on a core already partly busy (6 to 100%). N7R7 batch 32 takes 171 ms per call (capture 028: 176 ms), its binary64-only mode 166 ms (169 ms). The changed lanes show in stability capture 033 (51 points more accurate, none less) and the speed in amplitude capture 016 (12 to 14% faster). It ran alone. Capture 028 (TIRust ac7128e, load 239 to 271): the precision rescue's double-double C0 masters run the hand-written triangle kernel too. In double-double its r-functions now take the discriminant $k^2 - 4$ of real arguments from the exact threshold polynomial instead of the rounded k, which next to a two-particle threshold left

about 10^{-24} on 2610 random triangles the kernel is then never 16 times less accurate than the generated double-double code, and more accurate on 31. Binary64 is unchanged bit for bit (TIRust's binary64-only arrays of all 27500 stability points are byte-identical to those of TIRust 7022322). The rescued lanes change in their last bits (stability capture 032 changes the default mode's errors on 20 of the 27500 points, in their last digits, and no status, flag, binary64-only value or summary statistic; amplitude capture 015′s hexagon arrays change, its pentagon arrays do not, and both stay bit-identical between sharing on and off). TIRust's rows are a median 1.8% slower than in capture 027, COLLIER's 1.8% and IREGI's 1.0%; 25 of the 38 batch-1 and batch-32 measurements started on a core already partly busy (6 to 100%). TIRust's rows are within −3 and +10% of capture 027 (batch-1 N1R0 +10%, 0.05 µs; the dotted extended N4R6 row +7 to +9%, its binary64-only mode included). N7R7 batch 32 takes 176 ms per call (capture 027: 181 ms) with its binary64-only mode at 169 ms, so the rescue's share went from 12 to 7 ms. It ran alone. Capture 027 (TIRust 2db8ff8, load 201 to 212): the precision rescue's double-double D0 masters run the hand-written box kernels, now generic over binary64 and double-double, instead of the generated expressions; double-double C0 stays on the generated code (next to two-particle thresholds the triangle kernel keeps about 10^{-24} in double-double where the generated code keeps 10^{-32}). Binary64 is unchanged bit for bit: TIRust's binary64-only arrays of all 27500 stability points are byte-identical to those of TIRust 7022322. The rescued lanes change in their last bits (stability capture 031 changes the default mode's errors on 19 of the 27500 points, in their last digits, and no status, flag or binary64-only value; amplitude capture 014′s hexagon arrays change, its pentagon arrays do not, and both stay bit-identical between sharing on and off). TIRust's rows match capture 026 at the median (−4 to +6% outside N7R7), COLLIER's are 0.1% faster and IREGI's 0.3% slower; 14 of the 38 batch-1 and batch-32 measurements started on a core already partly busy (6 to 100%). N7R7 batch 32 sends 12 lanes to the precision rescue: it takes 181 ms per call (capture 026: 193 ms, capture 025: 219 ms) while its binary64-only mode takes 166 to 168 ms in all three, so the rescue's share went from 51 to 27 to 12 ms. N7R7 batch 1 is 9 to 10% faster than in capture 026 by default and contracted (2 to 4% in the binary64-only modes), back at capture 025′s level, as is IREGI's N4R6 batch 32. It ran alone. Capture 026 (TIRust c8b8264, load 141 to 225): in the precision rescue's double-double scalar masters, a complex product with a real factor skips the products with that factor's zero imaginary part, with bit-identical results. Most of their products have a real factor (rational constants, real invariants); products of two complex factors, as with complex masses, keep all four (amplitude section). Results are unchanged (stability capture 030 matches capture 029 in every TIRust status, error and diagnostic on all 27500 points; amplitude capture 013 is bit-identical to capture 012). Only lanes sent to the precision rescue run this code. TIRust's rows are a median 0.6% slower than in capture 025, COLLIER's 0.6% and IREGI's 0.5%. 13 of the 38 batch-1 and batch-32 measurements started on a core already partly busy (14 to 100% in the half second before; `cpu_busy_before` in the summary): there TIRust's rows are a median 1.2% slower than in capture 025, elsewhere 0.4%. TIRust's rows are within 5% of capture 025 except N7R7: batch 1 +4 to +11% (core 32% busy), batch 32 −12% by default, −4% contracted and −1 to +8% in the binary64-only modes, which never run the rescue (core fully busy). It ran alone. Capture 025 (TIRust 7022322, load 63 to 87): the small-Gram expansions skip the infrared coefficients on lanes without infrared poles, build their scalar factors once per expansion and compute only the coefficients their results depend on (amplitude section). Results are unchanged (stability capture 029 matches capture 028 in every TIRust status, error and diagnostic on all 27500 points; amplitude capture 012 is bit-identical to capture 011). TIRust's rows are a median 0.6% faster than in capture 024, COLLIER's 0.4% and IREGI's 0.2%, and within 7% of capture 024 except batch-1 N1R0 (−8.5%, 0.05 µs). It ran alone. Capture 024 (TIRust 06daf37, load 85 to 165): the amplitude store also shares the Cartesian moments of subsets that an integral gathers inside a larger one (amplitude section). These rows evaluate each integral alone, without the store, so nothing they compute changes: their errors are identical row by row (stability capture 028 matches capture 027 in every TIRust status, error and diagnostic on all 27500 points; amplitude capture 011 is bit-identical to capture 010). TIRust's rows are a median 1.4% faster than in capture 023, COLLIER's 2.3% and IREGI's 1.9% (capture 023: load ≈300), and within 8% of capture 023 except batch-1 N1R0 (+12%, 0.06 µs). It ran alone. Capture 023 (TIRust 0b607e9, load ≈300): when TIRust assembles the Cartesian moments of the subsets that start from the same propagator, it builds the products of their momenta once per call instead of once per subset, with identical results (stability capture 027 matches capture 026 in every TIRust status, error and diagnostic on all 27500 points; amplitude captures 009 and 010 are bit-identical to capture 008). These products grow with N and rank: TIRust's N5R6 rows are 4 to 11% faster than in capture 022, its N6R6 rows 1 to 13% and its N7R7 rows 6 to 31%. Its other rows are a median 0.6% slower (COLLIER's rows 2.9%, IREGI's 2.3%; capture 022: load ≈200) and within 7% of capture 022 except batch-1 N1R0 (−16%, 0.10 µs). It ran alone. Capture 022 (TIRust bdf47ab, load ≈200): the small-Gram expansions find the coefficients each step of their recursion reads through precomputed neighbour tables instead of recomputing each one's index, with identical results (stability capture 026 matches capture 025 in every TIRust status, error and diagnostic on all 27500 points; amplitude capture 008: the hexagon amplitude's time down 25%). These rows rarely need the expansions: TIRust's rows are a median 0.5% slower than in capture 021, COLLIER's 0.2%, IREGI's 0.8% faster (capture 021: load ≈220), and TIRust rows are within 7% of capture 021 except batch-1 N1R0 (+9 to +27%, at most 0.14 µs) and batch-1 binary64-only 2^{-20} N7R7 (+16%; its contracted mode +3%). It ran alone. Capture 021 (TIRust f06c5f3, load ≈220): a triangle of an auxiliary box of the small-Gram expansions whose expansion and small-Cayley variant cannot converge (COLLIER's convergence factors, raised to the number of passes of the largest boost, above 2^{-18} and 2^{-30}) goes straight to the alternative PV reduction, with identical results (stability capture 025, amplitude capture 007); TIRust's rows are a median 2% faster than in capture 020, COLLIER's 3%, IREGI's unchanged (capture 020: load ≈290); TIRust rows are within 7% of capture 020 except batch-1 N1R0 (−7 to −8%, 0.04 µs), the batch-32 2^{-20} binary64-only rows of N4R6 and N5R5 (−9% and −7%, after +7 to +9% in capture 020), batch-1 binary64-only N5R6 (−7%), batch-32 binary64-only contracted N7R7 (−10%) and the batch-1 N7R7 modes (−10 to −18%: every engine's batch-1 N7R7 time is within 1.2% of capture 018, so capture 020′s were host noise). Capture 021 ran alone; a first run alongside the stability capture had its extended rows disturbed (IREGI's up to 1.8 times slower than in capture 020) and was repeated. Capture 020 (TIRust a51d171, load ≈290): the small-Gram expansions store their coefficients in arrays indexed by rank instead of hash maps, with identical results (stability capture 024, amplitude capture 006); COLLIER's rows are a median 3% slower than in capture 019 (load ≈200), TIRust's 2% and IREGI's 1%; TIRust rows are within 7% of capture 019 except four batch-32 binary64-only rows of N4R6 to N5R6 (+7 to +9%), the batch-1 dotted N3R4 triangle (−11%, back to its capture-018 value, as is IREGI's) and five batch-1 N7R7 modes (+11 to +19%, host noise). Capture 019 (TIRust 939bb35): the small-Gram expansions obtain the sums over the pinched triangle's coefficients from a three-term recursion instead of binomial sums, and the variant for a small Cayley determinant also runs order by order; no status changes and arrays move only at the rounding level (stability capture 023, amplitude capture 005); its batch-1 jumps against capture 018 (dotted N3R4 +10 to +12%, default N7R7 +14%) were host noise. Capture 018 (TIRust f45d36f) no longer reruns accepted box and pentagon lanes for accuracy beyond the request (stability capture 022, amplitude capture 004). Capture 017 (TIRust a71e29d): the small-Gram expansions of the binary64 checks run order by order in Δ (the triangles enter once instead of in every pass) and stop after four orders when even their best observed rate cannot reach the tolerance; statuses and arrays are unchanged up to rounding (stability capture 021; amplitude capture 003 for the physical points where they dominate). These rows rarely need the expansions. In capture 017, at a higher host load (≈420 against ≈205) every code is slower than in capture 016, COLLIER by 7 to 22% and TIRust by up to 34% (N3R3 batch 32); the same-session ratios are unchanged within that noise (separate sessions; no ratio across them). Capture 016 (TIRust 5a6d415): with the precision rescue disabled, heptagon lanes whose cyclic orders disagree after passing the gate skip the binary64 retry that computes every box and triangle off PV, which kept none of them. IREGI was not rerun on N7R7 batch 32 (about half an hour per capture); its cells show capture 014′s measurement and that capture's own ratio. The binary64-

only modes start with capture 015. Since capture 014 the double-double rescue tier uses fused multiply-accumulate kernels (Dekker products and one exact sum per component, in safe code: the baseline build no longer runs each double-double product through out-of-line `fma` calls; the scalar masters' part is oneloopmaster PR #16). Since capture 026 a double-double complex product with a real factor skips the products with that factor's zero imaginary part, bit-identically. Since capture 027 its D0 masters run the hand-written box kernels (binary64's, made generic), since capture 028 its C0 masters the hand-written triangle kernel. N7R7 batch 32 takes 0.23 s (capture 013: 0.35 s, capture 011: 0.67 s, capture 010: 1.10 s; COLLIER's 0.15 s returns NaN on lane 0). Its 12 lanes refused by the rotation test go to the precision rescue directly (since capture 011; before, first through binary64 retries that do not succeed there), whose double-double tier does not evaluate them a second time with rotated propagators (since capture 012); with capture 014's kernels the rescued lanes agreed with 256 bits to 1.3×10^{-26} (not measured since). About 0.1 s of the batch is spent in the three binary64 passes of the rotation test (development profile on the same host). Its 20 lanes kept in binary64 reach E/A 9.7×10^{-11}, above the 2^{-34} request: their cyclic orders share the error, which the rotation test cannot see. Capture 009 (not published) measured a version that also gave accepted hexagon lanes the per-box choice: the N6R6 batch of 32 took 151.9 ms (capture 008: 17.3 ms). Since capture 006 the undotted box rows take two binary64 evaluations (the loop-rotation test; N4R4 35.6 µs, capture 005: 16.6 µs; N4R6 58.0 µs, 24.3 µs), still below COLLIER's; the dotted boxes (unequal powers) keep one. In N7R7 batch 32 the 12 lanes refused by the rotation test (a third cyclic order, since capture 005; 8 of them were off by more than the tolerance, up to E/A 1.2×10^{-9}) are rerun at higher precision (1.08 s, capture 005: 1.34 s, capture 004 with one rescued lane: 107 ms). Lane 0, whose E/A the table shows, is unchanged. The COLLIER driver reads its accuracy flags per lane (since capture 005). Capture 003 added the N7R7 row (the opening shifts and masses plus $q_6 = (1/4, 3/8, 1/8, 5/8)$, $m_6^2 = 4$; reference: TIRust at 256 bits confirmed by quad IREGI IBP to 6.3×10^{-17}, quad PV fails, COLLIER returns NaN, `validation/mre/collier-n7-nan`) and NinjaRS. TIRust's undotted N≥4 rows include the loop-rotation test (boxes since capture 006), the N≥5 rows the Gram-free pentagon reduction, which keep physical pentagons accurate in binary64 (capture 002, without them: N5R5 65 µs, N6R6 216 µs). Captures 001 (TIRust `c063a78`) to 015 remain bound unchanged in the report data (009 not published).

Table 1: Single phase-space point — median of paired block means

Input	TIRust (direct)	IREGI (best mode)	TIRust / IREGI	COLLIER 1.2.10	TIRust / COLLIER
N=1, R=0 1 moment × 4 slots	**557 ns** E/A 0	**804 ns** PV · E/A 0	**×0.698** 99% CI 0.6862–0.7087	**442 ns** COLI · E/A 0	**×1.259** 99% CI 1.2279–1.2746
N=1, R=6 210 moments × 4 slots	**3.16 µs** E/A 1.6×10^{-20}	**24.02 µs** Hybrid · E/A 3.8×10^{-19}	**×0.132** 99% CI 0.1312–0.1328	**4.87 µs** DD · E/A 1.6×10^{-20}	**×0.656** 99% CI 0.6463–0.6644
N=2, R=1 5 moments × 4 slots	**1.14 µs** E/A 4.4×10^{-16}	**3.11 µs** PV · E/A 2.2×10^{-16}	**×0.367** 99% CI 0.3637–0.3729	**1.18 µs** COLI · E/A 1.1×10^{-15}	**×0.972** 99% CI 0.9514–0.9787
N=2, R=6 210 moments × 4 slots	**5.20 µs** E/A 4.4×10^{-16}	**394.44 µs** IBP · E/A 2.2×10^{-16}	**×0.013** 99% CI 0.0131–0.0133	**9.41 µs** DD · E/A 2.2×10^{-16}	**×0.552** 99% CI 0.5465–0.5598
N=3, R=2 15 moments × 4 slots	**4.31 µs** E/A 4.4×10^{-15}	**25.85 µs** PV · E/A 1.9×10^{-15}	**×0.169** 99% CI 0.1661–0.1697	**8.56 µs** COLI · E/A 7.0×10^{-15}	**×0.504** 99% CI 0.5030–0.5080
N=3, R=3 35 moments × 4 slots	**5.05 µs** E/A 4.4×10^{-15}	**68.88 µs** PV · E/A 1.9×10^{-15}	**×0.073** 99% CI 0.0731–0.0738	**11.05 µs** COLI · E/A 7.0×10^{-15}	**×0.453** 99% CI 0.4462–0.4703
N=3, R=6 210 moments × 4 slots	**9.67 µs** E/A 4.4×10^{-15}	**2.78 ms** IBP · E/A 1.9×10^{-15}	**×0.0035** 99% CI 0.0035–0.0035	**51.97 µs** COLI · E/A 7.0×10^{-15}	**×0.186** 99% CI 0.1855–0.1874
N=4, R=4 70 moments × 4 slots	**35.34 µs** E/A 1.6×10^{-14}	**531.69 µs** PV · E/A 1.0×10^{-14}	**×0.066** 99% CI 0.0660–0.0675	**50.16 µs** COLI · E/A 9.7×10^{-15}	**×0.703** 99% CI 0.6947–0.7154
N=4, R=6 210 moments × 4 slots	**57.77 µs** E/A 1.6×10^{-14}	**12.79 ms** IBP · E/A 1.0×10^{-14}	**×0.0045** 99% CI 0.0045–0.0046	**271.22 µs** COLI · E/A 9.7×10^{-15}	**×0.214** 99% CI 0.2089–0.2159
N=5, R=5 126 moments × 4 slots	**138.22 µs** E/A 4.7×10^{-12}	**8.90 ms** PV · E/A 9.7×10^{-12}	**×0.016** 99% CI 0.0153–0.0158	**307.40 µs** COLI · E/A 1.1×10^{-11}	**×0.450** 99% CI 0.4288–0.4552
N=5, R=6 210 moments × 4 slots	**185.08 µs** E/A 1.8×10^{-11}	**47.22 ms** IBP · E/A 3.7×10^{-11}	**×0.0039** 99% CI 0.0038–0.0040	**1.11 ms** COLI, perm. · E/A 4.3×10^{-11}	**×0.167** 99% CI 0.1651–0.1731
N=6, R=6 210 moments × 4 slots	**494.25 µs** E/A 8.6×10^{-12}	**207.04 ms** IBP · E/A 1.8×10^{-11}	**×0.0024** 99% CI 0.0023–0.0025	**2.03 ms** COLI · nonfinite/refused · diagnostic	No ratio
N=7, R=7 330 moments × 4 slots	**2.63 ms** E/A 2.0×10^{-11}	**3.80 s** IBP · E/A 2.9×10^{-6}	**×0.0007** 99% CI 0.0007–0.0007	**5.29 ms** COLI · nonfinite/refused · diagnostic	No ratio
N=1, R=6 ν_1=2 210 moments × 4 slots	**3.84 µs** E/A 4.5×10^{-16}	**35.79 µs** IBP · E/A 9.0×10^{-17}	**×0.108** 99% CI 0.1048–0.1118	Outside public API	No ratio
N=2, R=2 ν_1=2 15 moments × 4 slots	**2.60 µs** E/A 2.6×10^{-16}	**27.98 µs** IBP · E/A 1.3×10^{-16}	**×0.093** 99% CI 0.0917–0.0939	Outside public API	No ratio
N=3, R=4 ν_1=2 70 moments × 4 slots	**9.95 µs** E/A 5.7×10^{-15}	**540.47 µs** IBP · E/A 5.4×10^{-15}	**×0.018** 99% CI 0.0183–0.0185	Outside public API	No ratio
N=4, R=6 ν_1=2 210 moments × 4 slots	**43.88 µs** E/A 6.0×10^{-14}	**15.14 ms** IBP · E/A 7.8×10^{-14}	**×0.0029** 99% CI 0.0029–0.0029	Outside public API	No ratio
N=4, R=6 ν_1=3 210 moments × 4 slots	**68.05 µs** E/A 8.1×10^{-14}	**18.72 ms** IBP · E/A 9.9×10^{-14}	**×0.0036** 99% CI 0.0036–0.0036	Outside public API	No ratio

Input	TIRust (direct)	IREGI (best mode)	TIRust / IREGI	COLLIER 1.2.10	TIRust / COLLIER
N=3, R=2 D=6 15 moments × 4 slots	**4.68 µs** E/A 1.4×10^{-14}	**78.63 µs** IBP · E/A 1.1×10^{-14}	**×0.059** 99% CI 0.0585–0.0598	Outside public API	No ratio

Table 2: Batch of 32 points, time per point — median of paired block means

Input	TIRust (direct)	IREGI (best mode)	TIRust / IREGI	COLLIER 1.2.10	TIRust / COLLIER
N=1, R=0 1 moment × 4 slots	**256 ns** /point E/A 0	**818 ns** /point PV · E/A 0	**×0.311** 99% CI 0.3056–0.3159	**414 ns** /point COLI · E/A 0	**×0.616** 99% CI 0.6089–0.6209
N=1, R=6 210 moments × 4 slots	**2.71 µs** /point E/A 1.6×10^{-20}	**23.90 µs** /point Hybrid · E/A 3.8×10^{-19}	**×0.113** 99% CI 0.1112–0.1143	**4.65 µs** /point DD · E/A 1.6×10^{-20}	**×0.578** 99% CI 0.5754–0.5875
N=2, R=1 5 moments × 4 slots	**832 ns** /point E/A 4.4×10^{-16}	**3.11 µs** /point PV · E/A 2.2×10^{-16}	**×0.268** 99% CI 0.2628–0.2784	**1.16 µs** /point COLI · E/A 1.1×10^{-15}	**×0.720** 99% CI 0.7101–0.7366
N=2, R=6 210 moments × 4 slots	**4.69 µs** /point E/A 4.4×10^{-16}	**406.76 µs** /point IBP · E/A 2.2×10^{-16}	**×0.011** 99% CI 0.0111–0.0119	**9.00 µs** /point DD · E/A 2.2×10^{-16}	**×0.522** 99% CI 0.4959–0.5333
N=3, R=2 15 moments × 4 slots	**4.10 µs** /point E/A 4.4×10^{-15}	**25.85 µs** /point PV · E/A 1.9×10^{-15}	**×0.156** 99% CI 0.1544–0.1606	**8.51 µs** /point COLI · E/A 7.0×10^{-15}	**×0.477** 99% CI 0.4681–0.4869
N=3, R=3 35 moments × 4 slots	**4.55 µs** /point E/A 4.4×10^{-15}	**68.73 µs** /point Hybrid · E/A 1.9×10^{-15}	**×0.066** 99% CI 0.0631–0.0669	**11.08 µs** /point COLI · E/A 7.0×10^{-15}	**×0.412** 99% CI 0.3925–0.4146
N=3, R=6 210 moments × 4 slots	**8.94 µs** /point E/A 4.4×10^{-15}	**2.75 ms** /point IBP · E/A 1.9×10^{-15}	**×0.0032** 99% CI 0.0032–0.0033	**51.30 µs** /point COLI · E/A 7.0×10^{-15}	**×0.173** 99% CI 0.1665–0.1762
N=4, R=4 70 moments × 4 slots	**35.36 µs** /point E/A 1.6×10^{-14}	**535.30 µs** /point PV · E/A 1.0×10^{-14}	**×0.066** 99% CI 0.0654–0.0689	**52.05 µs** /point COLI · E/A 9.7×10^{-15}	**×0.681** 99% CI 0.6615–0.6957
N=4, R=6 210 moments × 4 slots	**57.63 µs** /point E/A 1.6×10^{-14}	**12.84 ms** /point IBP · E/A 1.0×10^{-14}	**×0.0045** 99% CI 0.0045–0.0045	**271.41 µs** /point COLI · E/A 9.7×10^{-15}	**×0.212** 99% CI 0.2106–0.2161
N=5, R=5 126 moments × 4 slots	**142.11 µs** /point E/A 4.7×10^{-12}	**8.80 ms** /point Hybrid · E/A 9.7×10^{-12}	**×0.016** 99% CI 0.0159–0.0165	**310.64 µs** /point COLI · E/A 1.1×10^{-11}	**×0.457** 99% CI 0.4465–0.4658
N=5, R=6 210 moments × 4 slots	**184.84 µs** /point E/A 1.8×10^{-11}	**47.54 ms** /point IBP · E/A 3.7×10^{-11}	**×0.0039** 99% CI 0.0038–0.0039	**1.08 ms** /point COLI, perm. · E/A 4.3×10^{-11}	**×0.172** 99% CI 0.1681–0.1755
N=6, R=6 210 moments × 4 slots	**498.44 µs** /point E/A 8.6×10^{-12}	**214.54 ms** /point IBP · E/A 1.8×10^{-11}	**×0.0023** 99% CI 0.0023–0.0024	**1.54 ms** /point COLI · nonfinite/refused · diagnostic	No ratio
N=7, R=7 330 moments × 4 slots	**5.36 ms** /point E/A 2.0×10^{-11}	Not rerun in this capture capture 014: 4.00 s/point · IBP	capture 014 (its own session): ×0.0018	**4.85 ms** /point COLI · nonfinite/refused · diagnostic	No ratio
N=1, R=6 ν_1=2 210 moments × 4 slots	**3.32 µs** /point E/A 4.5×10^{-16}	**35.63 µs** /point IBP · E/A 9.0×10^{-17}	**×0.093** 99% CI 0.0920–0.0939	Outside public API	No ratio
N=2, R=2 ν_1=2 15 moments × 4 slots	**2.17 µs** /point E/A 2.6×10^{-16}	**32.58 µs** /point IBP · E/A 1.3×10^{-16}	**×0.066** 99% CI 0.0661–0.0671	Outside public API	No ratio
N=3, R=4 ν_1=2 70 moments × 4 slots	**9.63 µs** /point E/A 5.7×10^{-15}	**569.77 µs** /point IBP · E/A 5.4×10^{-15}	**×0.017** 99% CI 0.0168–0.0174	Outside public API	No ratio
N=4, R=6 ν_1=2 210 moments × 4 slots	**42.76 µs** /point E/A 6.0×10^{-14}	**15.33 ms** /point IBP · E/A 7.8×10^{-14}	**×0.0028** 99% CI 0.0028–0.0029	Outside public API	No ratio
N=4, R=6 ν_1=3 210 moments × 4 slots	**68.42 µs** /point E/A 8.1×10^{-14}	**19.38 ms** /point IBP · E/A 9.9×10^{-14}	**×0.0035** 99% CI 0.0035–0.0036	Outside public API	No ratio
N=3, R=2 D=6 15 moments × 4 slots	**4.40 µs** /point E/A 1.4×10^{-14}	**85.08 µs** /point IBP · E/A 1.1×10^{-14}	**×0.052** 99% CI 0.0508–0.0520	Outside public API	No ratio

TIRust and NinjaRS. NinjaRS (alphal00p/ninjars, a Rust port of Ninja's integrand reduction) integrates a given numerator, not the tensor array. Each row therefore gets a dense random numerator of its rank, and TIRust is timed with the contraction of its complete array with it inside every timed call. This is not like for like: TIRust's array serves every numerator of the topology, NinjaRS repeats its reduction per numerator. NinjaRS runs `case-bench` on the same lanes (pinned, one fresh process and ≈20 ms block per round, interleaved with the other candidates). Its fast engine, like original Ninja, marks cuts whose two momenta span a Euclidean plane `Unstable`; the opening inputs (spatial unit vectors) have such planes throughout, so these rows use NinjaRS's generic engine, as its documentation prescribes. E/A of NinjaRS: its lane-0 value against the reference array contracted exactly with the same numerator. On the N6R6 and N7R7 rows NinjaRS's generic engine fails its lane-0 cut construction (`CutFailed`, singular parameter; the opening points contain coplanar quadruples such as 0, e_x, e_y, e_x+e_y), so those times have no accepted value. Reference build: upstream 41a4c00 with the serde_json `float_roundtrip` repair (upstream pull request), without which NinjaRS reads about 3% of binary64 inputs one ulp off.

Table 3: TIRust (array and contraction with a dense random numerator) against NinjaRS (integrand reduction of that numerator); single point

Input	TIRust (array only)	TIRust (array + contraction)	NinjaRS	TIRust (contracted) / NinjaRS
N=1, R=0 1 moment × 4 slots	**557 ns** E/A 0	**543 ns** E/A 0	**269 ns** fast engine · contracted E/A 1.5×10^{-16}	**×2.019** 99% CI 1.9962–2.0851
N=1, R=6 210 moments × 4 slots	**3.16 µs** E/A 1.6×10^{-20}	**3.86 µs** E/A 1.6×10^{-20}	Outside API (rank > N+2, dotted or D≠4)	No ratio
N=2, R=1 5 moments × 4 slots	**1.14 µs** E/A 4.4×10^{-16}	**1.18 µs** E/A 4.4×10^{-16}	**1.10 µs** fast engine · contracted E/A 5.5×10^{-16}	**×1.069** 99% CI 1.0535–1.0872
N=2, R=6 210 moments × 4 slots	**5.20 µs** E/A 4.4×10^{-16}	**6.03 µs** E/A 4.4×10^{-16}	Outside API (rank > N+2, dotted or D≠4)	No ratio
N=3, R=2 15 moments × 4 slots	**4.31 µs** E/A 4.4×10^{-15}	**4.37 µs** E/A 4.4×10^{-15}	**47.76 µs** generic engine (fast: 1 Unstable lanes) · contracted E/A 7.9×10^{-15}	**×0.092** 99% CI 0.0906–0.0922
N=3, R=3 35 moments × 4 slots	**5.05 µs** E/A 4.4×10^{-15}	**5.18 µs** E/A 4.4×10^{-15}	**76.27 µs** generic engine (fast: 1 Unstable lanes) · contracted E/A 6.0×10^{-15}	**×0.068** 99% CI 0.0628–0.0695
N=3, R=6 210 moments × 4 slots	**9.67 µs** E/A 4.4×10^{-15}	**10.50 µs** E/A 4.4×10^{-15}	Outside API (rank > N+2, dotted or D≠4)	No ratio
N=4, R=4 70 moments × 4 slots	**35.34 µs** E/A 1.6×10^{-14}	**35.58 µs** E/A 1.6×10^{-14}	**257.39 µs** generic engine (fast: 1 Unstable lanes) · contracted E/A 8.3×10^{-14}	**×0.139** 99% CI 0.1350–0.1395
N=4, R=6 210 moments × 4 slots	**57.77 µs** E/A 1.6×10^{-14}	**58.37 µs** E/A 1.6×10^{-14}	**1.51 ms** generic engine (fast: 1 Unstable lanes) · contracted E/A 1.6×10^{-12}	**×0.039** 99% CI 0.0384–0.0388
N=5, R=5 126 moments × 4 slots	**138.22 µs** E/A 4.7×10^{-12}	**139.35 µs** E/A 4.7×10^{-12}	**747.03 µs** generic engine (fast: 1 Unstable lanes) · contracted E/A 8.5×10^{-12}	**×0.187** 99% CI 0.1851–0.1884
N=5, R=6 210 moments × 4 slots	**185.08 µs** E/A 1.8×10^{-11}	**188.19 µs** E/A 1.8×10^{-11}	**1.82 ms** generic engine (fast: 1 Unstable lanes) · contracted E/A 2.3×10^{-11}	**×0.102** 99% CI 0.1003–0.1053
N=6, R=6 210 moments × 4 slots	**494.25 µs** E/A 8.6×10^{-12}	**494.54 µs** E/A 8.6×10^{-12}	**1.37 ms** generic engine (fast: 1 Unstable lanes) · no value	No ratio
N=7, R=7 330 moments × 4 slots	**2.63 ms** E/A 2.0×10^{-11}	**2.65 ms** E/A 2.0×10^{-11}	**3.65 ms** generic engine (fast: 1 Unstable lanes) · no value	No ratio

Table 4: TIRust (array and contraction with a dense random numerator) against NinjaRS (integrand reduction of that numerator); batch of 32, time per point

Input	TIRust (array only)	TIRust (array + contraction)	NinjaRS	TIRust (contracted) / NinjaRS
N=1, R=0 1 moment × 4 slots	**256 ns** /point E/A 0	**258 ns** /point E/A 0	**127 ns** /point fast engine · contracted E/A 1.5×10^{-16}	**×2.027** 99% CI 1.9919–2.0694
N=1, R=6 210 moments × 4 slots	**2.71 µs** /point E/A 1.6×10^{-20}	**3.40 µs** /point E/A 1.6×10^{-20}	Outside API (rank > N+2, dotted or D≠4)	No ratio
N=2, R=1 5 moments × 4 slots	**832 ns** /point E/A 4.4×10^{-16}	**852 ns** /point E/A 4.4×10^{-16}	**614 ns** /point fast engine · contracted E/A 5.5×10^{-16}	**×1.374** 99% CI 1.3656–1.3954
N=2, R=6 210 moments × 4 slots	**4.69 µs** /point E/A 4.4×10^{-16}	**5.53 µs** /point E/A 4.4×10^{-16}	Outside API (rank > N+2, dotted or D≠4)	No ratio
N=3, R=2 15 moments × 4 slots	**4.10 µs** /point E/A 4.4×10^{-15}	**4.09 µs** /point E/A 4.4×10^{-15}	**39.32 µs** /point generic engine (fast: 32 Unstable lanes) · contracted E/A 7.9×10^{-15}	**×0.104** 99% CI 0.1021–0.1068

Input	TIRust (array only)	TIRust (array + contraction)	NinjaRS	TIRust (contracted) / NinjaRS
N=3, R=3 35 moments × 4 slots	**4.55 µs** /point E/A 4.4×10^{-15}	**4.71 µs** /point E/A 4.4×10^{-15}	**69.13 µs** /point generic engine (fast: 32 Unstable lanes) · contracted E/A 6.0×10^{-15}	**×0.069** 99% CI 0.0673–0.0730
N=3, R=6 210 moments × 4 slots	**8.94 µs** /point E/A 4.4×10^{-15}	**9.72 µs** /point E/A 4.4×10^{-15}	Outside API (rank > N+2, dotted or D≠4)	No ratio
N=4, R=4 70 moments × 4 slots	**35.36 µs** /point E/A 1.6×10^{-14}	**35.99 µs** /point E/A 1.6×10^{-14}	**254.63 µs** /point generic engine (fast: 32 Unstable lanes) · contracted E/A 8.3×10^{-14}	**×0.142** 99% CI 0.1293–0.1445
N=4, R=6 210 moments × 4 slots	**57.63 µs** /point E/A 1.6×10^{-14}	**58.88 µs** /point E/A 1.6×10^{-14}	**1.43 ms** /point generic engine (fast: 32 Unstable lanes) · contracted E/A 1.6×10^{-12}	**×0.041** 99% CI 0.0405–0.0424
N=5, R=5 126 moments × 4 slots	**142.11 µs** /point E/A 4.7×10^{-12}	**140.81 µs** /point E/A 4.7×10^{-12}	**736.77 µs** /point generic engine (fast: 32 Unstable lanes) · contracted E/A 8.5×10^{-12}	**×0.192** 99% CI 0.1904–0.1998
N=5, R=6 210 moments × 4 slots	**184.84 µs** /point E/A 1.8×10^{-11}	**185.64 µs** /point E/A 1.8×10^{-11}	**1.75 ms** /point generic engine (fast: 32 Unstable lanes) · contracted E/A 2.3×10^{-11}	**×0.105** 99% CI 0.1039–0.1081
N=6, R=6 210 moments × 4 slots	**498.44 µs** /point E/A 8.6×10^{-12}	**500.45 µs** /point E/A 8.6×10^{-12}	**1.36 ms** /point generic engine (fast: 32 Unstable lanes) · no value	No ratio
N=7, R=7 330 moments × 4 slots	**5.36 ms** /point E/A 2.0×10^{-11}	**5.42 ms** /point E/A 2.0×10^{-11}	**3.52 ms** /point generic engine (fast: 32 Unstable lanes) · no value	No ratio

TIRust binary64 only. The same driver with the precision rescue disabled (`TensorWorkspaceOptions::precision_rescue = Disabled`), at the default accuracy request 2^{-34} and at 2^{-20} (`set_accuracy_precision(39)`); their accuracy is the stability section's curves "TIRust binary64 only". Every lane is computed in binary64; a lane its checks refuse returns its best binary64 array, uncertified and flagged ("flagged" in the cell), and counts as complete, as COLLIER's flagged lanes do. With the rescue disabled a refused lane goes through TIRust's binary64 retries and the best-effort search over cyclic orders instead of the precision rescue. Against NinjaRS each mode is timed with the contraction of its array with the row's numerator, as "TIRust (array + contraction)" above; NinjaRS rows as above. Same session, TIRust in the numerator, acceptance and ratios as in the tables above. Capture 016: on every row but N7R7 no lane is refused, so both modes cost what the default costs and their ratios to COLLIER and NinjaRS are the default's. N7R7: single point 3.3 ms at 2^{-34} and 2.3 ms at 2^{-20} batch of 32 0.27 s at 2^{-34} (capture 015: 0.67 s), whose 12 lanes refused by the rotation test go through the a priori method choice and the best-effort search over cyclic orders and return flagged (since this capture without the retry that computes every box and triangle off PV, which kept none of them; the default rescues them at 106 bits: 0.23 s), and 65 ms at 2^{-20}, where no lane is refused. COLLIER takes 0.15 s but returns NaN on lane 0, and NinjaRS fails its lane-0 cut there, so N7R7 has no ratio.

Table 5: TIRust binary64 only (precision rescue disabled) against COLLIER; single point

Input	TIRust binary64, request 2^{-34}	TIRust / COLLIER	TIRust binary64, request 2^{-20}	TIRust / COLLIER	COLLIER 1.2.10
N=1, R=0 1 moment × 4 slots	**539 ns** E/A 0	**×1.210** 99% CI 1.1987–1.2248	**565 ns** E/A 0	**×1.273** 99% CI 1.2474–1.3055	**442 ns** COLI · E/A 0
N=1, R=6 210 moments × 4 slots	**3.15 µs** E/A 1.6×10^{-20}	**×0.650** 99% CI 0.6419–0.6586	**3.15 µs** E/A 1.6×10^{-20}	**×0.645** 99% CI 0.6400–0.6501	**4.87 µs** DD · E/A 1.6×10^{-20}
N=2, R=1 5 moments × 4 slots	**1.14 µs** E/A 4.4×10^{-16}	**×0.962** 99% CI 0.9487–0.9698	**1.14 µs** E/A 4.4×10^{-16}	**×0.959** 99% CI 0.9574–0.9699	**1.18 µs** COLI · E/A 1.1×10^{-15}
N=2, R=6 210 moments × 4 slots	**5.24 µs** E/A 4.4×10^{-16}	**×0.557** 99% CI 0.5518–0.5597	**5.30 µs** E/A 4.4×10^{-16}	**×0.565** 99% CI 0.5579–0.5681	**9.41 µs** DD · E/A 2.2×10^{-16}
N=3, R=2 15 moments × 4 slots	**4.30 µs** E/A 4.4×10^{-15}	**×0.503** 99% CI 0.4989–0.5066	**4.34 µs** E/A 4.4×10^{-15}	**×0.506** 99% CI 0.5029–0.5192	**8.56 µs** COLI · E/A 7.0×10^{-15}
N=3, R=3 35 moments × 4 slots	**5.03 µs** E/A 4.4×10^{-15}	**×0.455** 99% CI 0.4481–0.4616	**5.12 µs** E/A 4.4×10^{-15}	**×0.463** 99% CI 0.4522–0.4754	**11.05 µs** COLI · E/A 7.0×10^{-15}
N=3, R=6 210 moments × 4 slots	**9.67 µs** E/A 4.4×10^{-15}	**×0.186** 99% CI 0.1845–0.1874	**9.66 µs** E/A 4.4×10^{-15}	**×0.186** 99% CI 0.1849–0.1867	**51.97 µs** COLI · E/A 7.0×10^{-15}
N=4, R=4 70 moments × 4 slots	**35.07 µs** E/A 1.6×10^{-14}	**×0.700** 99% CI 0.6907–0.7109	**35.25 µs** E/A 1.6×10^{-14}	**×0.704** 99% CI 0.6997–0.7144	**50.16 µs** COLI · E/A 9.7×10^{-15}
N=4, R=6 210 moments × 4 slots	**58.78 µs** E/A 1.6×10^{-14}	**×0.217** 99% CI 0.2123–0.2197	**57.66 µs** E/A 1.6×10^{-14}	**×0.212** 99% CI 0.2056–0.2203	**271.22 µs** COLI · E/A 9.7×10^{-15}
N=5, R=5 126 moments × 4 slots	**139.17 µs** E/A 4.7×10^{-12}	**×0.450** 99% CI 0.4306–0.4593	**138.86 µs** E/A 4.7×10^{-12}	**×0.450** 99% CI 0.4384–0.4608	**307.40 µs** COLI · E/A 1.1×10^{-11}
N=5, R=6 210 moments × 4 slots	**183.67 µs** E/A 1.8×10^{-11}	**×0.167** 99% CI 0.1652–0.1689	**183.22 µs** E/A 1.8×10^{-11}	**×0.167** 99% CI 0.1643–0.1694	**1.11 ms** COLI, perm. · E/A 4.3×10^{-11}
N=6, R=6 210 moments × 4 slots	**505.62 µs** E/A 8.6×10^{-12}	No ratio	**497.33 µs** E/A 8.6×10^{-12}	No ratio	**2.03 ms** COLI · nonfinite/refused · diagnostic
N=7, R=7 330 moments × 4 slots	**2.66 ms** E/A 2.0×10^{-11}	No ratio	**1.64 ms** E/A 2.0×10^{-11}	No ratio	**5.29 ms** COLI · nonfinite/refused · diagnostic
N=1, R=6 ν_1=2 210 moments × 4 slots	**3.81 µs** E/A 4.5×10^{-16}	No ratio	**3.76 µs** E/A 4.5×10^{-16}	No ratio	Outside public API
N=2, R=2 ν_1=2 15 moments × 4 slots	**2.56 µs** E/A 2.6×10^{-16}	No ratio	**2.54 µs** E/A 2.6×10^{-16}	No ratio	Outside public API
N=3, R=4 ν_1=2 70 moments × 4 slots	**10.24 µs** E/A 5.7×10^{-15}	No ratio	**9.93 µs** E/A 5.7×10^{-15}	No ratio	Outside public API
N=4, R=6 ν_1=2 210 moments × 4 slots	**43.54 µs** E/A 6.0×10^{-14}	No ratio	**44.24 µs** E/A 6.0×10^{-14}	No ratio	Outside public API
N=4, R=6 ν_1=3 210 moments × 4 slots	**68.26 µs** E/A 8.1×10^{-14}	No ratio	**68.47 µs** E/A 8.1×10^{-14}	No ratio	Outside public API
N=3, R=2 D=6 15 moments × 4 slots	**4.65 µs** E/A 1.4×10^{-14}	No ratio	**4.64 µs** E/A 1.4×10^{-14}	No ratio	Outside public API

Table 6: TIRust binary64 only (precision rescue disabled) against COLLIER; batch of 32, time per point

Input	TIRust binary64, request 2^{-34}	TIRust / COLLIER	TIRust binary64, request 2^{-20}	TIRust / COLLIER	COLLIER 1.2.10
N=1, R=0 1 moment × 4 slots	**252 ns** /point E/A 0	**×0.603** 99% CI 0.5987–0.6129	**256 ns** /point E/A 0	**×0.618** 99% CI 0.6063–0.6214	**414 ns** /point COLI · E/A 0
N=1, R=6 210 moments × 4 slots	**2.67 µs** /point E/A 1.6×10^{-20}	**×0.569** 99% CI 0.5661–0.5802	**2.76 µs** /point E/A 1.6×10^{-20}	**×0.592** 99% CI 0.5803–0.6003	**4.65 µs** /point DD · E/A 1.6×10^{-20}
N=2, R=1 5 moments × 4 slots	**831 ns** /point E/A 4.4×10^{-16}	**×0.711** 99% CI 0.6989–0.7200	**841 ns** /point E/A 4.4×10^{-16}	**×0.714** 99% CI 0.7044–0.7436	**1.16 µs** /point COLI · E/A 1.1×10^{-15}
N=2, R=6 210 moments × 4 slots	**4.78 µs** /point E/A 4.4×10^{-16}	**×0.529** 99% CI 0.5123–0.5451	**4.69 µs** /point E/A 4.4×10^{-16}	**×0.522** 99% CI 0.4845–0.5333	**9.00 µs** /point DD · E/A 2.2×10^{-16}
N=3, R=2 15 moments × 4 slots	**4.06 µs** /point E/A 4.4×10^{-15}	**×0.476** 99% CI 0.4685–0.4896	**4.01 µs** /point E/A 4.4×10^{-15}	**×0.470** 99% CI 0.4637–0.4820	**8.51 µs** /point COLI · E/A 7.0×10^{-15}
N=3, R=3 35 moments × 4 slots	**4.70 µs** /point E/A 4.4×10^{-15}	**×0.414** 99% CI 0.4070–0.4279	**4.60 µs** /point E/A 4.4×10^{-15}	**×0.415** 99% CI 0.3967–0.4396	**11.08 µs** /point COLI · E/A 7.0×10^{-15}
N=3, R=6 210 moments × 4 slots	**8.96 µs** /point E/A 4.4×10^{-15}	**×0.175** 99% CI 0.1674–0.1815	**8.90 µs** /point E/A 4.4×10^{-15}	**×0.173** 99% CI 0.1662–0.1755	**51.30 µs** /point COLI · E/A 7.0×10^{-15}
N=4, R=4 70 moments × 4 slots	**35.55 µs** /point E/A 1.6×10^{-14}	**×0.684** 99% CI 0.6657–0.7012	**35.68 µs** /point E/A 1.6×10^{-14}	**×0.686** 99% CI 0.6808–0.7025	**52.05 µs** /point COLI · E/A 9.7×10^{-15}
N=4, R=6 210 moments × 4 slots	**57.59 µs** /point E/A 1.6×10^{-14}	**×0.212** 99% CI 0.2098–0.2179	**57.60 µs** /point E/A 1.6×10^{-14}	**×0.211** 99% CI 0.2094–0.2167	**271.41 µs** /point COLI · E/A 9.7×10^{-15}
N=5, R=5 126 moments × 4 slots	**142.46 µs** /point E/A 4.7×10^{-12}	**×0.455** 99% CI 0.4421–0.4791	**139.80 µs** /point E/A 4.7×10^{-12}	**×0.452** 99% CI 0.4432–0.4567	**310.64 µs** /point COLI · E/A 1.1×10^{-11}
N=5, R=6 210 moments × 4 slots	**185.84 µs** /point E/A 1.8×10^{-11}	**×0.173** 99% CI 0.1667–0.1743	**183.44 µs** /point E/A 1.8×10^{-11}	**×0.170** 99% CI 0.1650–0.1763	**1.08 ms** /point COLI, perm. · E/A 4.3×10^{-11}
N=6, R=6 210 moments × 4 slots	**502.38 µs** /point E/A 8.6×10^{-12}	No ratio	**468.60 µs** /point E/A 8.6×10^{-12}	No ratio	**1.54 ms** /point COLI · nonfinite/refused · diagnostic
N=7, R=7 330 moments × 4 slots	**5.20 ms** /point E/A 2.0×10^{-11} · 12 flagged	No ratio	**1.59 ms** /point E/A 2.0×10^{-11}	No ratio	**4.85 ms** /point COLI · nonfinite/refused · diagnostic
N=1, R=6 ν_1=2 210 moments × 4 slots	**3.28 µs** /point E/A 4.5×10^{-16}	No ratio	**3.24 µs** /point E/A 4.5×10^{-16}	No ratio	Outside public API
N=2, R=2 ν_1=2 15 moments × 4 slots	**2.16 µs** /point E/A 2.6×10^{-16}	No ratio	**2.15 µs** /point E/A 2.6×10^{-16}	No ratio	Outside public API
N=3, R=4 ν_1=2 70 moments × 4 slots	**9.64 µs** /point E/A 5.7×10^{-15}	No ratio	**9.54 µs** /point E/A 5.7×10^{-15}	No ratio	Outside public API
N=4, R=6 ν_1=2 210 moments × 4 slots	**43.08 µs** /point E/A 6.0×10^{-14}	No ratio	**43.18 µs** /point E/A 6.0×10^{-14}	No ratio	Outside public API
N=4, R=6 ν_1=3 210 moments × 4 slots	**68.33 µs** /point E/A 8.1×10^{-14}	No ratio	**68.73 µs** /point E/A 8.1×10^{-14}	No ratio	Outside public API
N=3, R=2 D=6 15 moments × 4 slots	**4.38 µs** /point E/A 1.4×10^{-14}	No ratio	**4.38 µs** /point E/A 1.4×10^{-14}	No ratio	Outside public API

Table 7: TIRust binary64 only (array and contraction with the dense random numerator) against NinjaRS; single point

Input	TIRust binary64 + contraction, 2^{-34}	TIRust / NinjaRS	TIRust binary64 + contraction, 2^{-20}	TIRust / NinjaRS	NinjaRS
N=1, R=0 1 moment × 4 slots	**593 ns** E/A 0	**×2.194** 99% CI 2.1489–2.2409	**564 ns** E/A 0	**×2.087** 99% CI 2.0708–2.1708	**269 ns** fast engine · contracted E/A 1.5×10^{-16}
N=2, R=1 5 moments × 4 slots	**1.16 µs** E/A 4.4×10^{-16}	**×1.058** 99% CI 1.0353–1.0942	**1.17 µs** E/A 4.4×10^{-16}	**×1.065** 99% CI 1.0298–1.0802	**1.10 µs** fast engine · contracted E/A 5.5×10^{-16}
N=3, R=2 15 moments × 4 slots	**4.41 µs** E/A 4.4×10^{-15}	**×0.092** 99% CI 0.0914–0.0936	**4.40 µs** E/A 4.4×10^{-15}	**×0.092** 99% CI 0.0908–0.0962	**47.76 µs** generic engine (fast: 1 Unstable lanes) · contracted E/A 7.9×10^{-15}
N=3, R=3 35 moments × 4 slots	**5.23 µs** E/A 4.4×10^{-15}	**×0.069** 99% CI 0.0643–0.0697	**5.16 µs** E/A 4.4×10^{-15}	**×0.068** 99% CI 0.0651–0.0689	**76.27 µs** generic engine (fast: 1 Unstable lanes) · contracted E/A 6.0×10^{-15}
N=4, R=4 70 moments × 4 slots	**35.57 µs** E/A 1.6×10^{-14}	**×0.138** 99% CI 0.1359–0.1395	**35.62 µs** E/A 1.6×10^{-14}	**×0.138** 99% CI 0.1365–0.1400	**257.39 µs** generic engine (fast: 1 Unstable lanes) · contracted E/A 8.3×10^{-14}
N=4, R=6 210 moments × 4 slots	**59.14 µs** E/A 1.6×10^{-14}	**×0.039** 99% CI 0.0377–0.0397	**58.52 µs** E/A 1.6×10^{-14}	**×0.039** 99% CI 0.0380–0.0391	**1.51 ms** generic engine (fast: 1 Unstable lanes) · contracted E/A 1.6×10^{-12}
N=5, R=5 126 moments × 4 slots	**140.29 µs** E/A 4.7×10^{-12}	**×0.188** 99% CI 0.1849–0.1892	**139.07 µs** E/A 4.7×10^{-12}	**×0.187** 99% CI 0.1782–0.1891	**747.03 µs** generic engine (fast: 1 Unstable lanes) · contracted E/A 8.5×10^{-12}
N=5, R=6 210 moments × 4 slots	**188.49 µs** E/A 1.8×10^{-11}	**×0.103** 99% CI 0.1005–0.1051	**186.89 µs** E/A 1.8×10^{-11}	**×0.101** 99% CI 0.0999–0.1048	**1.82 ms** generic engine (fast: 1 Unstable lanes) · contracted E/A 2.3×10^{-11}
N=6, R=6 210 moments × 4 slots	**494.46 µs** E/A 8.6×10^{-12}	No ratio	**499.19 µs** E/A 8.6×10^{-12}	No ratio	**1.37 ms** generic engine (fast: 1 Unstable lanes) · no value
N=7, R=7 330 moments × 4 slots	**2.67 ms** E/A 2.0×10^{-11}	No ratio	**1.63 ms** E/A 2.0×10^{-11}	No ratio	**3.65 ms** generic engine (fast: 1 Unstable lanes) · no value

Table 8: TIRust binary64 only (array and contraction with the dense random numerator) against NinjaRS; batch of 32, time per point

Input	TIRust binary64 + contraction, 2^{-34}	TIRust / NinjaRS	TIRust binary64 + contraction, 2^{-20}	TIRust / NinjaRS	NinjaRS
N=1, R=0 1 moment × 4 slots	**255 ns** /point E/A 0	**×2.023** 99% CI 1.9743–2.0665	**255 ns** /point E/A 0	**×2.010** 99% CI 1.9766–2.0350	**127 ns** /point fast engine · contracted E/A 1.5×10^{-16}
N=2, R=1 5 moments × 4 slots	**863 ns** /point E/A 4.4×10^{-16}	**×1.393** 99% CI 1.3610–1.4411	**854 ns** /point E/A 4.4×10^{-16}	**×1.394** 99% CI 1.3737–1.4164	**614 ns** /point fast engine · contracted E/A 5.5×10^{-16}
N=3, R=2 15 moments × 4 slots	**4.08 µs** /point E/A 4.4×10^{-15}	**×0.104** 99% CI 0.1009–0.1059	**4.12 µs** /point E/A 4.4×10^{-15}	**×0.105** 99% CI 0.1022–0.1067	**39.32 µs** /point generic engine (fast: 32 Unstable lanes) · contracted E/A 7.9×10^{-15}
N=3, R=3 35 moments × 4 slots	**4.67 µs** /point E/A 4.4×10^{-15}	**×0.068** 99% CI 0.0670–0.0722	**4.66 µs** /point E/A 4.4×10^{-15}	**×0.068** 99% CI 0.0669–0.0724	**69.13 µs** /point generic engine (fast: 32 Unstable lanes) · contracted E/A 6.0×10^{-15}
N=4, R=4 70 moments × 4 slots	**35.54 µs** /point E/A 1.6×10^{-14}	**×0.139** 99% CI 0.1283–0.1436	**35.68 µs** /point E/A 1.6×10^{-14}	**×0.140** 99% CI 0.1284–0.1417	**254.63 µs** /point generic engine (fast: 32 Unstable lanes) · contracted E/A 8.3×10^{-14}
N=4, R=6 210 moments × 4 slots	**58.32 µs** /point E/A 1.6×10^{-14}	**×0.041** 99% CI 0.0405–0.0413	**58.29 µs** /point E/A 1.6×10^{-14}	**×0.041** 99% CI 0.0402–0.0420	**1.43 ms** /point generic engine (fast: 32 Unstable lanes) · contracted E/A 1.6×10^{-12}
N=5, R=5 126 moments × 4 slots	**139.94 µs** /point E/A 4.7×10^{-12}	**×0.191** 99% CI 0.1892–0.1933	**142.31 µs** /point E/A 4.7×10^{-12}	**×0.194** 99% CI 0.1912–0.1955	**736.77 µs** /point generic engine (fast: 32 Unstable lanes) · contracted E/A 8.5×10^{-12}

Input	TIRust binary64 + contraction, 2^{-34}	TIRust / NinjaRS	TIRust binary64 + contraction, 2^{-20}	TIRust / NinjaRS	NinjaRS
N=5, R=6 210 moments × 4 slots	**185.52 µs** /point E/A 1.8×10^{-11}	**×0.106** 99% CI 0.1041–0.1081	**187.47 µs** /point E/A 1.8×10^{-11}	**×0.108** 99% CI 0.1036–0.1102	**1.75 ms** /point generic engine (fast: 32 Unstable lanes) · contracted E/A 2.3×10^{-11}
N=6, R=6 210 moments × 4 slots	**498.19 µs** /point E/A 8.6×10^{-12}	No ratio	**470.29 µs** /point E/A 8.6×10^{-12}	No ratio	**1.36 ms** /point generic engine (fast: 32 Unstable lanes) · no value
N=7, R=7 330 moments × 4 slots	**5.22 ms** /point E/A 2.0×10^{-11} · 12 flagged	No ratio	**1.58 ms** /point E/A 2.0×10^{-11}	No ratio	**3.52 ms** /point generic engine (fast: 32 Unstable lanes) · no value

Lanes 1–31 of each batch are binary-exact perturbations of the reference point (shifts×(256+l)/256, masses²+l/64). TIRust and same-session original-IREGI IBP agree on all of them with maximum E/A 1.6×10^{-10}. IREGI modes that STOP, refuse or exceed the accuracy gate are retained in the snapshot (`iregi_modes`). Reproduce with `python3 validation/benchmark/direct-engine-20261030/build.py` and `run.py --cpu N`, then `summarize.py`.

1.1 Amplitudes: work shared between integrals

The integrals of an amplitude contain the same boxes and triangles. Each amplitude here is an N-gon of the binary64 stability study with every pinch of at least two propagators, each at rank equal to its number of propagators (renormalizable vertices, Feynman gauge), on the study's first 32 physical phase-space points; one call evaluates every integral at every point. TIRust (commit `e76dd38`) uses `AmplitudeWorkspace`: with `share_subsets` on, every scalar master, every binary64 PV sub-integral's coefficients, and every box or triangle that its checks send to the double-double partial rescue or to the small-Gram expansions is computed once per call and reused by every integral (and every pass and retry) that needs it, under an exact key (method settings, rank, the subset's propagators and mu^2 bit for bit). Since capture 011 the same holds for the Cartesian moments of subsets that some integral gathers inside a larger one (a pentagon integral's Gram-free pentagon, gathered again by the hexagon), keyed by every value the gather reads (its coefficients, momenta, origin and rank, bit for bit). Boxes and triangles of the rescue and the expansions are computed at the amplitude's highest rank for their size, a pinched integral using the leading coefficients, with sharing on and off alike, so the arrays are bit-identical with sharing off; the integrals share the amplitude's loop-momentum routing. COLLIER 1.2.10 (COLI, TenRed 6, requested accuracy 10^{-10}, critical 10^{-3}) evaluates the same integrals, each in its own zero-origin routing, with its global cache initialized once and `InitEvent_cll(1)` before each phase-space point (the cache learns the call sequence of the first events and later returns results by position) or without it. Both codes empty their reuse at every phase-space point in effect: COLLIER by `InitEvent_cll`, TIRust by keys that contain each point's kinematics, its store being emptied at every call (here a batch of 32 points). All four candidates run as long-lived processes on CPU 155 (load average 222.76 at start), two warm-up calls each, then 7 rounds in seeded random order; ratios are medians of same-round ratios with 99% bootstrap intervals (TIRust or the shared/cached variant in the numerator). Original IREGI and NinjaRS are not measured here: neither reuses work between integrals (their per-integral timings are in the tables above).

Table 9: Time per amplitude call (32 phase-space points) and same-session ratios

Amplitude	TIRust shared	TIRust separate	shared / separate	COLLIER cache	COLLIER no cache	cache / no cache	TIRust shared / COLLIER cache
gg -> gggg, top-loop hexagon, rank 6 57 integrals	**165.59 ms** per call · 7 blocks	**296.61 ms** per call · 7 blocks	**×0.558** 99% CI 0.5536–0.5826	**84.65 ms** per call · 7 blocks	**250.30 ms** per call · 7 blocks	**×0.337** 99% CI 0.3330–0.3414	**×1.951** 99% CI 1.9208–2.0559
gg -> Hgg, top-loop pentagon (g g g H g), rank 5 26 integrals	**27.07 ms** per call · 7 blocks	**29.66 ms** per call · 7 blocks	**×0.916** 99% CI 0.8877–0.9385	**18.06 ms** per call · 7 blocks	**30.22 ms** per call · 7 blocks	**×0.596** 99% CI 0.5661–0.6259	**×1.499** 99% CI 1.4220–1.5315

gg -> gggg, top-loop hexagon, rank 6: TIRust reused 37529 and computed 3097 per-subset results per call (Cartesian moments: 443 reused, 442 gathered and stored) its 57 integrals' arrays, statuses and estimates are bit-identical with sharing on and off (1824 of 1824 integral lanes certified). COLLIER's coefficient sums per integral and point agree with and without its cache to 0 (relative); without its cache it sets its accuracy flag on 101 of 1824 calls (with the cache the flag is reset only per phase-space point, so it is not counted per call). **gg -> Hgg, top-loop pentagon (g g g H g), rank 5:** TIRust reused 9361 and computed 1251 per-subset results per call (Cartesian moments: 0 reused, 0 gathered and stored) its 26 integrals' arrays, statuses and estimates are bit-identical with sharing on and off (832 of 832 integral lanes certified). COLLIER's coefficient sums per integral and point agree with and without its cache to 2.8×10^{-16} (relative); without its cache it sets its accuracy flag on 6 of 832 calls (with the cache the flag is reset only per phase-space point, so it is not counted per call). The binary64 arrays of these integrals are what TIRust's checks certify (rotation test, method-choice retries, double-double boxes), and on these physical points the checks, not the binary64 pass, take most of TIRust's time (see the stability section for both codes' accuracy on the same points). Sharing removes the repeats of that work: 1.8 times faster on the hexagon amplitude (capture 015: 1.9; capture 014: 1.9; capture 013: 2.1; capture 012: 2.2; capture 011: 2.1; capture 010: 2.0; capture 005: 2.1; capture 003: 2.2; capture 002: 2.3; capture 001, with only the double-double boxes and the expansions shared and without rank floors: 1.8 times), 1.1 times on the pentagon amplitude (captures 003 and 001: 1.3 and 1.1 times). COLLIER's cache reuses whole sub-integrals between calls (3.0 and 1.7 times faster; capture 015: 2.9 and 1.7; capture 014: 3.0 and 1.7; capture 013: 3.0 and 1.6; capture 012: 2.9 and 1.7; capture 011: 3.0 and 1.7; capture 010: 3.3 and 1.8; capture 008: 3.0 and 1.7). Capture 003 (TIRust a71e29d) runs the small-Gram expansions order by order in Δ and stops those that cannot converge (timing section): TIRust shared over COLLIER's cache ×9.1 on the hexagon and ×4.9 on the pentagon amplitude (capture 002, TIRust 8ec79e9: ×10.4 and ×6.6; separate sessions). Capture 004 (TIRust f45d36f) also drops the accuracy reruns of accepted box and pentagon lanes (stability section): ×7.3 and ×3.3. Capture 005 (TIRust 939bb35) speeds up the expansions further: ×6.6 and ×3.0. Capture 006 (TIRust a51d171) stores the expansions' coefficients in arrays instead of hash maps, with TIRust's arrays bit-identical to capture 005′s: ×4.7 and ×2.4. Capture 007 (TIRust f06c5f3) sends the triangles of the expansions' auxiliary boxes whose expansion cannot converge straight to the alternative PV reduction, with TIRust's arrays bit-identical to capture 006′s: ×4.3 and ×2.3 (a first run, with COLLIER's medians 4 to 10% slower than here and intervals up to four times wider, gave ×4.2 and ×2.2 and was repeated). Capture 008 (TIRust bdf47ab) finds the coefficients each step of the expansions' recursion reads through precomputed neighbour tables, with TIRust's arrays bit-identical to capture 007′s: ×3.3 and ×2.2 (TIRust's shared medians 353 ⟶ 263 ms and 40.6 ⟶ 39.8 ms; COLLIER's cached medians 81.8 ⟶ 81.7 ms and 17.7 ms in both). Capture 010 (TIRust 0b607e9) builds the momentum products of the Cartesian moments once per call (timing section), with TIRust's arrays bit-identical to capture 008′s: ×3.1 and ×2.1 (TIRust's shared medians 263 ⟶ 253 ms and 39.8 ⟶ 37.6 ms; COLLIER's cached medians 81.7 ⟶ 80.7 ms and 17.7 ⟶ 17.8 ms). It repeats capture 009 (same build), which ran on a core already about half busy (block times spread by up to 23% for TIRust and 37% for COLLIER) and gave ×3.2 and ×2.1 with a hexagon interval of ×2.4 to ×3.5. Capture 011 (TIRust 06daf37) shares the Cartesian moments of the hexagon's Gram-free pentagons, which their own pentagon integrals have already gathered, with TIRust's arrays bit-identical to capture 010′s: ×3.0 and ×2.1 (hexagon interval ×2.6 to ×3.4; TIRust's shared medians 253 ⟶ 251 ms and 37.6 ⟶ 38.7 ms; COLLIER's cached medians 80.7 ⟶ 84.2 ms and 17.8 ⟶ 17.5 ms). The pentagon amplitude gathers no subset twice. The hexagon's change is within the spread of its blocks (TIRust shared 233 to 270 ms, capture 010: 247 to 261 ms); measured in the same rounds, its shared over separate went from 0.49 to 0.47. Capture 012 (TIRust 7022322) speeds up the small-Gram expansions (timing section), with TIRust's arrays bit-identical to capture 011′s: ×2.7 and ×2.0 (hexagon interval ×2.3 to ×3.4; TIRust's shared medians 251 ⟶ 215 ms and 38.7 ⟶ 35.8 ms; COLLIER's cached medians 84.2 ⟶ 78.7 ms and 17.5 ⟶ 18.8 ms). Three of the seven hexagon blocks ran slow for each code (TIRust shared 237 to 268 ms against 213 to 215 ms, COLLIER cached 88 to 101 ms against 78 to 79 ms), which widens the interval; TIRust's fastest hexagon block went from 233 to 213 ms. The pentagon's change is within the spread of its blocks (TIRust shared 34.8 to 44.1 ms, capture 011: 36.0 to 41.0 ms). Capture 013 (TIRust c8b8264) skips, in the double-double scalar masters of the precision rescue, the products with a zero imaginary part (timing section),

with TIRust's arrays bit-identical to capture 012′s: ×2.6 and ×2.0 (hexagon interval ×2.2 to ×2.9; TIRust's shared medians 215 → 222 ms and 35.8 → 38.4 ms; COLLIER's cached medians 78.7 → 82.4 ms and 18.8 → 19.8 ms). It ran on a busier host (load 125 to 130; capture 012: 62 to 65). Both codes' shared or cached medians rose, while their hexagon medians without sharing or cache fell (TIRust separate 510 → 455 ms, COLLIER 296 → 249 ms). TIRust's fastest hexagon block went from 213 to 204 ms and its fastest pentagon block from 34.8 to 35.1 ms; both changes are within the spread of the blocks (TIRust shared 204 to 239 ms and 35.1 to 42.2 ms). Capture 014 (TIRust 2db8ff8) runs the double-double D0 masters of the precision rescue with the hand-written box kernels (timing section): ×2.3 and ×1.8 (hexagon interval ×2.2 to ×2.5; TIRust's shared medians 222 → 188 ms and 38.4 → 31.5 ms; COLLIER's cached medians 82.4 → 82.6 ms and 19.8 → 18.0 ms; load 195, capture 013: 125 to 130; the pentagon's core was 44% busy in the half second before). The blocks are tight (TIRust shared 186 to 205 ms and 30.9 to 32.1 ms). The rescued hexagon lanes change in their last bits (fingerprint 627118d8081d80dd, capture 013: d5982c2edce61827); the pentagon's arrays are unchanged. Without sharing, which repeats the rescue, the hexagon went from 455 to 369 ms, so sharing saves less than before (shared over separate 0.47 → 0.51). Capture 015 (TIRust ac7128e) runs the double-double C0 masters with the hand-written triangle kernel too (timing section): ×2.2 and ×1.7 (hexagon interval ×1.9 to ×2.2; TIRust's shared medians 188 → 189 ms and 31.5 → 31.6 ms; COLLIER's cached medians 82.6 → 85.1 ms and 18.0 → 18.5 ms; load 265 to 271, with the hexagon's core 36% and the pentagon's 50% busy in the half second before). The change, 2.4% of the hexagon call's instructions and ×0.96 in its cycles (development profile and an alternating A/B of ten rounds on the same host), is within this capture's spread: TIRust's fastest hexagon block went from 186 to 179 ms (blocks 179 to 193 ms), and its ratios improved mostly because COLLIER's medians rose. The rescued hexagon lanes change in their last bits again (fingerprint b89f51302382fec0); the pentagon's arrays are unchanged. Capture 016 (TIRust e76dd38) sends the lanes the a priori method choice does not keep to the precision rescue without the binary64 retries, and those it keeps without a clear agreement to its double-double tier first, and allocates less per call (timing section): ×1.95 and ×1.50 (hexagon interval ×1.92 to ×2.06, pentagon ×1.42 to ×1.53; TIRust's shared medians 189 → 166 ms and 31.6 → 27.1 ms; COLLIER's cached medians 85.1 → 84.7 ms and 18.5 → 18.1 ms; load 223 to 229, capture 015: 265 to 271, with the hexagon's core 44% and the pentagon's 74% busy in the half second before). TIRust's blocks no longer overlap capture 015′s (hexagon 163 to 173 ms against 179 to 193 ms, pentagon 26.5 to 27.8 ms against 29.6 to 34.1 ms). A development profile counts 11% fewer instructions per hexagon call ($1.909 \to 1.698\times10^{9}$), and an alternating A/B of ten rounds on the same host gives ×0.90 in the cycles of both amplitudes (hexagon ×0.84 to ×0.94, pentagon ×0.89 to ×0.91). Without sharing, TIRust's hexagon takes 297 ms (capture 015: 356 ms) and its pentagon 29.7 ms, below COLLIER's 30.2 ms without its cache (×0.98, interval ×0.94 to ×0.99). Sharing saves less than before (hexagon shared over separate 0.53 → 0.56): the retries it avoided repeating are gone. The hexagon's rescued lanes change (fingerprint fb38d648191764a4; 3097 integrals computed per call, capture 015: 3160), and its arrays stay bit-identical between sharing on and off; the pentagon's arrays are unchanged. The rotated evaluations of TIRust's rotation test start each box from another propagator, so those remain separate computations.

2 Binary64 stability: TIRust, original IREGI and COLLIER

Setup (as in the stability study of Hirschi and Peraro, arXiv:1604.01363, figs. 2–4): random flat phase-space points at $\sqrt{s} = 1000$ GeV with $p_T > 50$ GeV for every final-state gluon and $\Delta R > 0.5$ between gluons (seed 20261030); top-quark loops, $m_t = 173$ GeV, $m_H = 125$ GeV, $\mu = 1000$ GeV. Every code reads the same binary64 inputs, as a user would run it: massless legs are lightlike only to binary64 rounding ($|p^2| \approx 10^{-16}$ s), and each code treats them as exactly lightlike — TIRust by its lightlike tolerance, IREGI by its default on-shell snapping (as in MadLoop), COLLIER through exact zero invariants (passing the rounded invariants instead made COLI return NaN and DD wrong arrays on most pentagons and hexagons). Δ is the E/A of the complete returned array (all Cartesian moments, all four Laurent slots, dimensionally rescaled) against the reference. Each curve gives the fraction of points whose array is worse than Δ; points without an array (refused, nonfinite) count as worse than every Δ. The gray bar marks $\Delta = 10^{-3}$, the usual Monte-Carlo rescue threshold.

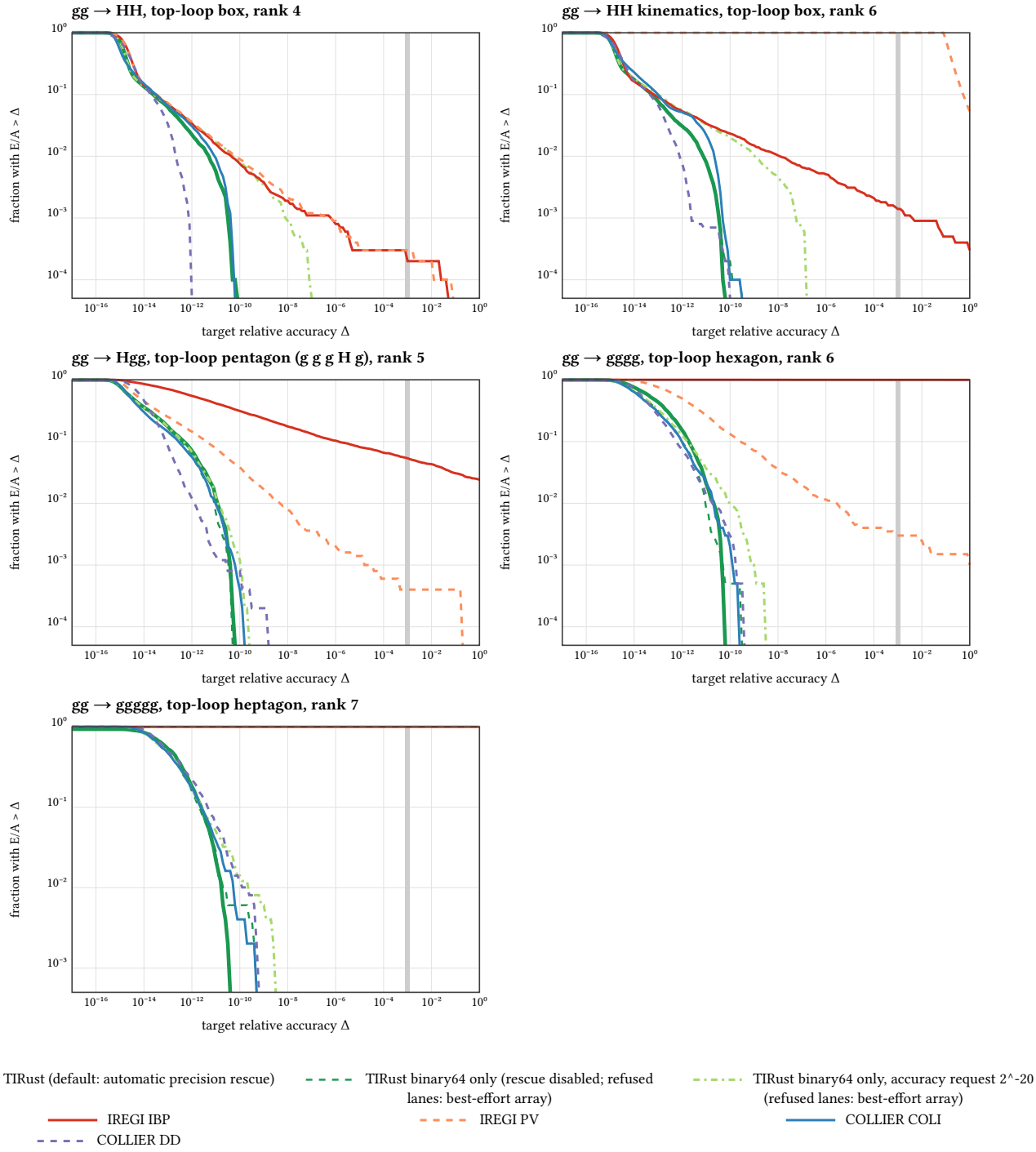


Figure 1: Cumulative binary64 stability profiles (fraction of points with E/A > Δ). TIRust (default) is the public binary64 API: lanes its own checks refuse (conditioning gates; for N≥4 with equal powers a loop-rotation test comparing two binary64 evaluations, three when the first two agree only narrowly) are rerun by the direct engine at 106, 256 or 512 bits before results are returned. “TIRust binary64 only” is the same facade with the precision rescue disabled (every lane in binary64), at the default accuracy request 2^{-34} and at 2^{-20} a lane its checks refuse is scored by the best binary64 array it kept, uncertified and flagged like COLLIER’s flagged lanes (table: best-effort), and counts as missing only without one. IREGI and COLLIER are binary64 throughout.

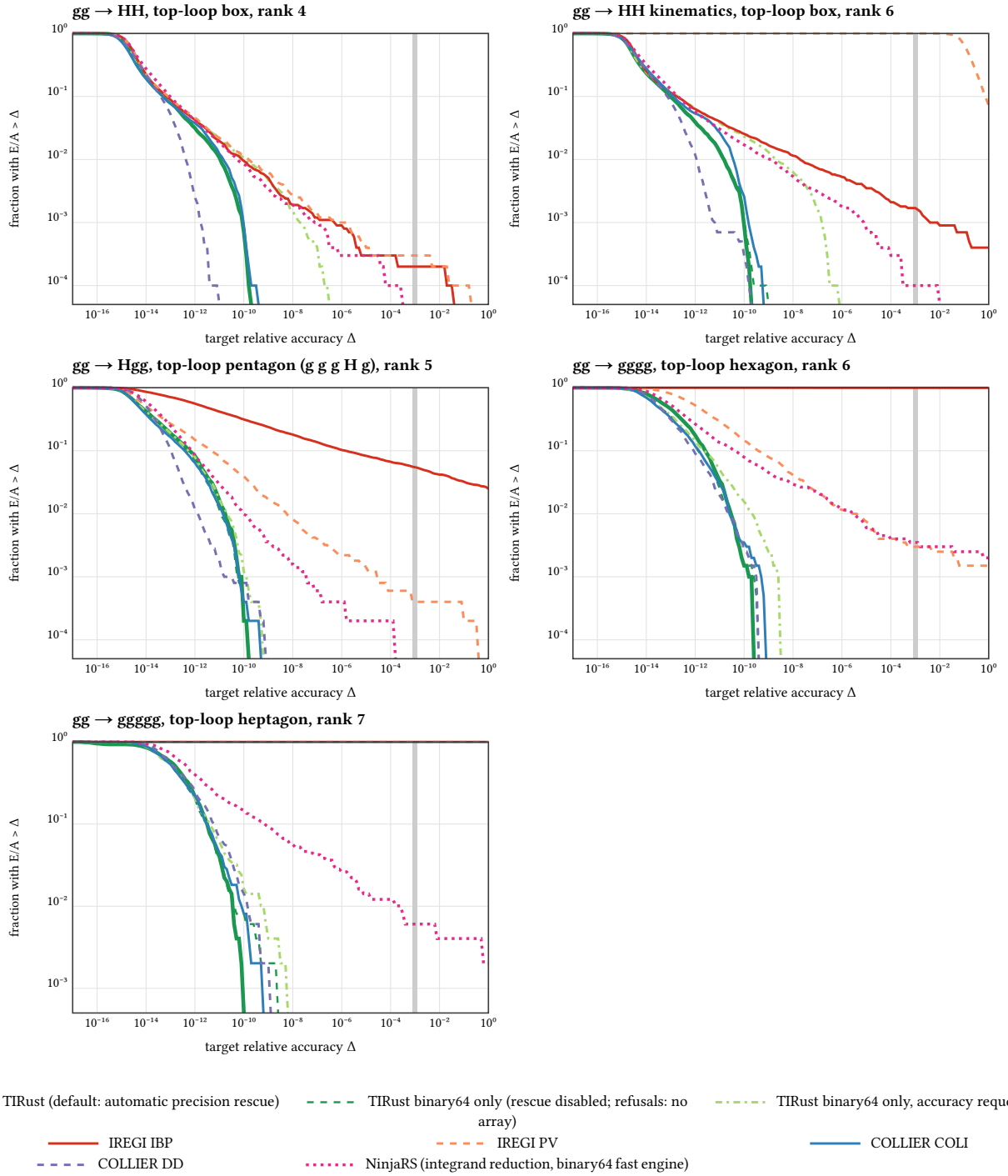


Figure 2: Contracted dense random numerator: fraction of points with $|I - I_{ref}|/\max|I_{ref}| > \Delta$ over the finite part, the (UV + IR) single pole and the double pole, $I = \Sigma_\alpha c_\alpha \int q^\alpha$ with $c_\alpha = (u + i\,v)/\mu^{|\alpha|}$, u, v uniform in [−1, 1], fixed per point. The array codes contract their arrays; NinjaRS (alphal00p/ninjars, integrand reduction, binary64 fast engine; lanes it marks Unstable count as without value) reduces the numerator directly.

Table 10: Per-point E/A: median, 99th percentile and the number of points worse than 10^{-3} (green: none; blue: below 1%; amber: 1% or more). “No array”: refused or nonfinite output; “flagged”: COLLIER accuracy flag or IREGI non-success status (values still scored).

Integral	TIRust (default: automatic precision rescue)	TIRust binary64 only (rescue disabled; refused lanes: best-effort array)	TIRust binary64 only, accuracy request 2^-20 (refused lanes: best-effort array)	IREGI IBP	IREGI PV	COLLIER COLI	COLLIER DD
box-r4 10000 points	**1.5×10^{-15}** p99 5.2×10^{-12} · >10^{-3}: 0/10000	**1.5×10^{-15}** p99 5.3×10^{-12} · >10^{-3}: 0/10000 · best-effort 2	**1.5×10^{-15}** p99 5.5×10^{-11} · >10^{-3}: 0/10000	**2.2×10^{-15}** p99 4.7×10^{-11} · >10^{-3}: 2/10000	**2.2×10^{-15}** p99 6.7×10^{-11} · >10^{-3}: 3/10000	**1.1×10^{-15}** p99 9.1×10^{-12} · >10^{-3}: 0/10000 · flagged 335	**2.0×10^{-15}** p99 2.1×10^{-13} · >10^{-3}: 0/10000 · flagged 93
box-r6 10000 points	**1.5×10^{-15}** p99 7.5×10^{-12} · >10^{-3}: 0/10000	**1.5×10^{-15}** p99 8.0×10^{-12} · >10^{-3}: 0/10000 · best-effort 4	**1.5×10^{-15}** p99 1.1×10^{-9} · >10^{-3}: 0/10000	**2.2×10^{-15}** p99 1.2×10^{-8} · >10^{-3}: 14/10000	**1.4×10^{-1}** p99 5.7×10^{0} · >10^{-3}: 10000/10000	**1.7×10^{-15}** p99 2.2×10^{-11} · >10^{-3}: 0/10000 · flagged 1499	**2.1×10^{-15}** p99 7.8×10^{-13} · >10^{-3}: 0/10000 · flagged 538
pentagon-r5 5000 points	**3.4×10^{-15}** p99 1.2×10^{-11} · >10^{-3}: 0/5000	**3.3×10^{-15}** p99 8.5×10^{-12} · >10^{-3}: 0/5000 · best-effort 17	**3.3×10^{-15}** p99 1.2×10^{-11} · >10^{-3}: 0/5000	**2.3×10^{-12}** p99 5.0×10^{3} · >10^{-3}: 268/5000	**6.9×10^{-15}** p99 4.4×10^{-9} · >10^{-3}: 2/5000	**2.9×10^{-15}** p99 9.8×10^{-12} · >10^{-3}: 0/5000 · flagged 77	**8.8×10^{-15}** p99 1.3×10^{-12} · >10^{-3}: 0/5000 · flagged 473
hexagon-r6 2000 points	**6.0×10^{-14}** p99 2.2×10^{-11} · >10^{-3}: 0/2000	**2.6×10^{-14}** p99 9.7×10^{-12} · >10^{-3}: 0/2000 · best-effort 57	**2.8×10^{-14}** p99 1.0×10^{-10} · >10^{-3}: 0/2000	**1.2×10^{65}** p99 2.3×10^{76} · >10^{-3}: 1993/2000	**9.9×10^{-13}** p99 2.7×10^{-6} · >10^{-3}: 6/2000	**2.1×10^{-14}** p99 2.0×10^{-11} · >10^{-3}: 0/2000 · flagged 210	**2.6×10^{-14}** p99 2.0×10^{-11} · >10^{-3}: 0/2000 · flagged 985
heptagon-r7 495 points	**1.4×10^{-13}** p99 1.9×10^{-11} · >10^{-3}: 0/495	**8.5×10^{-14}** p99 2.6×10^{-11} · >10^{-3}: 0/495 · best-effort 27	**8.7×10^{-14}** p99 2.3×10^{-10} · >10^{-3}: 0/495	**7.8×10^{85}** p99 1.4×10^{95} · >10^{-3}: 495/495	**1.2×10^{2}** p99 1.6×10^{5} · >10^{-3}: 495/495	**8.3×10^{-14}** p99 5.0×10^{-11} · >10^{-3}: 0/495 · flagged 160	**9.8×10^{-14}** p99 2.5×10^{-10} · >10^{-3}: 0/495 · flagged 409

Findings (capture 033, TIRust `e76dd38`). TIRust’s default returns an accurate array on every point of all five integrals, worst E/A 6.7×10^{-11} (capture 021: 5.0×10^{-11} capture 010: 9.6×10^{-9}). “TIRust binary64 only” (precision rescue disabled) reports every point with a binary64 array, as COLLIER does: lanes that pass TIRust’s checks are certified, the others return their best binary64 array flagged with an estimated error, so its curves have no plateau. At the default request ($2^{-34}\approx5.8\times10^{-11}$) binary64 certifies 9998/10000 rank-4 and 9996/10000 rank-6 boxes, 4983/5000 pentagons, 1943/2000 hexagons and 468/495 heptagons (capture 010: 9998, 9996, 4983, 1895, 436); one certified heptagon (2.3×10^{-10}) and, since capture 022, one rank-4 box (6.7×10^{-11}) exceed the request; no array is worse than 4.3×10^{-10} (capture 017: 1.2×10^{-9}, capture 010: 3×10^{-6} COLLIER COLI 5.0×10^{-10}). Binary64 against COLLIER (median, 99th percentile): rank-4 boxes 1.5×10^{-15}, 5.3×10^{-12} (COLI 1.1×10^{-15}, 9.1×10^{-12} DD 2.0×10^{-15}, 2.1×10^{-13}); rank-6 boxes 1.5×10^{-15}, 8.0×10^{-12} (COLI 2.2×10^{-11}, DD 7.8×10^{-13}); pentagons 3.3×10^{-15}, 8.5×10^{-12} (COLI 2.9×10^{-15}, 9.8×10^{-12}); hexagons 2.6×10^{-14}, 9.7×10^{-12} (capture 017: 6.6×10^{-11} COLI 2.1×10^{-14}, 2.0×10^{-11} DD 2.6×10^{-14}, 2.0×10^{-11}); heptagons 8.5×10^{-14}, 2.6×10^{-11} (capture 017: 1.2×10^{-10} COLI 8.3×10^{-14}, 5.0×10^{-11} DD 2.5×10^{-10}). Worst arrays: pentagons 4.8×10^{-11} (COLI 1.5×10^{-10}), hexagons 3.1×10^{-10} (COLI 2.1×10^{-10}, DD 3.4×10^{-10}), heptagons 4.3×10^{-10} (COLI 5.0×10^{-10}). Capture 023 (faster expansions, timing section) changes no lane’s status; no lane’s E/A moves by more than 1.4×10^{-12}, and the worst arrays and 99th percentiles of capture 022 are unchanged to two digits. Capture 024 (expansion coefficients stored in arrays, timing section) is identical to capture 023 in every TIRust status, error and diagnostic on all 27500 points; in the same way, capture 025 (nested triangles whose expansions cannot converge go straight to the alternative reduction, timing section) is identical to capture 024, capture 026 (precomputed neighbour tables in the expansions’ recursion, timing section) to capture 025, capture 027 (momentum products of the Cartesian moments built once per call, timing section) to capture 026, capture 028 (Cartesian moments shared between the integrals of an amplitude, timing section) to capture 027, capture 029 (small-Gram expansions that compute only the coefficients their results depend on, timing section) to capture 028, and capture 030 (double-double products with a real factor, timing section) to capture 029. Capture 031 (hand-written double-double boxes, timing section) differs from capture 030 only in TIRust’s default errors on 19 points that use the double-double rescue: no status, flag or binary64-only value changes, an E/A changes by at most 6×10^{-3} of itself except on one box ($1.8\longrightarrow1.3\times10^{-17}$), and one point of the box-r4 curves moves by 10^{-4}. Capture 032 (hand-written double-double triangles, timing section) differs from capture 031 only in TIRust’s default errors on 20 points: no status, flag, binary64-only value or summary statistic changes; the contracted E/A of three pentagon points at 10^{-15} to 10^{-14} moves by 2 to 9% (largest $1.4\longrightarrow1.3\times10^{-15}$) and every other changed E/A by at most 0.3% of itself. Capture 033 (the precision rescue right after the a priori method choice, timing section) differs from capture 032 only on the 51 points whose refused lanes the default now sends to the double-double tier instead of keeping them by a binary64 retry (1 rank-4 box, 14 rank-6 boxes, 31 pentagons, 5 hexagons): each one’s E/A falls from at most 5.2×10^{-12} (19 pentagons and all 5 hexagons above 10^{-14}) to below 10^{-28} (boxes) or 0 (pentagons and hexagons, equal to the reference after rounding), and each now counts as rescued at 106 bits; no binary64-only value, status or worst array changes, the only 99th percentile that moves is the contracted rank-6 boxes’ ($1.54\longrightarrow1.47\times10^{-11}$), and the default medians fall by up to 3.6% (pentagons). **How** (since capture 010, following COLLIER and Denner and Dittmaier, hep-ph/0509141): boxes get the loop-rotation test; an expansion that has not converged tries the next larger boost; every lane the binary64 pass refuses is retried first with COLLIER’s per-box choice (boxes and triangles whose a priori PV estimate $2^{-52}w^r$ exceeds the request, or that the gate refuses, by the small-Gram expansions, sects. 3.4, 3.5, or the alternative PV reduction, sect. 3.2; a converged expansion yields to the alternative PV where the two agree within its remaining change), and lanes it keeps without a clear agreement also try every box and triangle with $w>2$ off PV (since capture 033, with the precision rescue on, only when its double-double tier refuses them, and the lanes the choice does not keep go to the rescue directly instead of binary64 retries below the gate, by the expansions, by the alternative PV and with every box and triangle off PV); refused pentagon and hexagon lanes first evaluate every cyclic order of the propagators, whose binary64 errors differ widely, and keep an order three others agree with within 2^{-4} of the tolerance (until capture 021, accepted box and pentagon lanes where the estimate trips also took the per-box choice where its rotated copy agreed better). A lane no check certifies (best-effort) returns, since capture 018, the array closest to two others among every cyclic order evaluated with the expansions and without the gate, with that distance as its estimate. **What remains:** the hexagon median (2.6 against COLI 2.1×10^{-14}) and worst array; with every box and triangle computed exactly, binary64 hexagons reach a median of 1.2×10^{-15}, so the gap is in the boxes TIRust computes by its expansions, which do not yet reach COLLIER’s accuracy. With the precision rescue on, refused heptagon lanes go to it directly (their binary64 retries rarely succeed), and its double-double tier skips the rotated check for lanes whose binary64 rotation difference was at most 2^{-20}. Captures 019 (TIRust 3f7cd05, the fused double-double kernels of the timing section) and 020 (5a6d415, the binary64-only retry

change of the timing section) reproduce capture 018: every binary64-only array is bit-identical, and the rescued lanes take the same tiers and stay at the rounding level of their binary64 output (E/A ≤ 1.9×10^{-17}); capture 020 also leaves every default array of capture 019 unchanged. Capture 021 (a71e29d, the expansions order by order in Δ of the timing section) changes arrays only at the rounding level: the flagged counts, 99th percentiles and worst arrays of all three TIRust codes are those of capture 020 and the medians agree to three digits. Capture 022 (f45d36f) drops those reruns of accepted box and pentagon lanes for accuracy beyond the request: no lane changes status; binary64-only 99th percentiles rise from 5.1×10^{-13} (rank-4 boxes), 6.8×10^{-13} (rank-6 boxes) and 3.4×10^{-12} (pentagons) to the values above, still below COLLIER COLI's, while the amplitudes of the next section take less time. The default's pentagon 99th percentile, 1.2×10^{-11} (capture 021: 5.5×10^{-12}), is now above COLI's 9.8×10^{-12}. Timing capture 009 and stability capture 014 (TIRust 32a79e2, also the per-box choice for accepted hexagon lanes) are not published: that version took 8.8 times longer on the opening N6R6 batch of 32 without changing the hexagons. COLLIER (both branches) returns an accurate array on every point (worst 1.3×10^{-9}, DD pentagons); its flags are read per lane, and every unflagged COLLIER lane is within 5.5×10^{-11}. Original IREGI's PV fails on every rank-6 box and on every heptagon, its IBP on the hexagons and heptagons (E/A up to 10^{97}), all reported as successful (minimal reproducers in `validation/mre`). NinjaRS, reducing the contracted numerator, returns a value on every point with tails beyond 10^{-3} on one rank-6 box and a few hexagons and heptagons. Heptagon references are confirmed by COLLIER COLI and DD (≤10^{-9}) instead of quad IREGI, which fails there in both modes; COLLIER's heptagon curves are therefore conditioned on that agreement (5 of 500 points excluded); six hexagon points are confirmed the same way. Captures 001-019 remain bound unchanged in the report data (014 not published).

Table 11: Reference and TIRust precision rescue per integral.

Integral	256-bit reference confirmed by	Confirmation E/A median / max	Confirmed only above 10^{-12} (COLLIER sides with 256 bits)	Excluded	TIRust lanes rescued (bits: lanes)
box-r4	quad IBP: 10000	6.3×10^{-17} / 6.3×10^{-17}	0 (0)	0 / 10000	106: 3
box-r6	quad IBP: 10000	6.3×10^{-17} / 1.4×10^{-16}	0 (0)	0 / 10000	106: 17, 256: 1
pentagon-r5	quad IBP: 4352 quad PV: 648	2.2×10^{-16} / 1.9×10^{-11}	13 (13)	0 / 5000	106: 31, 256: 2
hexagon-r6	COLLIER COLI and DD: 6 quad PV: 1994	7.6×10^{-15} / 2.2×10^{-5}	122 (114)	0 / 2000	106: 7, 256: 1
heptagon-r7	COLLIER COLI and DD: 495	5.9×10^{-1} / 1.0×10^{0}	495 (0)	5 / 500	106: 16, 256: 16

Reference. The TIRust direct engine at 256 bits on the same binary64 inputs, accepted per point only when quadruple-precision original IREGI (IBP or PV, whichever agrees better; on-shell snapping as above) confirms it to E/A ≤ 10^{-9} otherwise the quad IBP array when both quad modes agree, else the point is excluded. Quad IREGI alone cannot be the reference: its IREGI-native normalization carries a binary64 π (every array is off by 6.3×10^{-17}), quad PV fails on the rank-6 box (rank above N), quad IBP on the hexagon, and quad PV keeps only 10^{-12}–10^{-10} on some hexagon points, where COLLIER (independent of both) agrees with the 256-bit array instead. Below the confirmation level Δ is not independently verified. TIRust lanes rescued at 256 bits equal the reference after rounding (Δ = 0). This capture: TIRust `e76dd38`. Binaries and the TIRust commit are in `validation/benchmark/stability-20261030/summary-033.json` reproduce with `make-points.py`, `build.py`, `run.py` and `summarize.py` in that directory.

3 Example tensor integrals: double-precision timing

Representative $N \leq 6$, cumulative rank $R \leq 6$ inputs. **Every timing in these two opening tables uses binary64 (f64).** Each call requests all Cartesian moments q^α with $|\alpha| \leq R$ and all four native Laurent slots. Colors identify timing components, not relative speed.

Exact inputs: metric $(+,-,-,-)$; use the first N shifts from $\left(0, e_x, e_y, e_z, \frac{e_t}{2}, e_x + e_y\right)$ and squared masses $(1,2,3,4,2,3)$; scale $\mu = 3$. Denominators are $(q+p_i)^2 - m_i^2 + i0$, powers ν_i; $D = 4 + 2h - 2\varepsilon$. Numerator components stay four-dimensional. **Requested accuracy:** for ordinary N5/R5, N5/R6 and N6/R6, TIRust targets $2^{-20} \approx 9.54 \times 10^{-7}$ and COLLIER targets 10^{-6}. Other rows use TIRust's 2^{-34} target and COLLIER's 10^{-10} target. These are requests, not achieved precision; no working-precision promotion is used.

Input integral	Powers / shift	TIRust excl. masters	Scalar masters	TIRust total	COLLIER · f64
N=1, R=0 f64 · 1 moment Generic	all 1 $h = 0$	826.6 ns / point · SE 1.20	139.0 ns / point · SE 0.89	965.6 ns / point · SE 1.52 δ_{rel} = 0	380.5 ns / point · SE 0.24 δ_{rel} = 0 · COLI
N=1, R=6 f64 · 210 moments Generic	all 1 $h = 0$	59.62 µs / point · SE 1.73	453.1 ns / point · SE 2.63	60.08 µs / point · SE 1.73 δ_{rel} = 0	4.37 µs / point · SE 0.001 δ_{rel} = 1.6e−20 · DD
N=2, R=1 f64 · 5 moments Generic	all 1 $h = 0$	6.04 µs / point · SE 0.19	664.6 ns / point · SE 1.62	6.70 µs / point · SE 0.19 δ_{rel} = 4.4e−16	1.09 µs / point · SE 0.001 δ_{rel} = 1.1e−15 · COLI
N=2, R=6 f64 · 210 moments Generic	all 1 $h = 0$	1.48 ms / point · SE 0.02	4.81 µs / point · SE 0.10	1.49 ms / point · SE 0.02 δ_{rel} = 0	8.36 µs / point · SE 0.003 δ_{rel} = 2.2e−16 · DD
N=3, R=2 f64 · 15 moments Generic	all 1 $h = 0$	1.30 ms / point · SE 0.001	8.90 µs / point · SE 0.09	1.31 ms / point · SE 0.001 δ_{rel} = 4.0e−15	8.20 µs / point · SE 0.06 δ_{rel} = 7.0e−15 · COLI
N=3, R=3 f64 · 35 moments Generic	all 1 $h = 0$	1.44 ms / point · SE 0.005	9.74 µs / point · SE 0.11	1.45 ms / point · SE 0.005 δ_{rel} = 4.0e−15	10.45 µs / point · SE 0.004 δ_{rel} = 7.0e−15 · COLI
N=3, R=6 f64 · 210 moments Generic	all 1 $h = 0$	5.62 ms / point · SE 0.02	30.01 µs / point · SE 1.12	5.65 ms / point · SE 0.02 δ_{rel} = 4.0e−15	49.97 µs / point · SE 0.03 δ_{rel} = 7.0e−15 · COLI
N=4, R=4 f64 · 70 moments Generic	all 1 $h = 0$	1.37 ms / point · SE 0.03	150.86 µs / point · SE 2.19	1.52 ms / point · SE 0.03 δ_{rel} = 0	48.97 µs / point · SE 0.03 δ_{rel} = 9.7e−15 · COLI
N=4, R=6 f64 · 210 moments Generic	all 1 $h = 0$	11.05 ms / point · SE 0.03	207.69 µs / point · SE 2.97	11.26 ms / point · SE 0.03 δ_{rel} = 1.7e−14	261.91 µs / point · SE 0.17 δ_{rel} = 9.7e−15 · COLI
N=5, R=5 f64 · 126 moments Generic	all 1 $h = 0$	4.57 ms / point · SE 0.01	817.01 µs / point · SE 2.96	5.39 ms / point · SE 0.01 δ_{rel} = 6.3e−12	293.29 µs / point · SE 0.45 δ_{rel} = 1.1e−11 · COLI
N=5, R=6 f64 · 210 moments Generic	all 1 $h = 0$	12.09 ms / point · SE 0.02	859.31 µs / point · SE 5.00	12.95 ms / point · SE 0.02 δ_{rel} = 2.5e−11	1.05 ms / point · SE 0.001 δ_{rel} = 4.3e−11 · COLI
N=6, R=6 f64 · 210 moments AffinePower	all 1 $h = 0$	417.34 ms / point · SE 0.96	1.78 ms / point · SE 0.009	419.12 ms / point · SE 0.97 δ_{rel} = 1.2e−11	2.82 ms / point · SE 0.01 finite δ_{rel} = 1.7e0 · 195 nonfinite · COLI Diagnostic output

Optimistic selection: fastest arithmetic mean among the fixed measured candidates returning a complete finite array with $\delta_{\text{rel}} \leq 10^{-3}$. If none qualifies, any observed duration remains visible as diagnostic. A missed 10^{-6} request or an internal accuracy flag alone does not suppress the timing. All candidate attempts are retained.

Phase split: TIRust excl. masters = total call wall time − nested scalar-master wall time. Means of the same timed calls add before rounding. SE is descriptive standard error across blocks, not a controlled-host confidence guarantee. Conversion, dispatch, stability checks and tensor assembly are included; cold preparation, first call, warm-up, result checks and I/O are excluded. Plans are reused; numerical results are recomputed.

COLLIER: 1.2.10, binary64, scalar masters included. Its selected public mode and all raw flags are recorded with each row. **DD** denotes the Denner–Dittmaier algorithm, still in binary64; it does not mean DoubleFloat or double-double arithmetic. Both tools' total cells show measured output error where available, defined on the next page. Amber denotes error above 10^{-3}, nonfinite or refused output, rather than failure to reach a stricter accuracy request.

4 Generalized requests outside COLLIER's direct API

Dotted and independently shifted requests, all in **binary64**. The same exact input family, full cumulative tensor outputs, optimistic selection and component timing definitions apply.

Input integral	Powers / shift	Outside direct API	TIRust excl. masters	Scalar masters	TIRust total
N=1, R=6 f64 · 210 moments Generic	(2) $h = 0$	Dotted propagator powers	69.40 µs / point · SE 0.88	253.9 ns / point · SE 4.84	69.66 µs / point · SE 0.88 δ_{rel} = 4.2e−16
N=2, R=2 f64 · 15 moments Generic	(2,1) $h = 0$	Dotted propagator powers	70.39 µs / point · SE 0.30	1.04 µs / point · SE 0.009	71.44 µs / point · SE 0.31 δ_{rel} = 5.5e−16
N=3, R=4 f64 · 70 moments Generic	(2,1,1) $h = 0$	Dotted propagator powers	1.54 ms / point · SE 0.02	14.55 µs / point · SE 0.30	1.55 ms / point · SE 0.02 δ_{rel} = 7.0e−15
N=4, R=6 f64 · 210 moments Generic	(2,1,1,1) $h = 0$	Dotted propagator powers	25.85 ms / point · SE 0.05	263.99 µs / point · SE 0.93	26.12 ms / point · SE 0.05 accuracy unavailable Refused output · diagnostic timing
N=4, R=6 f64 · 210 moments Generic	(3,1,1,1) $h = 0$	Dotted propagator powers	25.26 ms / point · SE 0.11	250.06 µs / point · SE 0.88	25.51 ms / point · SE 0.11 accuracy unavailable Refused output · diagnostic timing
N=3, R=2 f64 · 15 moments Generic	all 1 $h = 1$	Independent dimension shift	119.40 µs / point · SE 0.31	8.47 µs / point · SE 0.12	127.87 µs / point · SE 0.35 δ_{rel} = 1.7e−14

Direct-API exclusion: dots mean $\nu_i > 1$; independent shifts request $D = 4 + 2h - 2\varepsilon$ with $h \neq 0$. COLLIER 1.2.10's public tensor routines accept momenta, masses and rank, without propagator-power or general dimension-shift inputs. Internally used higher-dimensional objects and equivalent derivative/coincident-denominator constructions are different interfaces and were not benchmarked here. See the COLLIER manual.

Measured relative error: $\delta_{\text{rel}} = \frac{E}{A}$, with $E = \max_j\left(|x_j - x_j^{\text{ref}}|_{\frac{\perp}{w_j}}\right)$, $A = \max_j\left(|x_j^{\text{ref}}|_{\frac{\perp}{w_j}}\right)$ and $w_j = \mu^{d_j+4+2h-2\sum_i \nu_i}$, where d_j is the numerator degree. The maxima cover all cumulative moments and all four complex Laurent slots. This is a dimensionally rescaled array-relative discrepancy against the qualified saved finite-precision tensors, not an exact-integral error bound. The older componentwise mixed metric remains in the raw audit. Withheld or unavailable arrays have **accuracy unavailable**, never a zero error; nonfinite counts are shown separately. A cell marked "finite" compares only its available finite fields and cannot certify the incomplete array.

Measured path: public `TensorReductionPlan / TensorWorkspace`, selected from 38 initial algorithm/master-policy candidates and four PV trials added after dotted IBP refusals. The latter only permute denominator triples; the integral, loop variable and accuracy request stay unchanged. Engine names, candidate IDs and profiling scope are saved. These paths are not a formal default-path or equal-accuracy speed comparison. No automatic higher-precision rescue is permitted in this cohort.

TIRust protocol: singleton input lanes. For $N \leq 4$, three warm-ups and 11 blocks target 30 ms, capped at 10,000 calls per block; for $N \geq 5$, one warm-up and five one-call blocks. Plan/workspace construction and first-call specialization are recorded separately from warm calls. Complete finite arrays, repeated-output consistency and unchanged inputs are checked outside the timer.

Environment and provenance: 2026-10-26, AMD EPYC 9754, pinned logical CPU 18, one thread; Rust 1.99.0 release build. Exact build/binary hashes, all candidates, requested tolerances, numerical flags, raw calls and selection decisions are retained in `validation/benchmark/opening-double-20261026/`. Prior mixed-precision opening captures remain archived unchanged.

Reproduction (fresh output copy; see suite README):

```
python3 validation/benchmark/opening-double-20261026/run.py build
python3 validation/benchmark/opening-double-20261026/run.py run
python3 validation/benchmark/opening-double-20261026/run.py summarize
```

No cross-session IREGI/COLLIER speed ratio is inferred from these absolute times.

5 Current coverage

The documented N≤6/rank≤6 finite numerical checkpoint now passes the agreed ordinary, dotted and shifted-array gates. The original quadrature-bearing Dual witness passes complete384 DD/context/lifecycle checks and SAME_SOURCE 256/384 consistency at unchanged2^{-80}. FAILED 128/256 (628 fields) remains explicit. Performance is deferred.

Interface / domain	Executed scope or remaining gate	Status
Local PyO3 facade	Restored-source c65b wheel: 10 installed smoke cases / 2709 saved checks, 205 selected bodies / 128 declared coherent of the actual 210-source build. Its N6 smoke is rank one; older 65API/7marimo/typing stay historical.	Scoped
Dotted propagators	N1/N4 first-dot/triple and N5 rank-six first-dot results retain their scoped passes. Current N6 first-dot256/prior128 strict pairs now pass; failed pre-repair attempts remain visible.	Scoped
Example gallery	Seven historical runtime cases and earlier three HTML exports/six browser views keep their own source contexts. No new gallery or typing run accompanies the current 10-case smoke.	Historical
Tensor / API coverage	Current1f source24/c65 wheel10 remain inherited. Original-SAMPLE128/256/384 pass complete DD/context/lifecycle checks; strict 256/384 passes, while FAILED 128/256 (628 fields) remains visible. API point count is dynamic; this qualification is finite N≤6/rank≤6.	Scoped
Public rank-six Dual	Historical h0 PV/Hybrid jets retain scope. The shifted requested-PV / selected-IBP original witness passes complete384 and strict 256/384 gates. Failed 128/256 and all earlier resource/math attempts remain archived. No new default-Hybrid h1 invocation is inferred.	Scoped
Performance	Historical private selected TIRust/COLLIER: seven slower families and one short inconclusive family. The fresh scheduler cohort is ineligible because of host contention; no default adoption.	Measured

Evidence is local Linux/Python3.13 and source-specific. Other platforms, separate Community-host loading and broader tensor numeric/exceptional domains remain unqualified. Earlier C0/D0 identities retain their wheel004 context.

6 Rank-six feature checkpoint: validated

Validated numerical checkpoint. All named finite numerical gates have sealed evidence. The documented scope is N≤6/rank≤6; performance is deferred. All454 prior archival records and timings remain unchanged.

Feature gate	Sealed evidence and qualified scope	State
Repaired parent arrays	Current Cartesian N6/rank-six first-dot and D2 pass 3360 current/prior reference slots, 1680 strict prior128/current256 cross-source pairs and 2520 pole controls. Pair maxima≈4.47284×10^{-28} /2.63578×10^{-27}; same-source convergence is not claimed.	Sealed pass
Independent D6	Current256 passes both complete 840-slot defining-integral rules (maxima≈1.13558×10^{-11} /6.67806×10^{-15}) and 840 strict cross-source pairs (max≈1.77715×10^{-26}). All 84 degree-six UV slots, including 20 macroscopic values, remain intact.	Sealed pass
Quadrature-bearing Dual	Complete384 passes both2520-field DD rules (max≈3.97433×10^{-15}), raw poles,3parent/6child full contexts and lifecycle/terminal gates. SAME_SOURCE 256/384 passes2520 at unchanged2^{-80}, max≈3.89996×10^{-60}. FAILED 128/256 (628 fields) stays archived.	Sealed pass
Release source join	Current c65 wheel passes 10 installed cases / 2709 saved checks: 205 selected / 128 coherent of restored 210-source 1f build. Native 8978 equals historical safe 5ef. Its N6 smoke is rank one; ccee/2674 and older typing/ gallery stay historical.	Inherited pass
Feature domain	Generic cumulative rank≤6 with positive powers and agreed numeric/ batch/API mechanisms. The API has no fixed N ceiling. The documented N≤6/rank≤6 families are a finite checkpoint; arbitrary-N and every-chart qualification are not inferred.	Scoped
Performance	Optimization is deferred. Keep every historical timing record unchanged and every displayed ratio with TIRust in the numerator. No new speed or parity gate is inferred from validation durations.	Deferred

The ordinary parent capsules remain sealed. The earlier e37b bridge is historical. Historical403/ccee and its96 GiB tensor failures retain their own epoch. Current restored1f binds24 focused passes; currentc65 binds10 installed cases. The final capsule binds the actual 210-build-snapshot source aliases; historical canonical aliases are not substituted. Complete128/256/384 DD/context/lifecycle schedules pass. Full256/384 consistency passes at2^{-80}; the failed 128/256 gate remains separate. The failed 128/256 strict pair is inherited without duplicate capture/source bodies; full256/384 convergence and the additive final capsule are now sealed.

The numerical gates are complete on this finite checkpoint. Preserve all wider numerical-domain limits on the next page.

7 Retained failures and limits of the completed checkpoint

This checkpoint records supported routes, their earlier failures and the precise limits of the accepted numerical evidence. The saved b2/de3c schema audit, root scratch-override failure and corrected final archive source-path assertion remain separate from the ten failed native witnesses. Strict128/256 is also a separate comparison failure; it remains visible beside the qualified256/384 pass. No missing coefficient, failed lane or unrun comparison may become a zero or a successful result.

Retained domain	Required qualification boundary	Outcome
Ten full witnesses	Retain eight resource/time no-array attempts and two96 GiB nonfinite tensor failures. Focused/compiler and saved-checker histories remain separate; strict 128/256 (628 fields) remains failed. The latter each record one failed call; empty/recovery is unrun. A two-second/96-sample profile of the earlier12864 GiB run is not the exact256 failure stage.	Retained
Earlier scalar/tensor failures	Old Cartesian dot/D2256 E6 failures, rejected original D6 finite comparisons and earlier cost stops remain unchanged. New successful source epochs have separate manifests and actual comparisons.	Retained
Scalar Dual charts	Unsupported massless C0/D0 derivative neighborhoods remain explicit failure paths. A rejected implementation chart does not prove mathematical nonanalyticity, and no broader branch qualification is inferred from one passing anchor.	Scoped limit
Independent accuracy	The qualified256/384 precision consistency and two-rule agreement are numerical evidence. Retain FAILED 128/256 (628 fields) of2520 adjacent. No rigorous quadrature remainder or certificate of achieved working-bit accuracy follows. Cross-source pairs are labeled explicitly.	Scoped limit
Point count and kinematics	The tensor API has no fixed point-count ceiling. The independently tested finite envelope is N≤6/rank≤6; arbitrary-N, arbitrary-kinematics and broader derivative charts remain unqualified.	Scoped limit
Completion boundary	Broader powers/shifts, accepted Legacy/rescue and exceptional domains retain their existing mechanism-witness scope. They are not blanket new-grid requirements. Performance optimization remains separate future work.	Scoped limit

The historical optional-profile skip passed24 focused checks but failed both full96 GiB tensors. Restored1f source passes24 focused checks, including finite first-order R0 contexts and an honestly refused transverse E6 lane; currentc65 passes10 installed cases. The unchanged original-fixture mass-second128/256/384 arrays pass actual DD/context/lifecycle checks. The stricter128/256 gate failed628 fields; full256/384 consistency and its final capsule now pass without relabeling that failure. The observed requested-PV route selects genuine IBP (0PV/610IBP requests, runtime fallback0). Future qualification must retain actual profile/analytic rows rather than demand actual PV or zero analytic rows. In the retained nonfinite failures, accepted masters and converged E6 endpoints did not establish finite tensors; the first bad node remains unknown.

Final source bindings must identify which current core, scalar dependencies and frontend release were actually exercised. Reuse unchanged source/archive data without implying a complete hermetic dependency closure. Historical old wheel and test-source contexts remain distinguishable from the final installed smoke. Physical comparisons retain their rank-four graph/CLI scope; no new rank-six physical or broad external Community-host qualification is implied.

8 Historical da7 precision boundary: retained128/256 failure

HISTORICAL — da7 cutoff. The following384-active/no-array and pending statements describe the archived earlier milestone. The validated checkpoint above supersedes those runtime states, preserving every original comparison, failure and raw value.

Intermediate checkpoint — objective open. The original restored fixture agrees with the independent defining integral, while the stronger same-artifact 128/256 precision gate fails. Its threshold is unchanged.

Gate / scope	Observed result or pending admission	Outcome
Defining-integral agreement	Both complete 128/256 runs pass 2520 fields per DD rule at 1e-10 (all four maxima≤3.97433×10^{-15}). The DD rule pair also passes. Four retained 1806-control sets preserve raw reducer/reference poles without projection; the joint saved replay counts each dataset explicitly.	DD pass
Owned contexts / lifecycle	Each precision completes 3 parent / 6 child owning contexts with higher errors 0/1/2, bit-identical duplicate/recovery, isolated invalid null/context refusal and actual empty reset. Both terminal codes are 0; guards and cleanup pass. These are actual complete schedules.	Complete
Strict 128/256 consistency	FAILED 628 of 2520 fields at unchanged $2^{-80}\approx8.27181\times10^{-25}$. Maximum normalized difference≈2.92162×10^{-21}. The independent saved-array replay confirms the failure. Retain all failed fields and the original comparison; no relaxed gate or exclusion.	Failed
Precision interpretation	Requested 53 uses heuristic admission 2^{-34} / E6 target 2^{-36}; Success does not promise 80-bit consistency. All 630 finite differences are consistent with combined estimates (saved 128 Q≈9.99×10^{-21}). Order 64 / 2400 boxes are unchanged; tracked imaginary precision 94→222 bits. This fits precision-dependent cancellation/E6 noise, not a rigorous bound.	Scoped
Higher precision	One 384-bit producer is now active on the same 749b / current 1f source, fixture, 53-bit request and mathematics, under 256 GiB / 7200 native seconds. No 384 result is admitted. Full 384 DD/context/lifecycle and strict 256/384 comparison at 2^{-80} remain pending; failed 128/256 stays visible.	Active
Preserved scope	All 451 prior raw values and nested publication history stay exact. Ten old native witnesses and the separate saved-schema failure remain visible; this precision-gate failure is separate. No new performance claim; TIRust remains the timing-ratio numerator.	Retained

The DD comparisons and the strict precision comparison answer separate qualification questions. Passing the former does not erase the failed latter. Source/error estimates are heuristic, not certified achieved-bit accuracy. No source formula, input or threshold is changed for the proposed continuation.

Sealed capsule cd656689 owns the actual complete 128/256 packet once. This intermediate binds that immutable capsule and its checkpoint. Future 384 capture files are excluded while mutable. The API has no fixed point-count ceiling; independently documented families remain N≤6 / rank≤6. Final feature completion remains pending.

Source: `validation/e6-dual-rank6-precision-boundary`. The adjacent checkpoint and root saved-replay packet preserve exact comparisons, source/guard context and harness history. Saved replay success retains the FAILED precision gate; it does not promote it to a pass.

9 Earlier exact-zero restoration (a7ccffb)

Historical a7 cutoff: full128 was preparing and 256 was held. The current precision-boundary page supersedes those runtime states, with no change to their archived evidence.

Intermediate checkpoint — objective open. The historical optional-profile bypass did not produce a valid full rank-six tensor. Its failures are retained; the restored route has a separate source and release epoch.

Evidence	Sealed evidence; objective remains open	State
Two 96 GiB failures	Both original-SAMPLE runs return 101 with a failed first-call record. Four N5 rows fail the nonfinite-output guard; valid parents are Unstable, failed arrays null and the invalid neighbor isolated. No 2520 tensor comparison or owned error context passes. Capsule 792765c1 preserves 526 nonbinary guards.	Retained
Restored source	Sealed 1f1a source capsule removes the 15-line bypass: 24 focused checks pass in 26.8588 s from build b069 / product 749b. The first-order Dual regression accepts finite R0 rows and retains the transverse E6NonConvergent refusal. Earlier 23-pass / failed-expectation invocation b804 and compiler boundaries remain intact. Compiler opt3 disables debug assertions and integer overflow checks; old debug403 enables them. Numerical precision/branch/error guards remain unchanged; no compiler-equivalence claim.	Sealed pass
Installed frontend	Sealed c65b wheel passes the same 10 installed cases / 2709 saved checks: 205 selected, 128 coherent of 210. Actual build 118.851 s and fresh install/import bind the restored source. Native 8978 matches historical safe 5ef byte-for-byte. N6 smoke is rank one; old typing/gallery remain separate.	Sealed pass
Retained generic route	The failed captures record requested PV selecting actual IBP: 0 PV / 610 IBP scalar requests, runtime fallback 0. Finite-master admission and converged E6 endpoints did not establish finite tensors. Actual profile/analytic rows must remain visible; old PV-only / zero-analytic expectations are obsolete.	Scoped
Full Dual witness	Restored original SAMPLE 128 is root-released with 192 GiB / 7200 native seconds; 256 is held. Full 2520 fields, same-source pair, DD/core agreement, owning error contexts and lifecycle remain pending. Active capture files are not bindings.	Pending
Coverage and timings	Ordinary dot/D2/D6 and DD reference are inherited without duplicate counts. No fixed N ceiling is added to the API; independently documented families remain N≤6 / rank≤6. No new performance measurements; TIRust remains the timing-ratio numerator.	Unchanged

Ten unsuccessful full-witness attempts remain visible: eight earlier resource/time cases and these two returned failed-call records. Empty and singleton recovery remain unrun in the latter. The first nonfinite evaluator node is unknown; exact-null geometry as its cause remains a hypothesis. Saved guard checks are archival integrity evidence, not independent tensor comparisons. Separate focused-test and compiler-boundary failures remain in the source history; they are not counted among these ten full witnesses.

This additive checkpoint preserves all 446 baseline raw values and nested publication history. It records restored-source checks and source epochs; it does not close the remaining numerical gates or promise a resource bound.

Sources: `validation/e6-dual-postfix-nonfinite-boundaries`, `validation/e6-dual-restored-zero-profile-core-tests` and `validation/python-interface/restored-zero-profile-wheel-refresh-smoke`. Root saved replays are archived in the adjacent restoration checkpoint; no new performance or full rank-six numerical qualification is inferred.

10 Earlier resource boundaries (c112865)

At the c112865 cutoff, the original-SAMPLE N6/rank-six/h1 mass-second witness had no admitted full tensor array. Four post-fix attempts had ended without completing a call; four earlier failures were retained. Two larger-budget 96 GiB attempts were then active. Both subsequently failed at the nonfinite tensor guards documented above. That diagnostic checkpoint preserved all 443 prior raw values and added no TIRust numerical or performance qualification.

Frozen attempt / model	Terminal evidence or precisely bounded interpretation	Outcome
Post-fix Float128/256	Both earlier 2400-second attempts complete generic construction (about 1778s each), then hit the native execution deadline without a completed call or array. They retain the source 210/product 4f430732 context and requested accuracy 53; these are resource limits, not numerical disagreements.	No array
Float128 /64 GiB	The 4200-budget attempt aborts in numerical allocation after 3733.628 native seconds; GMP cannot allocate 16 bytes. The final receipt retains 250 unchanged guarded bindings, empty cleanup and an absent owned process group. No array is returned.	No array
Float256 /64 GiB	The 4200-budget attempt aborts after 2912.259 native seconds; GMP cannot reallocate 16→40 bytes. The same 250-binding terminal guard/cleanup checks pass. This does not establish the exact evaluator, slot or refinement active at failure.	No array
Then-active 96 GiB attempts	At this historical cutoff, separate 128/256 requests were active: 7200 native / 7380 total / 120 cleanup seconds each, explicit PV, source 210 / product 4f430732 and unchanged fixture/shape/accuracy 53. Their later failed-call records are separately bound by 7927 above.	Historical
Two-second /96-sample profile	The sample concerns one numerical instant of the older Float128 /64 GiB run. It sees complex multiprecision arithmetic, copies and allocation. It does not identify the later Float256 failure stage, whole-run cost, peak memory, or 96 GiB sufficiency.	Diagnostic

The source model is finite but is not an exact peak bound. For master-batch capacity B>1, Symbolica reserves B×(root slots + shared-body slots), including inactive shared bodies, and retains the largest capacity. E6 with R active requests, order q and c components prepares 5Rq box rows, 11c input values and 6c allocated master outputs per box row. At c=3, that is 33 complex inputs and 18 allocated outputs per box row (including internal quantities), before other retained workspaces, graphs and allocator storage.

One active request through default orders 8–128 accounts for 1240 cumulative box rows and a largest refinement of 640 rows. These are structural formulas, not observations of the failed stage. Initially one-bit Float zeroes can acquire larger limbs when touched at working precision. First-touch expansion and allocator retention are plausible explanations of growth, not a proven fault cause, leak or plateau. The saved artifacts lack the exact live slot inventory.

Newly archived reference. The already completed DD defining-integral stream and all 2520 parent rule pairs pass (maximum$\approx 2.44997\times10^{-15}$). This archives reference qualification; it adds no TIRust numerical success. TIRust/DD comparisons remain pending.

The initial offline symbolization timeout remains failed; the later no-inline report is separate. Raw perf data is omitted by historical hash. At this cutoff, 96 GiB terminal results were not yet bound. Ordinary/core/wheel counts stay inherited. Final completion waits for actual full-Dual arrays, strict pairs and independent comparisons; performance remains deferred.

Source: `validation/e6-dual-postfix-resource-boundaries` and `validation/e6-dual-rank6-mass-second-defining-integral`. Memory formulas are source-based diagnostics, not a complete dependency closure or a certified byte/accuracy bound.

11 Earlier ordinary closure (dfcae98)

The documented ordinary N6/rank-six first-dot, D2 and genuine-D6 families now have complete current Float256 results and strict cross-source precision comparisons. The full rank-six feature goal remains open while the quadrature-bearing Dual witness runs. All 437 earlier raw report values and timings remain unchanged; no performance or arbitrary-N claim is added.

Executed domain	Complete sealed evidence and boundary	Outcome
Cartesian first dot / D2	Together: 3360 current/prior original-quad slots, 1680 strict cross-source pairs and 2520 pole controls pass. Current256 maxima are≈9.11420×10^{-16} and 2.67157×10^{-17}; pair maxima≈4.47284×10^{-28} and 2.63578×10^{-27}.	Pass
Genuine D6 current Float256	Both complete 840-slot defining-integral rules pass 10^{-10} (maxima≈1.13558×10^{-11} /6.67806×10^{-15}). All 840 strict old Float128/ current Float256 pairs pass 2^{-80} (max≈1.77715×10^{-26}). The 5332-gate independent audit and all 84 degree-six UV slots remain preserved.	Pass
Actual ordinary batching	The D6 parent uses four packed same-dimension N5/IBP rows, 124 scalar masters and 2400 pentagon-box rows. Requested and actual methods remain explicit; broader mixed-method tensor completion is not called pure IBP.	Scoped
Earlier final wheel	The 5ef source epoch passes 10 installed smoke cases with 204 selected files and 128 coherent core bodies; 2614 saved bindings pass. This wheel predates the resource-route change below and remains immutable.	Historical pass
Current wheel refresh	The ccee epoch repeats 10 cases after the route fix: 8 low/interop, 1 checked-Dual rejection, 1 N6 rank-one lifecycle/contraction. 205 selected files match 128 declared core bodies within the actual 210-source build; 2674 saved checks pass.	Pass
Still open	Original-SAMPLE N6/rank-six explicit PV Float128/256 arrays with owned full-component quadrature contexts, their strict pair, independent mass-derivative anchor agreement, and final coherent publication remain pending.	Open

Every precision pair above is explicitly CROSS_SOURCE: current Float256 is compared to a separately frozen prior Float128 result. No same-build precision convergence or achieved working-bit accuracy is inferred. Rejected original D6 finite values remain archived and are not the defining-integral gate. The earlier dot/D2256 E6 failures remain failed in their original source epochs.

Both wheel epochs use the same exact ten API cases. The N6 smoke has rank one; full rank-six mathematical validation is separate. Earlier 65API/7marimo and consumer typing remain historical contexts. There is no new notebook, typing, platform, hermetic dependency-closure or independent-integral claim from smoke.

Sources: `validation/ordinary-gh-n6-generalized-requalification`, `validation/ordinary-gh-n6-d6-requalification`, `validation/python-interface/final-rank6-wheel-smoke`, `validation/python-interface/final-rank6-wheel-refresh-smoke`.

12 Optional zero-profile route: correction and bounded tests

The retained 64GiB Hybrid and explicitPV preparations reach an optional exact-zero runtime specialization after their primary N5 child completes, then fail before a returned array. Saved stage ordering establishes this precise attribution. The earlier Hybrid 32 GiB failure occurs in primary mapping instead.

Stage	Actual source/test scope and retained limit	Outcome
Corrected old diagnosis	PV128 generic 12441-alias mapping and the primary N5 child finish before second 12318-alias preparation. Saved Hybrid64/PV128/PV256 logs show the same post-child order. The constructor eager-fallback stage precedes this later optional zero-profile; Hybrid32 fails earlier primary mapping.	Retained failure
Narrow route fix	For genuine E6 multi-component quadrature Dual workspaces, retain the generic route and skip optional zero-profile preparation. Full-Taylor zero classification and generic branch, precision, coupled-context and error guards remain; actual skipped lane groups are diagnosed.	Source pass
Concrete current tests	210 actual compiled source bodies; 24 distinct focused tests pass. The first stage has 23 passes and a 60-second timeout on the new real N5/D6 Dual test; the separate exact-test continuation passes in 146.707 total seconds. The original timeout remains failed.	Scoped pass
Coherent frontend	The current ccee 10-case wheel refresh binds the changed execution source and regression module. Its 205 selected files are tied to the fresh 210-source context; source identity is not a successful full N6/R6 Dual result.	Pass
Full original fixture	Post-fix original-SAMPLE explicit PV Float128/256 runs are active. No completed parent array or imported-context accuracy is admitted at this cutoff. The half-time replacement proposal remains held and unrun.	Pending
Independent anchor	The DD reference and rule-pair stream are ready in unsealed staging. They do not qualify the reducer until actual full parent/Taylor/Laurent comparison and owned propagated-context gates pass. No pending target packet is source-bound here.	Pending

The 24-test source capsule is a finite route witness, including one real small N5/D6 Dual regression. The initial new-test 60s timeout remains failed; its separate successful continuation does not rewrite that receipt. No tolerance, master branch, analytic domain or algebraic formula is relaxed by the fix. Ordinary and prior frontend evidence retain their own compiled source scopes.

All four earlier Hybrid/PV preparation/resource failures remain visible, irrespective of later success with the repaired generic route. Unsupported massless scalar derivative charts, conservative high-point Legacy admission, high-point explicit-Dual rescue, and arbitrary-N or arbitrary-chart qualification remain limits rather than blanket new-grid completion gates. Performance optimization stays deferred. All following sections retain their historical outcomes and original source contexts.

Source: `validation/e6-dual-optional-zero-profile-core-tests`. No active-capture mutable input or unsealed DD packet is a report binding.

13 Earlier ordinary/source follow-up (fccffab)

This follow-up preserves all 425 earlier raw result values and timings. Each new test or numerical result is tied to its actual source epoch. The generic rank-six goal remains open; independent D6 adjudication is on the next page. No performance measurement or fresh Python-wheel qualification is added.

Executed scope	Saved evidence and precise boundary	Outcome
Current core	22 exact focused tests pass: 15 joint quadrature-budget, five zero-mass strategy and two existing MassTaylor ownership/projection guards. The emitted Cargo artifact was fresh; its 128 selected source bodies include the current scalar-master repair.	Tests pass
Joint-budget source	The separate earlier source capsule preserves 15 tests and three contexts, including the failed synthetic fixture and Arc compile stage. Higher Taylor components combine their own propagated quadrature and operand-roundoff estimates; no actual propagated-quadrature Dual integral is qualified.	Scoped
Zero-mass supplied G	Private N5/rank0 first-dot h=0 ordinary and scale-only Dual witnesses pass 102 exact gates (34+68), including mixed invalid lanes and empty/reset/recovery. Only μ=3+3t has nonzero jets; all kinematic jets vanish.	Scoped
Earlier Cartesian dot	Pre-repair CLI003 N6/rank6 first-dot Float128 passes 840 quad slots and 630 pole controls. Float256 fails E6 convergence at order128, error≈2.3794×10^{-4}, with no parent array or precision pair.	Partial
Earlier D2 shift	The same pre-repair epoch at h=−1 passes 840 Float128 quad slots and 630 pole controls. Float256 fails E6 terminal1 at order128, error≈2.499×10^{-4}; its 840 reference and 840 pair comparisons remain unrun.	Partial
Still open	Repaired Cartesian dot/D2 and D6 parent precision pairs, a genuine quadrature-bearing Dual with independent anchor, and the final current release remain the completion gates.	Open

The private zero-mass fallback preserves the authoritative supplied products and fresh complete geometry proof. The failed derivative in another positive mass remains an unsupported existing massless C0/D0 derivative chart. It does not establish mathematical nonanalyticity. This rank-zero ordinary/scale witness does not qualify zero-mass rank-six or general kinematic Duals.

The source-test counts overlap: the later 22-test cohort reuses the 15 earlier joint-budget tests. Neither count is added to the historical 120 as a claimed unique total. The installed 65-case facade retains its prior wheel context. Broader powers/shifts, Legacy/rescue and exceptional domains retain scoped qualification limits; no new broad grids are required here. Original failures and unavailable arrays remain unchanged. Source-test success does not imply numerical array agreement, achieved working precision, rigorous integration error or speed.

Sources: `validation/rank6-current-focused-core-tests`, `validation/propagated-dual-joint-budget-core-tests`, `validation/checked-g-zero-mass-fallback`, `validation/affine-power-generalized-memo-dot-rank6`, `validation/public-n6-rank6-d2-partial-evidence`.

14 Independent D6 adjudication and scalar-master repair

A defining Feynman-parameter calculation now resolves the original D6 finite reference contradiction at the fixed N6/rank-six point. Separately, a shared coefficient repair fixes a scalar-box precision loss and restores one failed E6 evaluation. These are distinct numerical claims with distinct source contexts; parent precision-pair runs remain outside this checkpoint.

Independent check	Executed numerical evidence	Outcome
Complete D6 array	Two fixed rules each check all 210 tensor moments / 840 Laurent slots against TIRust Float128 at 10^{-10}. Maximum errors: order12≈1.13558×10^{-11}, order16≈6.67806×10^{-15}. All 840 rule differences pass, maximum≈1.13491×10^{-11}.	Pass
Original finite values	All 210 earlier finite disagreements remain archived. The original scalar/time coefficients also violate an exact positive-F mean bound. The independent parameter calculation agrees with TIRust; the earlier raw reference mismatch is not silently rewritten or counted as a Rust accuracy failure.	Rejected reference
Ordinary D0 repair	Existing Symbolica coefficient_sum preserves the exact shared line coefficient. Ten regressions and exact chart identities pass. One copied Float256 D0 value agrees with independent order20/32 rules: maximum≈3.5524×10^{-15} at 10^{-10}; rule comparisons pass 10^{-12}.	Scoped pass
Actual E6 recovery	The old failed trace is retained. The repaired ordinary scalar path converges at order64 with 600 box rows and estimate≈7.39054×10^{-27}. This is actual convergence, not an independent E6 value or parent-tensor accuracy certificate.	Scoped pass
Common-node identity	All 600 common nodes preserve exact inputs; 13200 component checks replay. Of their D0 values, 480 are unchanged and 120 at the isolated pinch change. Failed compiler, preflight and broad-observer preparation stages remain saved.	Pass
Precision limits	The D6 oracle uses 106-bit DoubleFloat with exactly saved binary64 rule data. No rigorous remainder, Float256 parent pair, full Dual branch, native/generated parity, arbitrary-kinematics or performance claim is added.	Limits

D6 replay checks 40 windows, 5068 acknowledgements and exact limb/digest/pole data. Its 84 degree-six UV slots include 20 macroscopic all-even values. The missing-NumPy launch, MPFR stop and finite-reference failure remain saved. Rule agreement is numerical convergence evidence.

The ordinary-only D0 repair preserves the complete prior seeded-Dual helper. Its existing Dual regressions retain their own scope. The full imported-error Dual path still needs actual numerical qualification. All following sections retain their earlier immutable checkpoints and original outcomes.

Sources: `validation/ordinary-complex-box-correlation` and `validation/affine-power-generalized-current-d6-rank6`.

15 Earlier feature boundary (6a826a0)

The current implementation checkpoint remains incomplete. Rank-six numerical coverage has advanced through named public derivative and positive-power paths; arbitrary point counts and the full generalized domain are not implied. No performance measurement is added, and every preceding raw result remains unchanged. Failed requests keep their original outcomes and source contexts.

Executed domain	Evidence and precise boundary	Outcome
Current core	158 canonical source bodies, nine additional versions and four compile contexts. All 120 distinct focused tests pass across a test-only bridge: 95 completed first, then the remaining 25. Six compiler errors and the stale Legacy engine-tag observer failure remain retained.	Pass
Fresh Python facade	65 installed API cases, seven marimo runtime cases and clean consumer typing. All 158 declared core bodies match the wheel input context within 201 selected files; 1575 portable saved checks pass.	Pass
N5 first dot	Rank-six Float128/256: 5040 original-quad, 5040 mass-Taylor, 2520 precision pairs and 7140 pole/duplicate/scale checks, totaling 19740. Maximum reference error $\approx 2.77 \times 10^{-16}$, precision difference $\approx 3.55 \times 10^{-29}$.	Scoped
LegacyNoOp N5	PV/unit/h=0: 26 focused tests, 1680 complete quad slots and 840 strict 2^{-80} pairs. Maximum reference error $\approx 1.09627 \times 10^{-16}$; pair difference $\approx 2.65877 \times 10^{-32}$. Legacy admits only a proved no-op snapping profile here.	Scoped
Checked supplied G	Current default-Hybrid N6 rank-six unit/first-dot, h=0: 3360 original-quad slots and 1680 strict 2^{-80} pairs pass. Maximum errors $\approx 9.11420 \times 10^{-16}$ and 6.86405×10^{-30}. Exported acceptance diagnostics remain distinct from source-only inner proof guarantees.	Scoped
Remaining gates	Cartesian generalized rank-six closure, D6/dimension shifts, broader positive powers, numerical Dual with imported quadrature budgets, accepted Legacy/rescue domains and remaining exceptional cases stay open.	Open

The N5 dotted pair uses the earlier memo CLI and its 119-source context. Its two short preparation timeouts and three older unrun cases remain in the archive. Passing first-dot arrays do not qualify N5 triple powers or later G/MassTaylor sources. The Legacy pair likewise has a separate 151-source context; its old memory abort, compile error and pre-launch refusal remain. Neither pair establishes achieved working-precision bits or performance.

Mass differentiation is an explicit strategy with a nonzero differentiated-mass domain. Default Cartesian/formal-G reduction retains direct recurrence, so valid dimensionally regulated zero-mass dots are not silently removed from its domain. The private checked-G hook still lacks a same-domain zero-differentiated-mass fallback; the massive checks above do not close that gap. A current Cartesian N6 first-dot Float128 run passes all 840 reference slots, but Float256 fails its E6 convergence check at order128 and returns no parent array or precision pair. D6/unit Float128 returns an array but fails all 210 finite reference slots (maximum error ≈1.4213); its 420 IR and 126 lower-rank/84 rank-six UV controls pass. Float256 remains unrun after that required gate fails. The current guard on unsupported imported full-Dual quadrature remains active; the proposed extension is not included in this publication.

Sources: `validation/checked-g-generalized-core-tests`, `validation/python-interface/facade-wheel-004`, `validation/low-n-generalized-rank6-memo-dot-pair`, `validation/legacy-noop-rank6`, `validation/checked-g-generalized-numeric`. Source binding is per cohort; no unexecuted equivalence to later edits is asserted.

16 Public six-point rank-six mixed Duals

The public AffinePower facade now has complete fixed-family mixed-Dual results for explicit PV and the unchanged Hybrid default. Each parent contains all 210 moments, four native Laurent slots and six ordered Taylor components for the first squared mass and a simultaneous dilation. Sample lanes are separate from Taylor components; every component is checked before contraction.

Public path	Complete saved numerical checks	Outcome
Explicit PV	Float128/256: 30240 original-quad parent comparisons and 75600 parent/child precision pairs, including 60480 child pairs. Maximum errors ≈3.07559 × 10^{-15} and 5.47562 × 10^{-30}.	Pass
Default Hybrid	The same full counts and numerical gates pass independently. Both captures explicitly record `public_default` origin. Actual child methods, notes, batching and counters remain visible; requested Hybrid is not a pure-IBP claim.	Pass
Coupled contexts	Per precision and method: 15 complete parent/child contexts, 12 actual N5 child rows, repeated valid inputs around an invalid-mu lane, empty reset and singleton recovery. Every child explicitly reports KnownNoQuadrature with complete error arrays.	Pass
Typed DoubleFloat	106-bit metadata: 15120 quad, 15120 qualified Float128 controls and 5040 scale checks pass. Maximum normalized errors ≈3.07559 × 10^{-15}, 1.96691 × 10^{-25} and 8.03765 × 10^{-26}. Static/dynamic base outputs match exactly.	Scoped
Typed f64	Three valid rows hit the real `pv_node_12`, Taylor-component-two cancellation guard and return null arrays. Saved metadata passes, with zero numerical comparisons. The initial input-size helper refusal remains a separate pre-reduction failure.	Guarded

All cases use real massive unit powers, h=0 and requested accuracy53. The reference/scale tolerance remains 10^{-10}, precision/control pairs 2^{-80}. The six ordered components are (0,0), (1,0), (0,1), (2,0), (1,1), (0,2). Original unit/dot/triple arrays supply the mass/dilation responses; their finite coefficients retain their scoped empirical admission. Agreement between precision outputs is not a rigorous eighty-bit accuracy certificate.

The typed example repacks existing Symbolica static/dynamic Dual values into the public component interface. It does not claim a new typed convenience constructor, and it exports no hidden sensitivity context. The frozen observer does export and validate full contexts. Zero quadrature arrays never establish an origin by themselves: unsupported propagated origins remain failed. Both earlier private ThinSix mixed128 allocation failures remain failed.

Sources: `validation/public-n6-affine-mixed-dual`, `validation/public-n6-hybrid-typed-dual`. No arbitrary-direction, all-kinematic, rigorous-error, full-feature or speed claim follows.

17 Independent first and second mass finite differences

Five ordinary Float384 samples, requested accuracy160, provide an independent check of the specified squared-mass derivatives. They form two three-point central stencils sharing the center, at $h=2^{-12}$ and $h=2^{-18}$. This is not a fourth-order five-point stencil. Existing Symbolica Matrix.solve produces the exact weights, including the second Taylor coefficient's factorial convention.

Check	Complete fixed-family evidence	Outcome
Fine-step accuracy	6720 comparison records include coarse diagnostics; all 3360 fine-step target gates pass 10^{-10} against qualified Float256 AD and original-quad dot/triple anchors. Maximum original-reference errors: first $\approx 4.63100 \times 10^{-12}$, second $\approx 5.76765 \times 10^{-12}$.	Pass
Resolved convergence	420 transitions improve by at least factor16; the fixed nonpolynomial scalar finite first-mass witness resolves. Coarse errors reach roughly 2×10^{-8} and are not tested against the final fine-step accuracy threshold.	Pass
Explicit floor cases	1260 transitions remain floor/unresolved. The floor is max(32 × AD256/quad gap, 32 × AD128/256 gap, 2^{-80}). It classifies convergence and never relaxes the final 10^{-10} accuracy requirement.	Retained
Actual ordinary data	All 4200 parent complex slots, five parent rows, twenty N5 child rows and 25 full native contexts are observed. Input preservation, child origins, same-rank subsets and explicit KnownNoQuadrature status are checked.	Pass
Preparation history	The first Symbolica weight construction fails on a 0^0 matrix input and produces no weights. The reviewed repair uses literal one for the constant row, with the same Matrix solver and exact residual checks. The original failure remains unchanged.	Retained

The stdlib replay reproduces the complete original comparison, observer and convergence report from immutable supplied bytes. It executes no solver, compiler, native module or integral. The PV parent supplies the already qualified AD/reference/source context; separate Hybrid and typed capsules are unchanged. Native identities and process cleanup are historical metadata.

Finite differences and AD share integral software; the original IREGI anchors retain their separate independent admission. This real massive N6/rank-six, unit-power, h=0 PV family validates first/second derivatives in one mass direction. It does not certify arbitrary higher/mixed directions, branches, imported quadrature budgets or rigorous error bounds. No performance measurement or complete-goal claim is added.

Source: `validation/public-n6-rank6-mass-fd`. All following sections retain earlier source checkpoints and their original failures; later scoped successes above do not rewrite those outcomes.

18 Earlier standalone Python facade (wheel003)

A newly built standalone wheel uses one native host for the shared Symbolica, Numerica, FeynKit and Spenso PyO3 classes. Its installed tests exercise the public shared tensor facade, ordered Dual layouts and the explicit checked scalar-product option. These frontend results qualify their captured source context; the full generic rank-six mathematical gate remains open.

Executed check	Current wheel003 evidence	Outcome
Installed API	64 cases: 25 upstream numeric precision/Float, 29 existing native-object interoperability/numerical cases and ten facade cases. Consumer typing has zero errors and warnings.	Pass
Actual marimo runtime	Seven cases execute graph rendering, numerator preparation, batched contraction, dots, Duals, beyond-binary64 inputs and the existing benchmark explorer with the installed wheel.	Pass
Routing and guards	Eligible six-point profiling and primal-only Dual workspaces select `affine_power`. Checked ordinary, profiling and primal-only Dual calls reject inconsistent supplied products before scalar masters; empty/reset counters are checked.	Scoped pass
Source and saved replay	147 selected source files. The saved-only replay passes 1042 checks over source/evidence bindings, test cases, consumer typing, process cleanup metadata and retained failures.	Pass
Domain limits	This frontend cohort alone does not qualify complete checked rank-six or six-point mixed Dual arrays. Other platforms, Community-host loading and current browser exports remain unqualified. Existing N6 rank-one and N7 scalar cases are frontend regressions.	Limited

The first wheel build failed after external Nix Rust/Cargo paths disappeared. The exact preceding 1.99 toolchain was restored and protected with project GC roots; all 146 common frozen inputs agree across the failed and successful builds. The first environment installation then hit an offline cache miss for pinned marimo 0.24.0. Installing the same pins with downloads allowed succeeded. Both original failed receipts remain failed.

The aggregate eight-case dispatch phase reached its fixed deadline after seven exact pytest nodes had passed. Its final checked-Dual case was unfinished. Only that case was completed with its full per-case budget; it passed without changing numerical assertions or accuracy guards. The partial raw log and failed aggregate receipt remain archived. The 64 API cases count each exact successful node once.

The capsule records the wheel/native identities but contains no compiled wheel, native binary, environment or cache. Its 147-file source selection is not a full compiler/native/system closure. Portable replay starts no native host, integral or benchmark. Historical wheel002 and wheel004 evidence below retains its original contexts and measurements.

Source: `validation/python-interface/facade-wheel-003`. Linux CPython3.13 development build only; no new timing or performance gate.

19 Dotted rank-six tensors: N1 and N4

The first-propagator power-two and power-three families now have complete TIRust rank-zero through six arrays at Float128 and Float256 for N1 and N4. Every valid lane returns 210 Cartesian moments and four separated native Laurent slots. The fixed massive spacelike inputs use h=0 and requested Hybrid. This is actual dotted tensor evaluation, beyond the earlier reference anchors.

Check	Saved complete evidence	Outcome
Eight actual calls	Four N1 and four N4 power-two/power-three calls. Each batch has base, exact-duplicate and scale-two valid lanes around an invalid-mu neighbor. All 20160 requested complex slots are present.	Pass
Reference and precision	20160 original-quad comparisons at normalized tolerance 10^{-10} and 10080 Float128/256 pairs at 2^{-80}. Complete arrays are compared before any numerator contraction.	Pass
Batch and pole controls	6720 exact duplicates, 6720 scale identities, 10080 massive IR zeros, 2046 UV zeros and 12240 parity zeros. Together with reference/precision checks: 68046 main gates and zero numerical failures.	Pass
Mass-Taylor consistency	A separate ordinary-Dual path passes 10080 saved mass-Taylor identities per N, 20160 total. This is agreement between distinct TIRust paths; it is not an independent integral oracle.	Scoped pass
N5 completion gap	The first-dot Float128 call reaches its fixed 810-second preparation limit under 32 GiB, with no array. Its Float256 partner and both power-three cases remain unrun.	Open

The initial N4 aggregate controller stopped at a deliberate barrier after three completed cases. Only the previously unstarted fourth case was executed with the corrected per-capture budget. Its original partial receipt remains unchanged. The N5 timeout likewise remains failed, with input guards and owned-process cleanup recorded; missing slots are never successful zeros.

The source-bound capsule retains all 145 selected compiled source bodies for its actual frozen CLI. Its source context predates the later wheel003 CLI and does not claim equivalence to that binary. The portable saved-only replay reconstructs the full original Fraction review and separate mass-Taylor reports in a minimal copy without the external target tree or a license. It starts no integral evaluator.

The original finite reference coefficients are empirically admitted. Pole and parity identities do not independently certify every finite coefficient or provide rigorous error bounds. These h=0 real-mass families do not qualify shifted dimensions, arbitrary kinematics, complex masses, N5/N6 dotted closure, or the complete rank-six objective. No timing measurement is added.

Source: `validation/low-n-generalized-rank6-current-cli`. Actual CLI `f69615f9…`, coherent build002; historical failures retained.

20 Checked supplied products: six-point rank six

The opt-in checked Cartesian interface retains the supplied scalar products and proves compatibility before evaluating child engines. Its newer packed implementation now returns complete ordinary N6 rank-six arrays at Float128 and Float256, in addition to the earlier guard and rejection evidence. The numerical scope is explicit PV, unit powers, h=0, Preserve and native Cartesian components.

Check	Source-bound executed evidence	Outcome
Complete rank-six output	All 840 native complex slots at each working precision pass admitted original-quad comparisons at 10^{-10}. Both maximum normalized reference errors are approximately 2.10018×10^{-16}.	Pass
Working-precision pairs	All 840 Float128/256 pairs pass 2^{-80} normalized agreement, maximum 2.13539×10^{-32}. Requested precision is 53 for both runs. This does not establish achieved 80/128/256-bit accuracy.	Scoped pass
Guard and batch lifecycle	20 exact named normal tests plus an actual rank-one mixed-validity/full-array/contraction/recovery test. Its two valid rows use one packed child call with 12 child rows and one parent call; the invalid neighbor remains failed.	Pass
Earlier resource failures	The old inline rank-six run has no completed array and a sampled last RSS near 145.7 GiB. A separate constructor test hits its fixed 60-second limit before immutable-variant reuse. Both failures remain archived.	Retained
Unqualified domains	Generalized supplied-G powers/dimension shifts, numerical Dual derivatives, other conventions, default Hybrid and arbitrary kinematics remain outside this fixed numerical cohort.	Open

The compatible full-Taylor certificate is checked before child compilation. Zero cofactors are masked before singular pinches are prepared. Child outputs keep their supplied products and native UV/IR split; nonzero components and poles are not projected away. The accepted native convention adapter is covered by focused source/tests. The earlier normalization draft and its positive source review are preserved together with their retraction; that rejected packet was never compiled or numerically evaluated.

The source capsule reconstructs all 145 selected compiled-source hashes using 110 byte-identical wheel003 files and 35 own additions. Its checker imports no producer or native library and recomputes all 1680 quad records, 840 pair records and 840 strict pair decisions with exact Fraction arithmetic. Binary identities and historical cleanup are saved attestations, not fresh binary or process probes.

The old inline timeout retains its original cleanup errors; a later additive release records historical process-group absence. The RSS value is a sample, not a measured peak or a performance result. No new timing ratio, speedup, rigorous error bound or complete-feature claim is added.

Source: `validation/checked-supplied-g-unit-rank6`. Generalized and derivative runtime closure remain separate requirements.

21 Affine denominator descent: ordinary N6 rank six

The new dynamic positive-power engine uses a Symbolica-derived constant-one denominator identity to descend to packed child topologies. It retains the tensor rank, denominator powers, dimension and numerical component layout. The ordinary N6 rank-six unit-power family now has complete Float128/256 arrays; its dotted and D6 resource gates remain open.

Check	Frozen actual engine evidence	Outcome
Complete ordinary arrays	All 840 complex slots per Float precision pass admitted original-quad comparisons at 10^{-10}. Maximum normalized reference error is 2.10018×10^{-16}. Separate controls check 840 IR zeros and 420 power-counted UV zeros across both captures.	Pass
Precision and batching	840 Float128/256 pairs pass 2^{-80}, maximum approximately 1.80270×10^{-31}. One affine profile has four DAG edges and four packed N5 rows in one child batch; 120 master rows use four families and the parent has no scalar masters.	Scoped pass
Actual source tests	56 exact focused tests, a preparation-only guard and the core/CLI/public-example/PyO3 feature build pass. All 145 selected build-source copies are retained. PyO3 compile checking is separate from the installed wheel runtime cohort.	Pass
Generalized requests	Both first-dot Float128 and D6 unit-power Float128 reach their fixed 810-second preparation deadlines without tensor arrays. Their Float256 partners remain unrun; cleanup and unchanged-source guards pass.	Open
Method scope	The outer request is IBP; child execution is explicitly Passarino–Veltman completion. Raw method tags, preparation notes and diagnostics are retained. This is no pure-IBP or complete generalized-domain qualification.	Limited

The structure supports denominator descent without an imposed small point-count limit. Its actual validation envelope is narrower: this fixed ordinary N6 family does not establish arbitrary-N numerical behavior. Resource budgets, singular identity charts and exceptional kinematics still require explicit diagnostics and qualification. The requested rank remains independent of the number of denominators during descent.

The capsule retains three initial test-only Symbolica replacement-API compile errors, the successful corrected build, the listing-only test failure and its single module-path correction. No failed integral is relabeled successful. Both generalized timeouts retain their original terminal receipts and empty outputs; portable replay succeeds only when those failures remain intact.

Independent saved arithmetic checks all arrays and precision pairs. Native products are represented by their recorded omitted identities; the checker does not rehash absent executables or probe current processes. The original quad reference has 113-bit storage with its original approximately 16-digit constants and stopping criteria. Agreement at 2^{-80} between TIRust outputs is not a certified achieved-accuracy bound. No Dual, physical or timing claim is added by this capsule.

Source: `validation/affine-power-generalized-current-core`. Complete generic rank-six positive-power/dimension-shift closure remains open.

22 Current CLI and physical graph integration

The current source-bound CLI passes fourteen tests. Complete rank-four tensor and contracted physical checks pass for one top-loop $gg \to HH$ box and one $gg \to HHH$ pentagon. Both graphs have seventy Cartesian moments, with all four native Laurent slots retained before contraction. These are fixed-graph spot checks, separate from the rank-six synthetic families.

Executed check	Complete saved evidence	Outcome
Current CLI	14 actual tests pass from the wheel003 build context. The capsule binds the exact CLI/test products, inherited 147 build sources and twenty additional physical input/helper/controller files.	Pass
Box N4 / rank4	Float128/256: 1680 original-IREGI tensor comparisons against IBP, PV and Hybrid; maximum normalized error 1.144×10^{-13}. Largest MadLoop merged-coefficient error 4.766×10^{-15}.	Pass
Pentagon N5 / rank4	Float256/384: 1120 original-IREGI tensor comparisons against PV and Hybrid; maximum error 4.616×10^{-14}. Largest MadLoop merged-coefficient error 9.291×10^{-14}. Historical original-IBP disagreement is not retested here.	Pass
Exact precision pairs	Each graph passes 280 tensor and four contracted Laurent checks, 568 total, using saved decimal strings and exact Fraction arithmetic at 2^{-80}. Maximum normalized L1 differences: box 7.952×10^{-36}; pentagon 6.145×10^{-55}.	Scoped pass
Retained pentagon failure	Float128 / requested53 hits the conservative cancellation guard at `pv_node_852`, with no tensor array or subsequent reference comparison. Prospective Float256/384 runs pass at unchanged requested accuracy.	Failed128

The current CLI requests Hybrid and executes PV in these successful cases. The external comparison threshold remains dimensionally normalized 10^{-10}, using binary64 reference/comparison arithmetic. MadLoop uses isolated graph groups two and five, fixed equal gluon colors, the retained helicity-one inputs and graph factor $-1/2$. Its three Laurent coefficients combine UV and IR single poles; complete original-IREGI tensor checks retain their separation. Bare UV/IR poles and reduction-generated R1 are included; R2 and UV counterterms are excluded. The cohort reuses the documented numerator/routing correspondence.

Precision-pair checks use $Q = 600$ for the box and $Q = 750$ for the pentagon, tensor dimension $4 + r - 2N$, and contracted dimensions zero and minus one. The normalized L1 delta uses the larger of Q^d and higher-precision coordinate magnitudes. Pair agreement is not a certified eighty-bit accuracy bound.

The failed first controller preserves its unrun planned pentagon256 and 128/256 pair stages. The independent higher-precision receipts do not promote its failed128 result. The capsule's saved replay checks all 2800 external tensor slots, 12 MadLoop merged slots and the exact pairs without evaluating an integral or starting a compiler. No new physical numerator export, rank-six physical coverage or performance claim is added.

Source: `validation/current-cli-physical-integration`. The previous timing measurements and their failed cases remain unchanged.

23 Integration scope and remaining functional gates

This is an intermediate implementation checkpoint. The complete generic rank-six objective remains open. The latest source integration adds per-call reuse of repeated primal-amplitude bound expressions using the existing Symbolica operations. Its focused tests qualify that narrow change; they do not turn incomplete dotted or dimension-shifted requests into successes.

Area	Executed scope or required closure	Status
Primal-amplitude memo	Six focused tests pass: alias-environment isolation, repeated conditional reuse, lazy inactive zero denominators, conditional path unions, nonlinear-master rejection and dimensional/Taylor-axis normalization.	Focused pass
Source/build bridge	Both actual 119-source builds are retained. The saved checker recomputes equality of the entire production-module prefix between the builds; the later change is test-only. Core/CLI and the revised test executable compile successfully.	Pass
Positive powers / shifts	N1/N4 first-dot/triple arrays pass; current N5 first-dot and N6 first-dot/D6 preparations remain failed in their frozen cohorts. Their unavailable arrays and unrun precision partners remain visible.	Open
Agreed numeric / API	Public highpoint higher/mixed Duals, generalized checked supplied products, accepted Legacy runtime closure and remaining exceptional/ method coverage still require actual numerical qualification.	Open
Completion and speed	Commit the feature-full tested version with an updated report when the functional gates pass. Higher-rank optimization and low-rank COLLIER parity are later work; no new performance result is added here.	Pending

The initial memo stress fixture used forty nested conditional levels and timed out before completing. It remains failed. A smaller representative ten-level fixture explicitly records construction and cached traversal, with 21 unique amplitude recipes; all six focused tests pass. This replacement does not establish that the original forty-level construction now succeeds or supply a before/after runtime comparison.

The memo source follows the earlier complete tensor and physical capture contexts. Source binding is per capsule: the focused build/test bridge does not automatically confer their full numerical qualification on later edits. The capsule claims no complete dotted array, current-CLI equivalence, physical validation, achieved-precision bound or performance improvement.

Six new immutable cohorts are summarized before the historical sections. All 395 preceding top-level result values and their original JSON spelling are preserved, including unsuccessful cases and every historical timing. Saved-only replay reconstructs the declared evidence and existing numerical checks; it executes no compiler, native library, integral or benchmark.

Latest narrow source evidence: `validation/primal-amplitude-memo-current-core`. Current completion scope: `docs/data/rank6-current-checkpoint-20261026.json`.

24 Rank-six feature validation: scoped progress

The completion target is generic cumulative tensor rank≤6, including positive propagator powers, dimension shifts and the agreed numeric/API features. N≤6 is the current validation checkpoint; higher-point denominator descent needs separate qualification. This checkpoint adds evidence from ten frozen source cohorts. It does not establish their equivalence to later source changes or close the complete mathematical acceptance gate. All prior result values and timings remain unchanged; no performance measurements were added.

Evidence	Complete saved checks	Scope
N2 first/mixed jets	One massive spacelike rank-seven family: 330 moments and all four native Laurent slots. Static/dynamic Float128/256 jets, ordinary f64/ DoubleFloat controls, and an independent Float384 finite-difference route.	Scoped pass
Derivative identities	79200 original comparisons; 36960 precision pairs; 23760 mass identities; 29040 scale identities; 23760 finite-difference components.	Pass
Beyond 53 bits	Exact $1 + 2^{-54}$ inputs: 7920 secant slopes and 2640 raw precision pairs. DoubleFloat/Float retain the distinction; binary64 rejects the unrepresentable input before planning.	Pass
Low-N dotted anchors	Six original-quad N1/N4/N5 rank-six first-dot/triple arrays: 5040 complex slots and 5321 reference/control records.	Anchors
Remaining derivative scope	Typed f64/DoubleFloat Dual carriers are not covered by the N2 cohort. Full public highpoint Dual and broader tensor derivative domains remain unqualified.	Open

The N2 source capsule records exact rational inputs and outputs, component precision, signs, invalid lanes and the original failed build before its narrow borrow correction. The finite-difference route uses Symbolica-generated jets, points and weights. Initial comparison thresholds remain 10^{-10}, precision/DD controls 2^{-80} and the final finite-difference target 2^{-70}. This one-family historical rank-seven result supplies useful rank-six controls; it is not a blanket qualification of all N≤6 families.

The low-N quad arrays are reference anchors. Storage at 113 bits / 33 decimal digits does not establish 33-digit accuracy. Finite generalized coefficients are not independently certified merely because a reference returned a stable flag, and these six arrays alone do not qualify TIRust dotted reduction.

Saved sources: `validation/rank7-numeric-jets`, `validation/quad-original-iregi-low-n-rank6-dot-triple`. Their portable checkers reconstruct declared source snapshots and replay saved arithmetic; they execute no compiler or integral. Timings in their receipts are engineering diagnostics.

25 Six-point runtime: Hybrid and precision rescue

At one real-mass N6 rank-six unit-power point, all 210 cumulative Cartesian moments and four native Laurent slots are available. The newer Hybrid and specialized precision-rescue cohorts close named ordinary runtime paths. Earlier generic, dotted and dimension-shift failures remain visible in their original frozen context.

Path	Executed result	Outcome
Hybrid primary	Float128 / requested53: 840 complex coefficients pass admitted original quad IBP; all 840 equal explicit specialized PV exactly. Maximum normalized reference error 9.3341×10^{-17}.	Pass
Hybrid orchestration	13 focused tests: batched Unstable-only retries, coupled component/ context scatter, lane isolation, counters, lazy shared plans and actual low-N IBP work. Invalid-mu and empty/reset lifecycle assertions pass in the N6 test.	Pass
Public f64 rescue	Specialized SixPoint: output53, working128, verification256. Primary cancellation remains recorded; accepted output passes 840 quad comparisons and 630 separate zero-pole checks. Maximum reference error 1.8670×10^{-16}.	Scoped pass
Earlier restricted CLI	One explicit-PV DoubleFloat array passes 840 references. Five other attempts fail or remain incomplete: generic f64 unit, generic rescue, specialized f64, first-dot Float128 preparation and unit D6 Float128.	1 / 6 pass

Hybrid preserves the requested method and uses specialized PV as primary. On this point its outer generic IBP fallback has **zero** rows, batches and preparation attempts. Existing nested PV child retries remain visible. The focused tests verify fallback mechanics, including real low-N IBP execution; they do not prove a successful genuine N6 IBP fallback on this fixture.

The first N6 Hybrid invocation passed coefficient comparisons but failed an incorrect observer assertion about nested flags. Its failure is retained. The corrected native test passed; its controller then failed to parse a payload sharing libtest's test-name line. Additive saved-only extraction recovered the unique complete raw payload without another integral evaluation. Both original failed receipts remain failed. Independent replay checks the extracted arrays.

The precision-rescue CLI exports the final f64 array, not the internal 128/256 arrays. Its internal convergence acceptance is supported by the executed shared Rust policy and stage counters; it is not independently reconstructed from absent arrays. Converted scalar sensitivity context is explicitly unavailable rather than replaced with stale data. Neither cohort qualifies public highpoint Dual, dots, D6, complex/exceptional cases or speed.

Saved sources: `validation/six-point-hybrid-rank6`, `validation/public-six-point-precision-rescue-rank6`, `validation/n6-rank6-generalized-current-cli`. Historical process absence is replayed as saved metadata; portable checks do not recreate PID history.

26 Thresholds, complex masses and retained collinear failure

The orthogonal CLI corpus is **partial**: 12 of 14 Float128/256 calls return complete rank-six arrays. The two collinear zero-Gram N3 reductions fail with nonfinite coefficients in both PV and IBP fallback. Their 1,680 unavailable complex slots are not reported as successful zeros.

Domain	Saved outcome	Status
Threshold / complex	Three equal-mass N2 threshold positions and lower-half-plane complex squared masses for N2/N3/N4, each at Float128/256. Unit powers; all 210 moments and four native slots.	12 pass
Independent checks	11760 references; 5040 precision pairs; 1680 duplicates; 5880 IR and 80 UV controls; 4 scalar branch anchors. Total 24444.	Pass
Collinear oracle	A separate order48 parameter reference admits 210 rank-six moments after 15150 independent record checks. Maximum empirical convergence indicator 4.1716×10^{-14}.	Anchor only
Earlier collinear TIRust	Two complete reduction failures retained, no tensor arrays in this frozen CLI. The following page records a newer direct-core affine prepass on the same reference family.	Failed
Completion gates	Generalized N6 arrays, highpoint Dual, checked supplied-G / accepted Legacy runtime closure, exceptional reductions and agreed physical/API coverage remain open.	Pending

Maximum normalized reference error among available orthogonal arrays is 2.629×10^{-15}; the precision-pair maximum is 7.167×10^{-36}. Thresholds remain 10^{-10} and 2^{-80}. Below-threshold batches include duplicate valid samples around an invalid-mu neighbor: valid arrays agree exactly and invalid output is null. Requested Hybrid executes actual PV through the Generic engine in this frozen cohort. Its successful rows use no fallback or precision rescue.

The original N4 complex quad result is rejected despite its stable flag and zero finite scalar. N4 instead uses the independently audited order32 parameter oracle. The separate collinear order48 anchor retains six failed coarse convergence decisions and all 330 originally stored moments, while admitting only the 210 rank-six moments. Both parameter-reference qualifications use empirical convergence, not rigorous error enclosures.

The captured CLI predates later supplied-G, Legacy and Hybrid source changes. It records requested precision, without per-component precision or numeric before/after input witnesses. It therefore does not qualify signed-zero/input preservation, threshold derivatives, Duals, broad exceptional coverage or later source revisions. No benchmark timings or physical acceptance are added.

Saved sources: `validation/orthogonal-rank6-current-cli`, `validation/collinear-rank6-parameter-oracle`. Every cohort retains its failures and declared source context. Full N≤6/rank≤6 acceptance remains open.

27 Rank-six numeric types and the collinear core fix

The newer collinear core uses a Symbolica-derived constant-denominator identity before a singular parent Gram inverse or master instruction is evaluated. It retains the original inputs and batches all accepted lanes through the selected child recipe. Its numerical evidence is separate from the earlier failed CLI cohort and the independent parameter oracle.

Executed family	Complete checks and limits	Outcome
N1–N5 numeric	21 complete captures from 24 actual attempts. Ordinary f64/DD controls and specified static/dynamic first/mixed Dual layouts; Float128/256 mixed components include N5.	Scoped
Saved component math	446880 complex checks: 216720 quad, 60480 precision pairs, 55440 DD, 40320 f64 and 73920 scale identities.	Pass
Precision guards	Unrescued N4 f64 mixed Dual and ordinary N5 f64 requested53 / DD requested96 remain guarded failures. The next page qualifies a separate public N4 rescue. No tolerance is relaxed.	3 retained
Collinear N3	Float128/256: all 840 coefficients per precision against the independent anchor; 840 precision pairs. Selected first-dilation Dual: all 840 primals and 840 jets per precision; 1680 pairs.	Pass[1]
Focused core	16 affine, 10 component-sensitivity and 4 checked-facade tests pass. Geometry/Taylor rejection occurs before masters; batch isolation, metadata, demand masks and mixed products are checked.	Scoped
Symbolic stress	Two fully symbolic general-N rank-four preparation tests exceed their fixed 90-second limits. Three initial build/controller failures remain preserved.	Retained

[1] Actual source-bound Rust assertions use the independent order48 anchor and existing Symbolica dilation response, with normalized tolerance 10^{-10} and precision-pair threshold 2^{-80}. Invalid-scale neighbors, equal duplicate outputs, unchanged inputs and one grouped child batch are checked. Numerical arrays were not exported: portable replay verifies the captured source, tests and execution records; it cannot independently recompute absent raw arrays. General Dual uncertainty, arbitrary directions and general-N reduction are not qualified by this one collinear profile.

The separate numeric-type capsule retains one real massive spacelike unit-power, h=0, PV family per N. DD/Float controls keep the 2^{-80} gate. The N5 mixed Float precision-pair maximum is 5.424×10^{-31}. The saved Fraction checker imports no producer. Its author previously wrote the four N2 prefix adapters; that overlap is disclosed rather than presented as author-independent evidence. The other twenty captures have another producer. Five saved mutation probes reject semantic corruptions.

Each cohort reconstructs its own captured sources. The current-core capsule retains 349 compiled source bodies and the later formatting-only transition; the numeric cohort uses its earlier frozen example. These are correctness checks, with no new physical or timing qualification and no full-goal claim.

Sources: `validation/rank6-numeric-types`, `validation/affine-collinear-current-core`. Generalized high-point tensors, public high-point mixed Duals and remaining API domains stay open.

28 Public box rank-six mixed Dual rescue

The public `TensorRescueWorkspace::new_dual` now has an actual binary64 box check with all 210 moments, four Laurent slots and six ordered Taylor components. It evaluates a static input and a dynamic three-lane batch with two valid scales and an invalid-scale neighbor, then checks empty-state reset. Higher-precision evaluation preserves the represented input exactly.

Check	Executed evidence	Outcome
Complete mixed output	Three valid lanes each return all 840 × 6 complex values. Independent saved arithmetic checks 15120 quad responses, 15120 qualified Float128 controls and 5040 scale identities.	35280 pass
Reference accuracy	Maximum normalized quad error 1.673×10^{-16}; Float128-control error 9.572×10^{-17}. Both retain the 10^{-10} gate.	Pass
Real fallback	All three primary lanes hit the original recurrence cancellation guard in Taylor component 2. Working128 and verification256 report convergence; the invalid lane is not rescued.	Pass
Numeric integrity	30240 real/imaginary coordinates are exactly binary64-representable with recorded 53-bit storage. Static/dynamic component ordering, actual counters and empty reset are checked.	Pass
Unexported context	Internal 128/256 arrays and generic detailed stage reports are not exported. Accepted scalar-sensitivity context is explicitly unavailable.	Limited

The independent checker verifies returned values, status/counter associations and the original primary failure. Internal higher-precision convergence is observed through the executed policy and its status; the absent internal arrays cannot be independently compared. Nearest rounding of the hidden verification array likewise cannot be reconstructed from an external Float128 control. The latter's 3326 exact-rounding mismatches are retained as diagnostics, including values near roundoff zeros; normalized agreement is unchanged.

The first saved observer incorrectly required exact nearest rounding from that external control. Its failure is preserved. The corrected observer retains the numerical tolerance, exact binary64 representation, precision, metadata and failure gates. The earlier example-build type error and its one-line harness repair are also retained. Neither correction changes a reduction formula.

This is one massive spacelike, unit-power, h=0 PV family. It qualifies the specified public box rescue, not high-point mixed Duals, arbitrary branches, generalized tensors, rigorous error bounds, physical numerators or performance. Its actual 106-source context and fresh minimal replay are retained separately from the older ordinary binary64 failure and the later unfinished descent code.

Source: `validation/public-four-point-dual-rescue-rank6`. All prior unsuccessful requests remain unsuccessful in their original cohorts.

29 Shared Python facade and native operand precision

Check	Captured scope	Outcome
Installed wheel002	60 tests; consumer typing has 0 errors and 0 warnings. Shared Symbolica, Numerica, FeynKit and Spenso classes use the public tensor facade.	Pass[1]
Native operand precision	Six new upstream regressions plus 19 existing Float/ComplexFloat tests. The installed reproducer retains the high-precision imaginary component when the real component has fewer bits.	Pass[1]
Vendor reproduction	Pinned upstream 9cdc2949 plus six committed patches reproduce all 219 retained files, totaling 9,601,877 bytes. Final observer exclusions leave the captured wheel arithmetic unchanged.	Pass
Earlier failures	The trap wheel and two failed trait-observer stages remain archived. The final stage changes observer exclusions only; numerical assertions, tolerances and the installed wheel remain unchanged.	Retained

[1] One local Linux CPython3.13 development wheel. Its 60 tests comprise 25 upstream numeric tests, 29 existing interoperability/numerical tests, four generic-facade tests, one six-point rank-one frontend test and one seven-point scalar frontend test. They exercise explicit Dual layouts, profiling, engine tags, packed batching, sparse contraction, invalid-lane isolation, empty/reset state, snapshots and recovery. They do not constitute 60 independent physical integral comparisons or qualify the complete high-point numerical domain.

The precision correction concerns native f64/complex operands in upstream ComplexFloat arithmetic. A value with two real bits and 128 imaginary bits retains the latter precision and agrees with the explicit 128-bit operand control. The public `.precision` property still reports the minimum component precision. Arbitrary Python-integer coercion is not qualified here. The source manifests and reproducer flags remain in their original spelling; their captured native-operand scope is not widened.

The source archive reconstructs the 116 inputs owned at wheel-build time. The Rust trait observer's `float.rs` read was outside those original guards; its later copy is retained as post-observation context. The successful observer excludes Rust-only `real_classify` and mutating `set_precision` Python uses immutable `with_precision` copies. This is not a complete compiler/native/system closure or a new wheel build. Other platforms, release performance, separate Community-host loading and broader numerical frontend acceptance remain unqualified.

Saved-only replay: `python3 -I -S -B validation/python-interface/facade-wheel-002/check.py`. Exact six-patch reproduction is bound by `validation/python-interface/complex-native-precision-vendor-reproduction.json`.

30 Complex tensor oracle and generalized N6 references

Reference family	Complete saved evidence	Admission
Complex N2/N3/N4	18 fixed grids, 88256 point rows and 23760 complex Laurent slots. Orders 12/20/32 and both vertex maps are retained.	Scoped
Independent saved audit	43,566 exact comparison records, 11,880 separate IR-zero checks and 15 negative controls pass. Includes 15,840 eligible N2/N3 quad comparisons and 27,720 order/map indicators.	Pass[1]
Original complex N4	Native stable flag is true, but the scalar finite coefficient is exactly zero despite a positive parameter-integral bound. The entire raw array remains diagnostic and enters no agreement gate.	Rejected
Generalized N6 / rank6	7 original-quad arrays / 5880 complex slots: unit powers, first/last dot2, first power3, mixed dots2/2 and unit powers in D2/D6.	Anchors[2]

[1] The ordinary complex fixture uses mu=3, real spacelike shifts and squared masses in the lower half-plane with Re(F)≥1. All 330 moments through rank7 and native UV1, IR1, IR2, finite coefficients are retained; the first 210 moments are the rank6 prefix. Symbolica prepares Gamma Laurent coefficients and source derivatives independently of reduction. Native $i\pi^2$ is included once, and dimensional normalization uses exact powers of Q=3. The actual kernel/transport and unchanged Python controls pass 15 and 17 tests.

Binary64 SciPy nodes and worker values are accumulated with longdouble real and imaginary sums. Maximum N2/N3 reference discrepancy is 7.008×10^{-13}; the largest retained order/map difference is 2.776×10^{-11}. All gates retain the mixed 10^{-10} tolerance. These are empirical convergence indicators, not rigorous error bounds or arbitrary-precision integrated accuracy. The order32 identity-map N4 scalar finite candidate is approximately −0.04470448539252041802 + 0.20873403285427626204i. Independent saved review admits this fixed empirical reference; it does not itself qualify agreement with TIRust. The failed initial E0599 build and bare-environment NumPy import attempt remain failed.

[2] The seven generalized reference arrays contain all 210 rank6 moments. Independent replay checks 5,166 records: 2,940 massive IR zeros, 1,386 convergent UV checks and 840 exact ordinary-unit prefix matches. All 84 degree-six D6 UV coefficients remain present. These necessary identities and controls do not independently prove every finite generalized coefficient or qualify TIRust/Dual results. Original limited-digit constants and stopping criteria remain despite quad storage. No validation duration becomes a comparative timing measurement.

Saved-only replay: `python3 validation/complex-parameter-tensor-oracle/check.py` `python3 validation/quad-original-iregi-generalized-n6-rank6/check.py`. The generic rank-six functional gate remains open. N≤6 is the present validation checkpoint; it does not establish arbitrary-N denominator descent. Broader complex, threshold, collinear and tensor-Dual cases need their own durable comparisons. Supplied-G, Legacy and new Hybrid source work is not qualified by this checkpoint. No new performance ratios are added.

31 Rank-seven correctness checkpoint

Current objective: generic rank-six tensors, with all agreed features including dots. N≤6 is the current validation checkpoint, not a proof of arbitrary-N denominator descent. Retain earlier rank-seven checks and the seven-point challenge. Further optimization is deferred. The functional gate remains open; historical timings retain their measured contexts.

Domain	Executed scope	Outcome
N1–N5 / rank7	Actual Cargo PV: three massive samples per topology, Float128/256, 39,600 complex reference coefficients and 19,800 precision pairs.	Pass[1]
Massless N2–N4	Complete rank7 arrays: 7,920 reference comparisons and 3,960 precision pairs. Separate UV, IR and IR² slots retained, including the scaleless bubble.	Pass[1]
N6 / rank7 batches	21,120 tensor and 128 sparse/dense contraction coefficients; empty, mixed failed lanes, reorder/chunks, cloned scratch and malformed-call recovery.	Pass[1]
N6 / ranks4–7	Separately requested plans, Float128/256: 5,888 reference coefficients and 2,944 precision pairs. All previous rank7 prefixes retained.	Pass[1]
N7 / rank0	Actual Cargo Float128/256 singleton agreement. Nine-call lifecycle stopped after six completed calls; later scratch/recovery checks remain open.	Partial
N6 / rank7 COLI	Native accuracy flag −2; 315 missing diagnostic coefficients and one finite discrepancy. All three pole probes return failure sentinels.	Rejected
Functional objective	Generic rank≤6: generalized tensors, numeric/Dual, supplied invariants and higher-N denominator descent still need qualification. Current tested checkpoint is N≤6; N7/rank7 is separate stress evidence.	Open

[1] Retained massive and massless families, native raw tensor components, requested accuracy 53 and working precision 128/256. Every moment includes all four native Laurent slots. Independent-reference gates remain 10^{-10}; precision and scaling gates remain 2^{-80}. Maximum reference discrepancies are 1.839×10^{-16} for massive N1–N5, 2.522×10^{-16} for massless N2–N4 and 1.265×10^{-16} at N6. Working precision does not assert full working-precision accuracy. These cohorts retain their exact captured source contexts, including the older N6 lifecycle context before the Cayley change. No kernel timing claim follows.

31.1 Complete arrays, batches and preparation

The N1–N5 corpus contains 30 successful sample lanes and passes 13,200 uniform scaling checks, 19,800 exact IR-zero checks and 996 convergent UV-zero checks. Original IBP supplies the reference arrays. Three original bubble PV requests produce no result; six stable triangle/box PV arrays disagree in 486 finite positions. Those failures remain visible and are never substituted as accuracy anchors.

The N6 rank-seven batch/contraction cohort passes at both precisions with 16 successful and eight explicitly rejected lanes. Sparse odd/mixed support and dense coefficients covering all 330 moments use the existing Symbolica contraction evaluator. The maximum contracted-reference discrepancy is 5.090×10^{-18}. All actual planar-child fallbacks are retained. Cloned scratch was exercised serially; concurrent execution is a separate gate.

The adopted routing and eligible five-point Cayley preparation uses Symbolica rational-polynomial matrices and exact identity certificates. Other syntax retains AtomField. Physical singularities preserve exceptional recurrence; conversion/certificate errors remain preparation failures. All 51 focused IBP tests pass. The subsequent literal N6 rank4–7 checks cover complete arrays in this source context. No local algebra solver or numerical specialization was introduced.

Typed-Dual malformed-input failures now clear earlier counters, profiling times and diagnostic messages. Two public regression tests pass, including a genuine mass derivative and valid recovery. Initial test-driver failures remain saved. This diagnostic fix does not change numerical formulas or qualify high-point Dual inputs.

31.2 Reference accuracy adjudicated

Original IREGI and its bundled OneLOop were built with uniform quad arithmetic, retaining the original complex-matrix sources, formulas, thresholds and limited-digit constants. Runtime type canaries and four scalar normalization controls pass. Stable IBP agrees with both complete TIRust arrays. The older binary64 IBP comparison still fails 90 slots; those failures remain preserved. Quad PV is unstable and disagrees in 139 positions. Quad Hybrid reports stability but disagrees in 19, and remains unqualified. Stored 113-bit arithmetic does not establish achieved 113-bit result accuracy.

The rejected COLI capture is not an independent accuracy witness. Its zero explicit Laurent-pole residual count does not turn its failed pole-parameter probes into passing checks. No high-rank kernel timings are inferred from validation-stage durations.

Saved-evidence replay, without new integral evaluation: `python3 validation/public-rank7-current-cargo-n1-n5/check.py`
`python3 validation/highpoint-six-point-rank7-lifecycle/check.py`
`python3 validation/highpoint-six-point-literal-ranks4-7-current-cargo/check.py`
`python3 validation/quad-original-iregi-small-n-rank7/check.py`. The separate rejection record is `validation/collier-n6-rank7-diagnostic/qualification.json`.

31.3 Massless poles and shared execution

The three massless points separately exercise a lightlike scaleless bubble, an on-shell triangle and a scattering box. Original quad IBP supplies all three reference arrays. All six Float128/256 arrays pass; the largest normalized reference discrepancy is 2.522×10^{-16}. The largest precision-pair discrepancy is 5.891×10^{-37}. All six actual routes are analytic PV, with no fallback. In particular, the bubble's nonzero UV and IR single poles are retained separately even though their sum vanishes. No positive-mass IR-zero rule or zero projection is applied to these data.

This cohort uses the same frozen actual-Cargo artifact as the earlier N1–N5 arrays. It does not qualify later source edits or the whole massless domain. The independent reviewer reconstructs all 11,884 comparison records. Per-case terminal records pass; controller phase order and maximum concurrency are executor-attested because absolute UTC start/finish records were not saved. Elapsed durations are diagnostics, not performance measurements.

The additive `TensorReductionPlan` / `TensorWorkspace` facade shares eligibility and contraction with the existing generic, six-point and seven-point engines. Six ordinary tests and a separately selected actual Float128 scalar delegation test pass. The latter checks N6/N7 direct-wrapper equality, invalid-lane isolation, contractions, nested execution counters and empty-call reset. It introduces no recurrence or scalar formula. Explicit Dual layouts, profiling and nondefault policies retain generic execution; requested algorithms and compiled scalar-master defaults are preserved. These core tests alone do not qualify new CLI or Python wheel builds.

The earlier separate rank-seven IBP captures complete at N1/N3 and pass 15,840 reference comparisons, 7,920 precision pairs and 5,280 scaling identities. Both N2 captures time out during workspace preparation and produce no coefficient array. The corpus remains incomplete. Original cleanup failures and later process-absence observations are retained separately. No timing gain or complete-method qualification follows from the available results.

The private seven-point probe completes ranks one and two at one regular positive-mass point, passing 160 complex reference comparisons and 80 precision pairs. Its earlier rank-seven attempts produce no arrays: Float128 reaches its preparation deadline, and Float256 is deliberately stopped before its deadline after the same preparation/resource problem. Both original cleanup failures remain recorded; later process-absence observations are additive. These are private positive-rank witnesses and failed rank-seven attempts, not public rank-seven qualification.

A targeted Symbolica compiler candidate reuses an exact SSA condition's zero/nonzero fact inside its dominating branch. The numeric condition value is preserved, and branch facts are restored before siblings and function scopes. Seven new and eleven relevant existing tests pass, including mixed higher Duals and instruction-major batching. The initial optimized test build times out; a test-only opt0/debug0 build completes with the same sources and features. These engineering records establish focused compiler semantics; the subsequent numerical N2 pass and N7 resource failure are reported below.

Additional saved-evidence replay: `python3 validation/public-rank7-massless-ir/check.py`
`python3 validation/shared-tensor-facade/check.py`
`python3 validation/public-rank7-ibp-owned-artifact-n1-n3/check.py`.

32 Shared CLI and additional rank-seven IBP

Check	Executed scope	Outcome
N2 / rank7 / IBP	Three massive rows at Float128/256: 7,920 reference coefficients, 3,960 precision pairs, 2,640 scaling checks and 3,960 IR-zero checks. Actual method IBP; no fallback.	Pass
CLI / low points	Thirteen exact CLI tests and four smoke calls: full bubble reference arrays, DOT numerator preparation/contraction, DoubleFloat, precision rescue and invalid-neighbor isolation.	Pass
CLI / N6 / rank3	All 140 complex Laurent coefficients agree with the retained quad reference prefix. The shared facade selects the six-point engine and reports nested counters.	Pass
CLI / N7 / rank0	All four coefficients agree with the retained quad reference. Twenty-one packed children, 105 master requests, contraction and invalid-neighbor handling are checked.	Pass

The new N2 comparison follows the scoped Symbolica condition-fact change. Its largest normalized reference discrepancy is 1.839×10^{-16}; the largest precision discrepancy is 7.579×10^{-37}. Scaling and IR residuals are exactly zero in the captured representation. The two old N2 timeouts remain failed historical attempts. The new complete arrays close this particular retained-family gap, without qualifying the whole IBP domain.

The CLI now uses the library's shared tensor plan, workspace and contraction helper. Graph import and Symbolica numerator preparation remain separate. Default Hybrid, precision rescue and scalar-master policy are preserved. Output exposes the selected engine, nested counters and per-lane failures; unavailable rescue observations are null. Explicit Dual layouts continue through the generic engine. No high-point Dual domain is implied.

The first CLI test run fails a test-only diagnostic-wording assertion; the one-assertion correction passes all thirteen tests. The first N7 scalar smoke times out before a result. A separately bounded attempt with the same executable, inputs and accuracy gates passes on a free core. Both original failures remain visible. Actual build and validation durations are engineering records, not comparable kernel timings. No new performance ratios are added.

The compact CLI evidence inherits the exact 77-file source snapshot of the N2 capture and retains its corrected test source separately. These successful artifacts precede the later component-Taylor and preparation-instrumentation changes. They do not silently qualify those later sources.

Saved-only reproduction: `python3 validation/public-rank7-ibp-n2-postfix/check.py` `python3 validation/cli-shared-facade/check.py`.

33 Generalized reference anchors and the N7 blocker

Domain	Saved evidence	Outcome
Original IREGI / generalized rank7	Twelve complete N2/N3 arrays: ten new first/last-dot, first-power-three and D2/D6 anchors, plus two unit-power controls. 15,840 complex slots retained.	Anchors
TIRust / N7 / rank7	Actual current-Cargo private Float128 attempt fails allocation under a 32 GiB address-space cap during workspace preparation. Zero output coefficients. Float256 and pair gates unrun.	Failed
Public N7 tensor support	The public guard remains rank zero. Private rank-one/two evidence and the new resource failure do not authorize positive-rank promotion.	Open

The generalized reference build applies only the already reviewed factorial-pair binomial-recurrence repair to its generated copy. The original source and binary remain untouched. The two unit-power controls match the earlier ordinary quad array and each other exactly. Independent checks cover 22,321 saved records, including UV power counting, positive-mass IR zeros and reflection identities. These are reference anchors for subsequent Rust comparisons. They are not a proof of every generalized finite coefficient or a new Rust/Dual qualification. Uniform quad storage retains the original limited-digit constants and stopping criteria; achieved 33-digit accuracy is not asserted.

The new N7 failure uses a freshly built main-source test executable and a 333-file source snapshot. Its Float128 child aborts after about 612 seconds; the timeout flag is false and owned cleanup succeeds. Its one memory snapshot is not a lifetime peak. A single read-only debugger sample is denied by system tracing policy and produces no stack profile. The failed attempt's coarse log does not uniquely identify the expensive substage. Later fine instrumentation and any optimization are separate work, with no passing array claimed here.

This preserves the distinction between an unsupported public domain, a failed private resource attempt and a numerical disagreement. All remain visible in the acceptance envelope. No preparation duration is interpreted as a speedup against original IREGI or COLLIER.

Saved evidence: `python3 validation/quad-original-iregi-generalized-rank7/check.py validation/highpoint-seven-point-main-rank7-resource-failure/qualification.json`.

34 Independently validated planar generalized scalars

Check	Executed scope	Outcome
N5 / planar scalars	D6 unit powers, plus one positive dot at each of the five positions in D4 and D6. Eleven requests, Float128/256 and retained D4 derivative controls.	Pass[1]
Full saved cohort	196 successful lanes, 26 explicit invalid lanes with null output, 880 native Laurent/Taylor components and 2,064 independent decisions.	Pass[1]
Derivatives and routing	Mass Dual versus dotted scalar, ordinary finite differences, common translations, scale changes, repeated samples and input preservation.	Pass[1]
Original quadrature grids	All direct accuracy checks pass, but 24 refinement decisions fail. The original inconclusive result and three failed optional imports remain preserved.	Retained

[1] One massive positive-F planar family: shifts `0, ex, ey, ez, ex+ey+ez`, squared masses `4, 9, 16, 25, 36` and scale energy three. The parameter denominator satisfies $F \geq 4$. Symbolica prepares the polynomial, Jacobian, input jets and derivatives independently of reduction. The numerical oracle integrates the defining parameter integral, calling no reduction or scalar-master backend.

The public certified-planar path uses its full-Taylor geometry certificate and constant-one denominator relation. These requests reduce to masters with at most four propagators; they do not evaluate a genuine full-rank D6 pentagon. Every dot position is checked in both dimensions. Native UV1, IR1, IR2 and finite slots remain separate.

Initial orders 12/16/20 give 2,040 passing and 24 failing decisions. The separately bounded oracle-only refinement uses orders 24/28/32, retaining all thirty Rust captures. It passes every decision with unchanged thresholds: oracle differences at most 1.065×10^{-13} then 5.398×10^{-16}, below 10^{-12}; reducer/oracle discrepancy at most 2.749×10^{-18}, below 10^{-10}. Precision-pair discrepancy is at most 7.854×10^{-32}, below 2^{-80}. Mass-Dual/dot discrepancy is 2.943×10^{-32} and finite-difference/dot discrepancy 5.906×10^{-21}.

An independent saved-data audit reconstructs every exact squared error, normalization, threshold and decision, with eight negative controls. Its portable replay preserves both the original failures and refined success. Earlier optional-import failures evaluate no quadrature; a pinned environment continuation is recorded separately. No Rust evaluation is repeated to obtain the refined oracle agreement.

This closes the named scalar slice. Generalized tensors, other endpoints, lower-span charts, higher Dual orders and genuine D6 pentagons require their own coverage. Numerical convergence estimates are not rigorous enclosures. No validation duration is interpreted as a performance benchmark.

Saved-only replay: `python3 validation/certified-planar-generalized-scalar-qualification/check.py`. The self-contained capsule retains 75 captured source files, raw arrays, all six successful quadrature outputs and both independent audits.

35 Method coverage and the seven-point challenge

Check	Executed scope	Outcome
Additional ordinary methods	N1/N2/N3 Hybrid, N4 IBP/Hybrid and N5 Hybrid: six Float128/256 pairs with complete rank7 arrays, including every rank6 prefix.	Pass[1]
Preparation resource fix	Narrow Symbolica variable discovery and no-leaf activation short-circuit; nineteen focused tests, including 21,204 declared inputs and reordered mixed Taylor components.	Focused
N5 requested IBP	Actual PV completion is reported. Its 128-bit output agrees with the reference, but the strict pure-IBP method gate fails; 256-bit partner unrun.	Partial
N7 / rank7 diagnostic	Instrumented main-source Float128 attempt aborts allocation under 32 GiB during master-activation preparation. No tensor coefficients; no public promotion.	Failed

[1] The six successful pairs pass 15,840 quad-reference comparisons, 7,920 precision pairs, 7,920 IR-zero checks and 15,840 retained-PV controls. Maximum normalized reference discrepancy is 1.839×10^{-16}. These are one-row retained families, not a complete kinematic domain. At N5, an epsilon-dependent IBP dimension divisor prevents completion from the available A0–D0 Laurent orders. The explicitly reported PV completion passes 1,320 reference and 1,320 previous-PV checks. This is a correct-output diagnostic, not pure-IBP qualification. True generalized D6 scalars are separate.

The instrumented seven-point attempt makes the preparation challenge explicit. Its child workspace prepares successfully. The parent spends about 214 seconds in distance conditioning, 197 in output monitors and 107 in denominator monitors, then aborts during master activation. These are measurements of one failed diagnostic, not kernel benchmark comparisons or a complete memory peak.

Source inspection finds two avoidable resource costs. Rational-polynomial conversion supplies 21,204 declared inputs although each distance uses only a few shifts. Existing Symbolica variable discovery constructs the same polynomial using its actual variables; the full evaluator input order, precision and Taylor shape remain unchanged. The exact certificate uses the same approach and retains declared-symbol validation. Separately, a parent with no own scalar-master leaves needs no master-activation graph traversal. Other stability monitors and every path with leaves remain intact.

The actual Cargo build and nineteen focused tests pass, including unknown-symbol rejection, wide late-input/mixed-jet equality, exact certificate contexts, no-leaf output/monitor preservation and existing active-master controls. The initial preflight and new-test API compile failures are retained separately. This establishes the focused preparation fix; it does not turn the failed N7 attempt into a successful array. No further long N7 run is required before closing the present N≤6/rank-six checkpoint. Higher-N denominator descent needs separate checked implementation and finite representative validation.

Seven-point rank-seven completion would still need fresh full arrays at two precisions, an independent accuracy check, batched lifecycle/contractions and the remaining numeric/generalized domains. Public N7 remains scalar-only.

Saved-only replay: `python3 validation/public-rank7-method-completion-n1-n5/check.py`. Additional source and scope: `validation/sparse-polynomial-activation-resource-fix/qualification.json` `validation/highpoint-seven-point-fine-profile-resource-failure/qualification.json`.

36 Additional correctness milestones

These retained private results support the remaining implementation work. They do not expand the qualified public domain by themselves.

Scope	Complete saved checks	Status
Generalized N6/N7	N6 dotted, N7 dotted and N7 h=−1: 240 complex outputs at Float128/256; original agreement, scaling, repeated/chunked batch profiles and 80 exactly retained control values.	Private pass
N6 scalar Dual	19 tests and four captures: 528 parent Taylor slots, 156 packed child rows and 264 precision pairs. Independent first mass derivative/dotted anchor and coherent dilation jets.	Private pass
N7 scalar API	35 focused actual-Cargo tests; singleton Float128/256 checks pass. The partial lifecycle retains 28 moment and 28 unit-contraction slots, including rejected neighbors and reordered/chunked inputs.	Partial

The generalized preparation change uses Symbolica's existing rational-polynomial matrix operations, with exact reconstruction certificates and an AtomField fallback. It also distinguishes a singular matrix from a preparation failure. The retained cohort passes 7,415 independent saved-data checks, covering all 2,060 input slots and all complete outputs. The preparation adapter is now adopted into the main Cargo source and passes the ordinary-array checks above. This older generalized numerical cohort retains its private dependency context; it has not been rerun in the new Cargo context. Higher-rank generalized requests, genuine D6 domains, complex masses and Dual directions remain separate work.

The component-sensitivity prototype transports complete heuristic Taylor sensitivity operands rather than padding derivative entries with zeros. Its mass derivative agrees with independent dotted IBP/original anchors; its coherent scale jets pass homogeneity and precision checks. These are numerical sensitivity estimates, not rigorous derivative uncertainty enclosures. General independent mixed directions, finite differences and public N6/N7 Dual constructors remain unqualified. The rectangular mixed-direction follow-up is source-only.

The N7 lifecycle stopped at its fixed deadline during the independent-scratch call. Six preceding calls are retained; the remaining scratch, later empty and malformed-call recovery checks are not silently treated as passing. The first failed Float256 attempt also remains alongside its separately bounded successful follow-up. Positive-rank seven-point validation is ongoing.

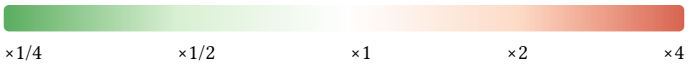
Exact-master preparation-cache experiments are preserved separately: their 38-request whole-output regression passes, while earlier typed schedules remain incomplete. No cache candidate or timing gain is adopted by this checkpoint. No new high-rank benchmark measurements are added here.

Sources and saved-evidence replay: `validation/routing-rational-cayley-full-core/` and `validation/highpoint-dual-sensitivity/`. The actual N7 source/build boundary is recorded in `validation/highpoint-seven-point-current-cargo/qualification.json`.

37 Historical kernel baseline: `4c32ece`

Table 12: **Massive bubble/triangle, binary64, hybrid mode.** TIRust and IREGI total medians in µs/point. **TIRust / IREGI** ratios are paired-block medians; intervals are 99% bootstrap confidence intervals. Batch 1 is latency; batch 32 is amortized cost per point. Each backend uses its own paired reference.

Case	Rank	Interpreter Batch 1	Interpreter Batch 32	Native Batch 1	Native Batch 32
Bubble	0	3.176 µs IREGI 1.988 µs · ×1.598 99% CI 1.585–1.607	1.708 µs IREGI 1.885 µs · ×0.906 99% CI 0.902–0.911	0.936 µs IREGI 0.943 µs · ×0.993 99% CI 0.987–1.003	0.903 µs IREGI 0.878 µs · ×1.031 99% CI 1.024–1.042
Bubble	1	9.552 µs IREGI 5.707 µs · ×1.670 99% CI 1.655–1.688	5.831 µs IREGI 5.748 µs · ×1.015 99% CI 1.008–1.019	4.663 µs IREGI 3.044 µs · ×1.534 99% CI 1.526–1.549	3.231 µs IREGI 3.029 µs · ×1.069 99% CI 1.062–1.081
Triangle	0	29.152 µs IREGI 8.496 µs · ×3.446 99% CI 3.415–3.454	18.229 µs IREGI 8.433 µs · ×2.160 99% CI 2.147–2.172	4.227 µs IREGI 4.968 µs · ×0.849 99% CI 0.844–0.858	8.405 µs IREGI 4.899 µs · ×1.715 99% CI 1.704–1.723
Triangle	1	38.505 µs IREGI 18.297 µs · ×2.101 99% CI 2.088–2.108	25.634 µs IREGI 18.224 µs · ×1.408 99% CI 1.400–1.413	8.809 µs IREGI 10.418 µs · ×0.850 99% CI 0.844–0.854	12.461 µs IREGI 10.213 µs · ×1.220 99% CI 1.206–1.228



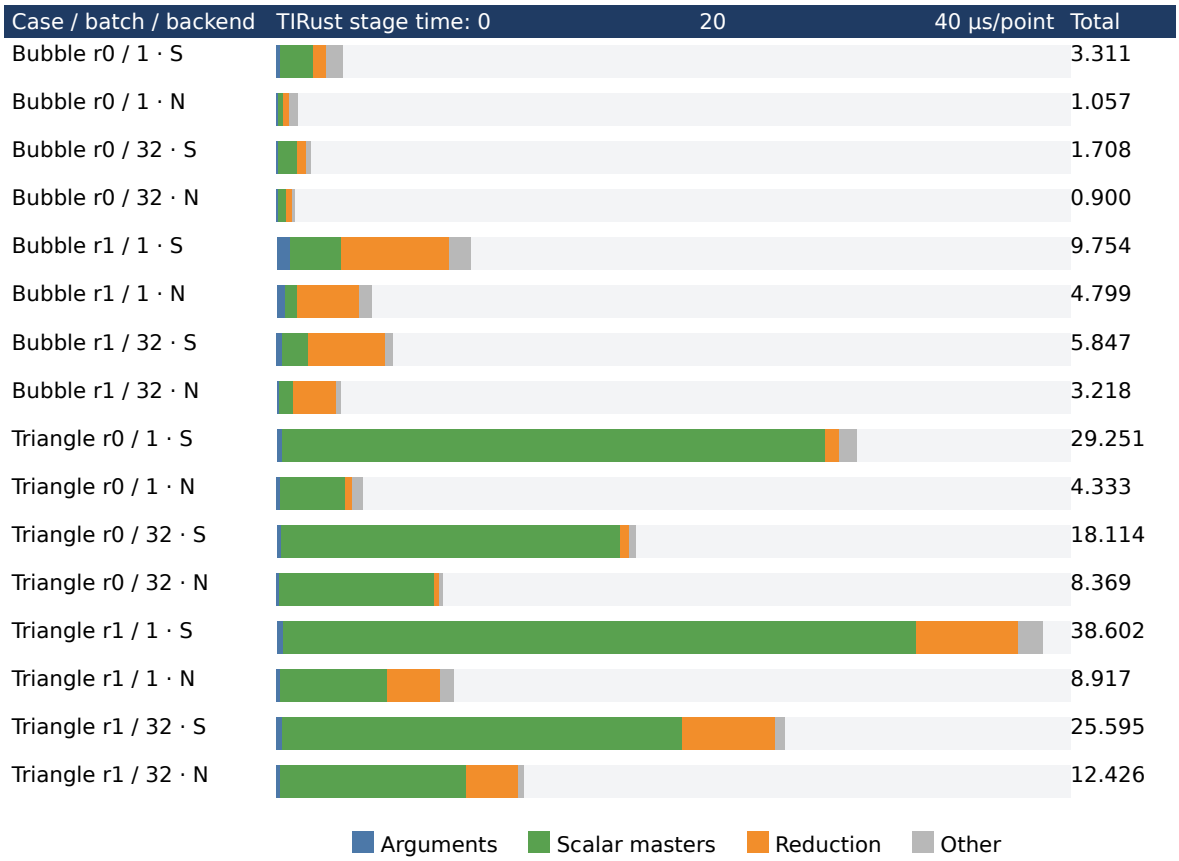
Both backends pass all eight numerical checks at 10^{-10}. The interpreter has seven regressions and one improvement; native has five regressions, one parity result and two improvements. **Absolute IREGI timings changed substantially between sessions**, despite identical inputs/reference binary and CPU settings. Each verdict uses its own paired run; no direct backend speedup is inferred across sessions. Both use the historical `4c32ece` revision; higher-rank performance remains unqualified.

The newer optimized interpreter snapshot `5c9e3dd` passes eight numerical checks, with six timing regressions and two noisy rows. The `80949db` snapshot gives five regressions/three noisy rows. Latest `6b24c03` passes all eight checks: seven regressions/one noisy row. Separate paired comparisons follow; performance acceptance is unmet.

Newer validation includes four integrated graphs. The top pentagon at Float 128/256 agrees with PV/hybrid and MadLoop; original IBP still fails. The latest physical timing comparisons remain inconclusive.

38 Historical stage profiles: 4c32ece

TIRust medians from separate instrumented passes, in µs/point. All bars use 0–40 µs/point; S denotes the Symbolica interpreter, N the generated native backend. Sessions are separate: compare each with its own reference in the preceding baseline table, not with the other backend's absolute bar length. Raw ns/batch are divided by 1000 × batch width.



"Other" is total median minus named stage medians, not a separately timed region. Scalar-master time includes complete master calls; reduction includes stability checks. IREGI has no measured stage split here. For native triangle rank 0, scalar masters cost 3.282 µs at batch 1, but 249.977 µs per batch 32, or 7.812 µs/point: this batch path has higher amortized master cost. The observation identifies a profiling target, not a cross-session speedup or a proven cause. Full stage values and samples remain in the raw data.

38.1 Preparation and measurement conditions

Backend	Bubble r0	Bubble r1	Triangle r0	Triangle r1
Interpreter	1.572 s	0.143 s	0.104 s	0.091 s
Native	0.011 s	0.012 s	0.001 s	0.013 s

Preparation values are single observations without repeated uncertainty, excluded from warm kernel ratios. Batch32 reuses the plan; numerator preparation is outside this tensor-array benchmark.

Both backends of the isolated pushed revision `4c32ece` ran on AMD EPYC 9754 CPU 0, with 5 warm-ups and 21 paired blocks targeting 30 ms, randomized order with seed 271828. TIRust used release opt-level3, thin LTO and one codegen unit; Fortran used `-O3`. Structural plans and allocation capacity were reused; numerical results were recomputed, and original recurrence caches were freed per sample. Subprocess transport is excluded. All source/binary provenance checks pass, and every ratio interval is no wider than 2% of its median. Raw samples, inputs, flags and hashes are retained in `validation/benchmark/baseline-4c32ece/` and `validation/benchmark/native-4c32ece/`.

A separate frozen-native policy experiment keeps `Whole`: all 18 C0/D0 comparisons across three orderings find `FixedRows4/8/16` slower, with per-comparison 99% intervals excluding parity. It uses 32 valid lanes, CPU 0 and 21 rotated blocks. This selects a batch policy, not IREGI speedup; see `validation/native-policy/`.

Native allocator adoption remains rejected. Historical 24 timing rows give two singleton parities/22 noisy rows; new D0 timings include two batch32 regressions. D0 exact old/new parity passes 40 configurations/1,488 complex outputs (f64/DD/Float256); interpreter differences remain. Separate candidate decisions appear in the optimization ledger and `validation/native-stack-batch/`.

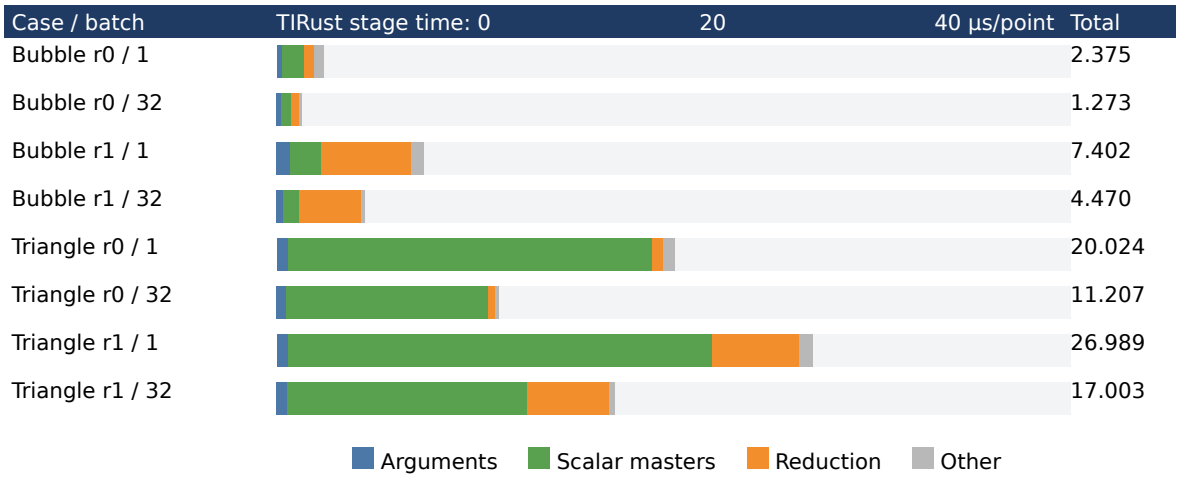
39 Latest kernel snapshot: `6b24c03`

All eight massive bubble/triangle checks pass 10^{-10}, with maximum normalized error 2.46×10^{-13}. Seven timing rows regress; the rank-one triangle singleton exceeds the unchanged 2% interval-width limit (2.357%) and remains **inconclusive**. This measured snapshot predates the standalone planar and finite-D0 certification APIs and later expression preparation. The performance requirement remains unmet.

Table 13: **Binary64 hybrid interpreter, µs/point.** Latency at batch 1 and amortized cost at batch 32. Ratios are **TIRust / IREGI**, using the same-run paired original-IREGI measurement.

Case	Rank	Batch 1	Batch 32
Bubble	0	2.225 µs IREGI 0.948 µs · ×2.344 99% CI 2.335–2.353 Regression	1.265 µs IREGI 0.896 µs · ×1.413 99% CI 1.407–1.416 Regression
Bubble	1	7.245 µs IREGI 3.048 µs · ×2.388 99% CI 2.372–2.403 Regression	4.498 µs IREGI 3.046 µs · ×1.480 99% CI 1.467–1.488 Regression
Triangle	0	19.618 µs IREGI 4.925 µs · ×3.986 99% CI 3.977–3.996 Regression	11.152 µs IREGI 4.963 µs · ×2.246 99% CI 2.237–2.255 Regression
Triangle	1	26.777 µs IREGI 10.356 µs · ×2.586 99% CI 2.570–2.631 Inconclusive: noise	16.967 µs IREGI 10.411 µs · ×1.627 99% CI 1.621–1.636 Regression

39.1 Separately instrumented component timings



Stages use a separate instrumented pass. “Other” is its total median minus named medians; stage sums are not the uninstrumented gate. Graph/numerator preparation is outside this tensor-array benchmark. The single plan-preparation observations remain in the data, not in speed ratios.

CPU 254, five warm-ups and 41 randomized paired blocks targeting 100 ms use identical inputs, complete tensor arrays and all native Laurent slots. Structural allocations are reused; numerical results are recomputed and reference caches cleared per sample. Transport is excluded. rustc 1.99 uses opt-level3, thin LTO and one codegen unit; GFortran 15.2 uses `-O3`.

The complete optimized workspace build exits successfully. All 414 immutable build-source hashes, both binaries, feature selection and harness match the retained provenance. Other project computations were held. CPU 254 was selected by a read-only load survey because unrelated work occupied CPU 0; external host activity remained uncontrolled. No system settings were changed. Historical sessions used different CPUs and timing protocols; only each session’s paired reference supports its verdict, with no causal revision speedup. Raw samples and reproduction: `validation/benchmark/baseline-6b24c03-licensed/`.

40 Direct TIRust / IREGI / COLLIER comparison

Private optimized prototypes compare the complete reduction call on the same massive Euclidean N2/N3 rank0/1 inputs. **B** is baseline TIRust with Symbolica masters; **S** is the same TIRust prototype selecting native A0/B0/C0 for plain f64. IREGI uses Hybrid and COLLIER uses COLI mode 1. Every ratio below has TIRust in the numerator; all forty results remain visible.

Table 14: **Same-cohort paired latency ratios and 99% CIs.** B and S denote TIRust variants, not separate reference libraries. Below one favors the numerator. Amber preserves the short-block family.

Family / batch	TIRust B / IREGI	TIRust S / IREGI	TIRust B / COLLIER	TIRust S / COLLIER	TIRust S / TIRust B
Bubble r0 batch 1	2.381 2.374–2.386 Short block	1.550 1.546–1.554 Short block	2.949 2.937–2.962 Short block	1.921 1.914–1.929 Short block	0.651 0.650–0.652 Short block
Bubble r0 batch 32	1.539 1.535–1.542 Slower	1.383 1.379–1.385 Slower	1.777 1.769–1.782 Slower	1.594 1.587–1.602 Slower	0.899 0.897–0.900 Gain
Bubble r1 batch 1	2.347 2.332–2.355 Slower	2.023 2.013–2.034 Slower	6.100 6.078–6.132 Slower	5.274 5.258–5.289 Slower	0.864 0.861–0.869 Gain
Bubble r1 batch 32	1.496 1.487–1.503 Slower	1.459 1.452–1.466 Slower	3.920 3.911–3.927 Slower	3.818 3.812–3.823 Slower	0.974 0.972–0.977 Gain
Triangle r0 batch 1	4.071 4.062–4.076 Slower	0.985 0.983–0.987 Gain	5.311 5.301–5.316 Slower	1.286 1.284–1.288 Slower	0.242 0.242–0.243 Gain
Triangle r0 batch 32	2.402 2.394–2.407 Slower	1.870 1.867–1.875 Slower	3.103 3.096–3.109 Slower	2.417 2.413–2.423 Slower	0.779 0.778–0.781 Gain
Triangle r1 batch 1	2.651 2.644–2.655 Slower	1.057 1.054–1.059 Slower	4.484 4.480–4.494 Slower	1.790 1.787–1.791 Slower	0.399 0.398–0.399 Gain
Triangle r1 batch 32	1.708 1.701–1.718 Slower	1.414 1.406–1.423 Slower	2.885 2.877–2.898 Slower	2.388 2.384–2.400 Slower	0.828 0.825–0.832 Gain

Selected versus COLLIER: seven regressions, one inconclusive family. Selected versus IREGI has six regressions, one gain (triangle rank 0 singleton) and one inconclusive family. Selected versus baseline has seven gains and one inconclusive family. Baseline/reference comparisons each retain seven regressions and one inconclusive family. No production/default adoption or complete performance acceptance follows.

Explicit posthoc analysis revision2. The original frozen analyzer marked five ratios short-block-inconclusive and 35 cohort-failed because one local short block set the aggregate status. A separately reviewed saved-data correction verifies clean termination and every original global check before assigning family gates. All forty sample arrays, medians, CIs, widths and numerical order-stratum statistics are unchanged; the conservative verdict remains archived. No new timing, threshold change, omitted family or repeated cohort.

One COLLIER block in bubble rank 0/batch 1 lasts 99.893971 ms, below 100 ms; all five ratios of that family remain inconclusive. Protocol: CPU 254, ten warm calls, 48 quartets per family containing all 24 permutations of the four engines twice, shared fixed iterations, 100 ms blocks, 10,000 paired bootstrap resamples, 99% CIs and 2% width gate. Every complete tensor/Laurent array and native pole probe passes before and after timing. Transport is excluded. This is a source-bound historical prototype, not the current production build. Sources: `validation/selected-native-four-engine/v3-frozen/` and `validation/selected-native-four-engine/posthoc-revision2/`.

41 Archival reference timings: IREGI and COLLIER

Identical massive Euclidean tensor integrals compare original IREGI Hybrid (mode 2) with COLLIER 1.2.10 COLI (mode 1). Each row requests the complete Cartesian tensor array and all four native Laurent slots. This is a reference-library comparison, separate from the TIRust performance gate. No TIRust/COLLIER ratio is inferred from different sessions. The direct four-engine cohort has its own separate source-bound table.

Table 15: **Archival binary64 reference timings, µs/point.** Batch 1 is latency; batch 32 is amortized cost per point. Qualified comparison outcomes are retained without reference/reference ratio displays.

Topology / rank	Batch	IREGI Hybrid	COLLIER COLI	Qualified outcome
Bubble rank 0	1	0.943	0.764	Inconclusive: short block
Bubble rank 0	32	0.927	0.766	COLLIER faster
Bubble rank 1	1	3.129	1.173	COLLIER faster
Bubble rank 1	32	3.122	1.172	COLLIER faster
Triangle rank 0	1	4.994	3.781	COLLIER faster
Triangle rank 0	32	4.963	3.825	COLLIER faster
Triangle rank 1	1	10.752	6.148	COLLIER faster
Triangle rank 1	32	10.864	6.145	COLLIER faster

The 99% interval width must be at most 2% of the paired median, and every block must reach 100 ms. Amber rows remain inconclusive. Raw ns/batch values are divided by 1000 × batch width for this display. Original paired ratios and intervals remain archival metadata in the JSON.

Numerical scope. All eight families pass the unchanged dimensionally normalized 1e-10 criterion; maximum error is 2.49e-13. The bubble rank0 singleton includes a 98.438 ms block below the 100 ms requirement and remains inconclusive; no row was rerun or omitted.

Cache policy. within-sample recurrence memoization in both; no cross-sample/event numeric reuse

COLLIER timing includes invariant setup and native conversion; IREGI includes mass setup and cache disposal. Both include the full tensor call and status checks.

Measurement protocol. CPU 254, five warm-ups and 41 balanced randomized paired blocks; common fixed iteration counts, registered minimum 100 ms. Both builds use GNU Fortran 15.2.0 -O3 without fast-math. Project computations were held; unrelated host activity remained uncontrolled. Transport and regulator probes are outside timing.

Ordinary non-dotted, fixed-dimension cases have zero IR poles; regulator probes are untimed. This archival timing cohort does not qualify higher points/ranks, singular or IR-divergent families or physical numerators. Raw observations and reproduction: `validation/collier-reference/timing-initial8/summary.json`.

42 Extended reference coverage: N2–N7

A separate correctness-only capture extends the ordinary, massive Euclidean reference matrix. Each cell keeps the native COLLIER admission status; numerical output from a rejected call cannot qualify a comparison. This table contains no additional timings and no new TIRust performance claim.

Table 16: **All 34 reference rows, including failures.** COLI-admitted means native status accepted and the complete native tensor/Laurent array agrees with IREGI within 10^{-10}. This table does not establish independent higher-point identities.

Propagators / rank	Batch 1	Batch 32
N2 / rank 0	COLI admitted	COLI admitted
N2 / rank 1	COLI admitted	COLI admitted
N3 / rank 0	COLI admitted	COLI admitted
N3 / rank 1	COLI admitted	COLI admitted
N4 / rank 0	COLI admitted	COLI admitted
N4 / rank 1	Unqualified	Unqualified
N4 / rank 2	Unqualified	Unqualified
N4 / rank 3	Unqualified	Unqualified
N5 / rank 0	COLI admitted	COLI admitted
N5 / rank 1	COLI admitted	COLI admitted
N5 / rank 2	COLI admitted	COLI admitted
N6 / rank 0	COLI admitted	Unqualified
N6 / rank 1	Unqualified	Unqualified
N6 / rank 2	Unqualified	Unqualified
N7 / rank 0	Unqualified	Unqualified
N7 / rank 1	Unqualified	Unqualified
N7 / rank 2	Unqualified	Unqualified

17 COLI-admitted, 17 unqualified rows. The six N5 rank 0–2 rows have maximum normalized disagreement 4.76×10^{-12}; the N6 rank 0 singleton reaches 2.60×10^{-13}. Existing N4 rank 1–3 rejections remain visible. Other higher-point calls retain COLI accuracy flags −1/−2 and DD error flag −10. Both policies and all raw statuses remain in the evidence; library code and rejection criteria are unchanged.

Attempt004 retained 239 validated isolated responses. Attempt005 appends only its eight missing requests, with no repeated capture or timing. The final N7 rank 2/batch 32 case retains unstable original PV, a nominally successful IBP result disagreeing with Hybrid by 0.520 normalized, and rejected COLLIER outputs. The N6 rank 2 mass-deformed IREGI disagreement of 1.97 also remains unresolved.

The independent saved-artifact/lane-gather audit verifies 81,086 assertions, 54,224 admitted native complex pairs and 5,427 rejected null slots. This is distinct from independent mathematical identities. IR-divergent, dotted and shifted-dimension COLLIER coverage is unqualified. Sources: `validation/collier-reference/shared/attempt004/` and `validation/collier-reference/shared/attempt005/`.

43 Independent checks and retained failures

43.1 Massive N5 reference identities

Three retained positive-F massive, undotted N5 branches at ranks 0–2 pass independent prefix, pinch/denominator/trace and cyclic/reversed-routing checks. Batch 32 cycles those same three branches; it does not add 32 distinct kinematic configurations. These synthetic checks use the reference codes, separately from TIRust reduction, physical numerators and MadLoop.

Independent check	Residual records	Complex slots	Maximum normalized error
Rank prefixes	924	3696	0
Pinch / denominator / trace	756	3024	3.135×10^{-12}
Cyclic / reversed routing	360	1440	4.101×10^{-12}
Finiteness / pole zeros	3357	10071	0

All 5397 residual records / 18,231 complex comparisons pass the unchanged 10^{-10} gate. The saved-data audit independently verifies 511,509 assertions and 346 archive members, including 465 admitted native lanes and 13,428 unique native complex pairs. The earlier capture's audit-pending status is immutable; the separate completed audit supplement supplies this result. No higher-rank, singular, dotted, dimension-shift, physical or timing qualification follows. Source: `validation/collier-reference/shared/identities006/correctness001/audit001/`.

43.2 Private N7: exact tests only

A frozen private N7 source build passes 17 exact tests. Its ignored generic variant-constructor test remains unrun. Source/build/test receipt review is distinct from numerical integral evaluation: no N7 full-array agreement, precision convergence, public adoption or performance is established. Source: `validation/highpoint-thin-n7-exact/`.

43.3 Private N6 rank 7: failed Float128 comparison

The retained Float128/requested-53 singleton returns 330 moments / 1320 native complex Laurent slots, including 990 exact zero pole slots. Its unchanged original-IBP comparison fails 90 finite slots: 12 at rank 6 and 78 at rank 7. Maximum normalized error is 2.539×10^{-9} against 10^{-10}. A completed process and reported stable status do not qualify the complete array.

For this real positive-F point, the independent analytic real-zero identity has no TIRust violations; original IBP and Hybrid each violate it in 88 slots. This explains 88 of the failed original-IBP complex comparisons from the reference real residual alone. It does not certify the remaining imaginary values or overturn the failed complete-array gate. Original PV stays explicitly unstable. Two saved quadrature witnesses are limited diagnostics, without rigorous error bounds or a causal cache-bug conclusion.

A separate Float256 diagnostic saves the same 1320 slots and 990 exact zero poles. Across Float128/256, the maximum observed normalized difference is 5.421×10^{-32}, below 2^{-80}. Original IBP, PV and Hybrid still disagree in 90, 151 and 109 slots respectively; unstable PV remains unqualified. This observed refinement is diagnostic: complete-array accuracy and precision-convergence acceptance remain false, with no high-rank adoption, lifecycle or timing result. The immutable saved-data packet and its initial missing-file failure are preserved at `validation/checkpoint-audits/2026-10-24-thin-n6-rank7-128/`. The separate terminal capture and independent audit are at `validation/highpoint-thin-n6-rank7-phase/`.

One-component quadrature diagnostic. For moment 126, q_0^6, six fixed-rule calculations (orders 12/16/20, two maps) give $(0, -0.05166894942216114)$. The estimated normalized uncertainty is 3.113×10^{-15} and is nonrigorous. Discrepancy plus that estimated margin is 3.249×10^{-15} for TIRust Float128, within the 10^{-10} diagnostic gate; original IBP and Hybrid give 4.274×10^{-10} and 4.267×10^{-10}, outside it. PV remains natively rejected. This supports one finite component; complete-array acceptance stays failed. Source: `validation/temporal-sixpoint-oracle/q0-sixth-001/qualification.json`.

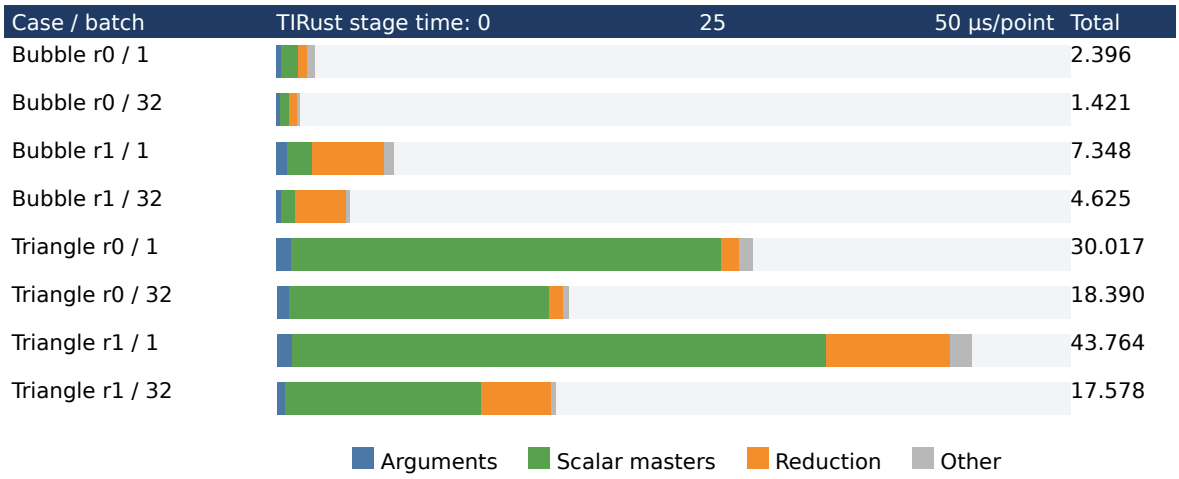
44 Historical kernel snapshot: `80949db`

All eight massive bubble/triangle checks pass the unchanged 10^{-10} criterion, maximum normalized error 2.46×10^{-13}. Five timing rows regress; three exceed the unchanged 2% interval-width limit and remain **inconclusive**. This optimized snapshot includes the adopted master stack allocation, generalized PV and lazy Cayley solve. Later core-allocation and experimental changes are excluded. The performance requirement remains unmet.

Table 17: **Binary64 hybrid interpreter, µs/point.** Latency at batch 1 and amortized cost at batch 32. Ratios are **TIRust / IREGI**, using the same-run paired original-IREGI measurement.

Case	Rank	Batch 1	Batch 32
Bubble	0	2.278 µs IREGI 0.946 µs · ×2.403 99% CI 2.389–2.420 Regression	1.380 µs IREGI 0.892 µs · ×1.550 99% CI 1.537–1.557 Regression
Bubble	1	7.243 µs IREGI 3.083 µs · ×2.342 99% CI 2.318–2.361 Regression	4.619 µs IREGI 3.077 µs · ×1.500 99% CI 1.490–1.518 Regression
Triangle	0	19.553 µs IREGI 4.967 µs · ×3.927 99% CI 3.911–3.944 Regression	17.604 µs IREGI 7.296 µs · ×2.374 99% CI 2.325–2.432 Inconclusive: noise
Triangle	1	43.420 µs IREGI 17.458 µs · ×2.500 99% CI 2.463–2.578 Inconclusive: noise	28.540 µs IREGI 17.009 µs · ×1.672 99% CI 1.490–1.762 Inconclusive: noise

44.1 Separately instrumented component timings



Stages use a separate instrumented pass. “Other” is its total median minus named medians; stage sums are not the uninstrumented gate. Graph/numerator preparation is outside this tensor-array benchmark. The single plan-preparation observations remain in the data, not in speed ratios.

CPU 0, five warm-ups and 21 randomized paired blocks targeting 30 ms use identical inputs, complete tensor arrays and all native Laurent slots. Structural allocations are reused; numerical results are recomputed and reference caches cleared per sample. Transport is excluded. rustc 1.99 uses opt-level3, thin LTO and one codegen unit; GFortran 15.2 uses `-O3`.

All 401 immutable build-source hashes, both binaries and the harness match the retained provenance. Other project computations were held; a pending native compiler was suspended. Host activity outside the project was not controlled. The accepted license contains no tracked key material. This is a separate session from the historical tables; its absolute timings establish no revision speedup. Raw samples and reproduction commands: `validation/benchmark/baseline-80949db-licensed/`.

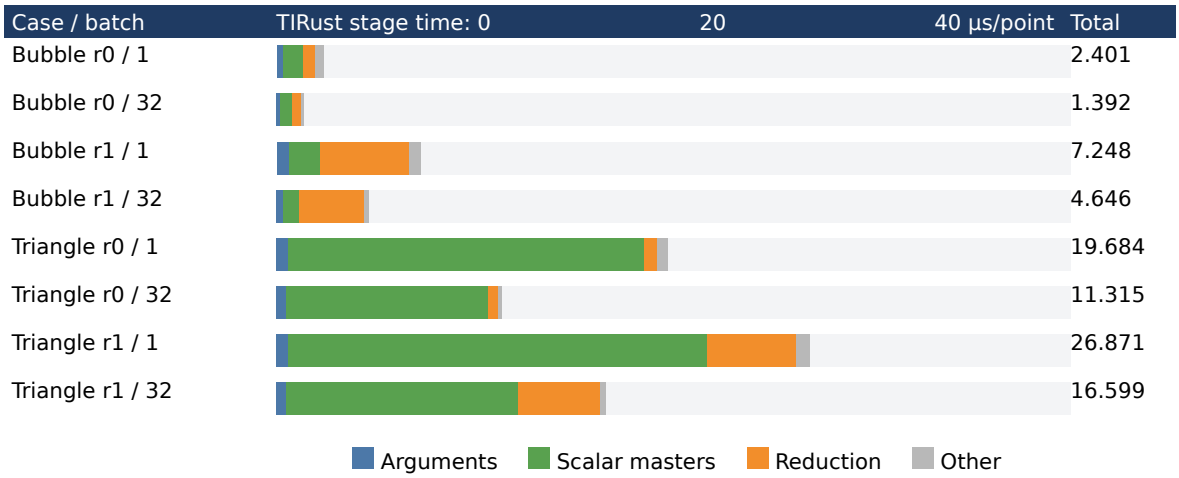
45 Historical kernel snapshot: `5c9e3dd`

All eight massive bubble/triangle checks pass the unchanged 10^{-10} criterion. Six timing rows regress; two exceed the allowed 2% relative confidence-interval width and remain **inconclusive**. Amber cells denote noise, not improvement. This commit predates default stack reuse, generalized PV and the vector-solve candidate. The overall performance requirement is unmet.

Table 18: **Binary64 hybrid interpreter, µs/point.** Latency at batch 1 and amortized cost at batch 32. Ratios are **TIRust / IREGI**, using the same-run paired original-IREGI measurement.

Case	Rank	Batch 1	Batch 32
Bubble	0	2.271 µs IREGI 0.955 µs · ×2.381 99% CI 2.364–2.393 Regression	1.389 µs IREGI 0.901 µs · ×1.541 99% CI 1.535–1.546 Regression
Bubble	1	7.146 µs IREGI 3.019 µs · ×2.367 99% CI 2.362–2.384 Regression	4.593 µs IREGI 3.020 µs · ×1.521 99% CI 1.516–1.532 Regression
Triangle	0	19.619 µs IREGI 4.975 µs · ×3.948 99% CI 3.932–4.027 Inconclusive: noise	11.310 µs IREGI 4.948 µs · ×2.281 99% CI 2.263–2.296 Regression
Triangle	1	26.724 µs IREGI 10.231 µs · ×2.610 99% CI 2.597–2.624 Regression	16.733 µs IREGI 10.279 µs · ×1.620 99% CI 1.608–1.649 Inconclusive: noise

45.1 Separately instrumented component timings



The stage sum belongs to a separate instrumented pass; “Other” is its total median minus named medians. These bars locate costs, not differences between revisions. Preparation is a single observation: bubble r0/r1 0.822/0.112 s; triangle r0/r1 0.067/0.061 s. It is excluded from the warm kernel ratios.

CPU 0, five warm-ups and 21 randomized paired blocks targeting 30 ms use identical inputs and all requested tensor slots. Structural allocations are reused; numerical results are recomputed. Transport is excluded. rustc 1.99 uses opt-level3, thin LTO and one codegen unit; GFortran 15.2 uses `-O3`. All 391 source hashes, both binaries and the harness remain unchanged under the retained provenance checks. Absolute timings from earlier sessions cannot establish a revision speedup. Raw blocks, inputs, manifests and reproduction commands: `validation/benchmark/baseline-5c9e3dd/`.

46 Physical pipeline measurements: `80949db`

Six preserved-input binary64 comparisons pass complete canonical tensors against all three original modes and all three isolated graph Laurent slots against MadLoop. **Every measured timing row remains inconclusive**: its 99% interval exceeds the unchanged 2% relative-width threshold. Both top-pentagon rows fail during preparation; no output or timing substitutes for them.

Table 19: **Equivalent complete numerical pipelines, ms/point, TIRust / MadLoop.** Repeated captured point; amortized batch32 cost does not establish throughput over varying phase-space samples.

Graph	Ranks TIRust/ML	Batch 1	Batch 32
Top triangle	2 / 3	0.877 ms MadLoop 0.108 ms · ×8.114 99% CI 7.808–8.575 Inconclusive: noise	0.729 ms MadLoop 0.105 ms · ×6.918 99% CI 6.817–7.061 Inconclusive: noise
Top box	4 / 4	11.689 ms MadLoop 0.971 ms · ×12.036 99% CI 11.854–12.303 Inconclusive: noise	5.088 ms MadLoop 0.978 ms · ×5.203 99% CI 5.087–5.421 Inconclusive: noise
Massless up box	4 / 4	7.242 ms MadLoop 0.904 ms · ×8.008 99% CI 7.642–8.225 Inconclusive: noise	3.620 ms MadLoop 0.888 ms · ×4.075 99% CI 3.997–4.116 Inconclusive: noise
Top pentagon	4 / 5	Failed before timing Preserved-input conditioning guard	Failed before timing Preserved-input conditioning guard

46.1 Scope and uncertainty

The timers include numerical numerator coefficients, reduction, conversion, contraction, status checks and returned-buffer/cache cleanup. Graph/model and external-wavefunction preparation, input packing, final comparison adapters and process transport are excluded from both. Bare poles and R1 are retained; R2 and UV counterterm graphs are excluded. Routing and rank differences are declared algorithmic choices: **no tensor-stage timing ratio is valid**.

Both engines use Hybrid, 53 working/requested bits, input preservation, CPU 0, five warm-ups and 21 blocks targeting 30 ms. Separate engine runs use an independent bootstrap ratio of medians (10,000 resamples). These intervals describe within-run variation and do not bound drift between engines. Project computations and producer writes were frozen; external host activity was not controlled. Stage samples and raw full responses remain available.

Attempt002 preserves all source/build identities. Attempt001 is retained as unqualified because producer inputs changed during execution. The optimized top-box numerical pass is specific to this captured harness path; earlier CLI binary64 guard rejections remain part of the ledger. Top-pentagon failure is the actual preserved-input derived-distance conditioning diagnostic. No failure is removed from the family count.

Sources, contracts, stage timings and reproduction: `validation/benchmark/physical-joint-quiet/`. The preceding three-block ggH smoke and 80 Python boundary/provenance tests are separately retained in `validation/benchmark/physical-joint-smoke/`.

46.2 Precision follow-up on the failed pentagon point

Promoting the same 95 represented binary64 input components exactly to Float256/512, with requested accuracy53 unchanged, gives convergence over all 280 complex tensor slots within 2.17×10^{-55}. Original PV/hybrid agree within 4.62×10^{-14} and the preserved MadLoop graph within 9.29×10^{-14}. Original IBP still differs by 0.60013; binary64/128 guard failures remain. The first 256-bit driver captured valid outputs before a rational-input comparison-parser failure; the retained separate comparison passes. This is one-point manual precision qualification, not automatic rescue or a timing gate. See `validation/physical/gghhh-guard/`.

47 Current implementation checkpoint

Production snapshot `06706e2` and private experiments have separate source and binary identities. Correctness evidence does not replace timing gates.

47.1 Ordinary scalar-master storage

The default interpreter now reuses root temporaries through Symbolica's existing allocator. All five families preserve every output, precision and signed zero in 40 configurations each across f64, DoubleFloat and Float256. Shared helpers and Dual preparation are unchanged.

Root temporary slots	A0	B0	dB0	C0	D0
Before	38	503	1,951	220	25,010
After	6	13	24	8	25

The default-stack workspace snapshot compiled with unchanged source hashes before the generalized-PV promotion below. Its current-guard acceptance passes 30 execution, 29 stability, six public mixed-precision and three scalar integration tests.

The optimized `80949db` snapshot also passes eight complete bubble/ triangle kernel checks at batches 1/32, maximum normalized error 2.46×10^{-13}. Three short blocks, zero warm-up and concurrent builds make this a correctness smoke only; its recorded timing ratios are unqualified.

The paired DD E6 experiment preserves all three statuses and 12 complex Laurent outputs exactly, with the independent 10^{-23} check passing in both variants. Each uses order 128 and 3,720 D0 rows.

Diagnostic observation	Before	After
Whole validation duration	46.682 s	43.822 s
Per-child peak resident memory	2,052,724 KiB	558,752 KiB

These are single-run diagnostics, with other compilation active; no timing gate or speed ratio is inferred. The paired experiment uses an earlier, stricter Float guard; fixed-precision DD is unaffected. Current-guard tests are separate. Sources: `validation/stack-reuse/production-factory/`.

47.2 Generalized PV and unresolved precision

The retained original-IREGI dotted-integral comparisons cover positive propagator powers and dimension shifts; they are separate from COLLIER's ordinary undotted slice. Generalized PV for $N \leq 4$ passes nine focused tests and 36 original-IREGI comparisons across 312 complex slots; maximum normalized error is 7.04×10^{-12} at the unchanged 10^{-10} gate. The promoted PV and lazy-solve core also passes five supplemental behaviors: supplied products, multinomial weights, complex scalar/tensor comparisons and Hybrid isolation. The latter uses working128/256, requested53; its binary64 fixture failure is retained. These checks use frozen former-default master archives; combined default-stack integration is separate. Higher points retain IBP/E6. Sources: `validation/generalized-pv/` qualification, promotion and supplemental manifests.

The exact small-routing E6 oracle exposes a remaining production failure:

Working bits	Requested bits	Public result
128	128	Unstable: nonfinite box output
256	128	Unstable: nonconvergent quadrature

Both runs finish within their bounds and expose no successful values. No oracle comparison or requested-128-bit agreement is claimed. The frozen binary predates default stack reuse. See `validation/shifted-pentagon/production-series-qualification.json`.

48 Validation coverage

Table 20: **Evidence ledger.** “Source audited” records code inspection; “verified” identifies executed checks described in that row.

Capability	Required evidence	Status
Independent reference oracle	Untouched/repaired Fortran twins 24 ordinary cases identical; 80 table entries exact; 7 generalized fixtures pass. New pentagon audits reveal reference defects.	Verified
Scalar-master component	OneLoopMaster A0/B0/C0/D0/DB0 Final27 scalar checks and13 consolidated D6 checks pass; all5 mixed physical D0 cases pass atFloat384. Unsupported derivative domains remain explicit.	Verified
IBP component	Powers, dimension shifts and singular identities 21 exact tests; the independent Gamma oracle covers 200 tensor requests and 40 generalized requests in all four slots.	Verified
PV component	Every tensor moment and native Laurent slot Canonical N1–4 ranks 0–7 pass. Generalized N≤4: nine focused + 36 original comparisons pass. Saved N5 rank-seven baseline: 1320 slots. Public N6/N7 tensor ranks pending.	Partial
Precision and Dual	binary64, DoubleFloat, Float, static/dynamic jets Selected f64/DD/Float128–512 and static/dynamic jets pass. Mixed-precision guards: 29 + 6 pass. Small-routing E6 requested at 128 bits: Unstable.	Partial
Execution / CLI / core oracle	Batches through reduction and scalar masters 30 execution and5 argument-guard tests pass. CLI19 smoke checks pass; frozen31 fixtures include30 binary64 plus1 explicit verified rescue.	Partial
Exceptional kinematics	Explicit errors and verified fallback N5 rank-six rescue passes; saved rank-seven baseline passes 1320 slots. Legacy: 22 primary + 8 rescues. E6 failures and broader domains remain visible.	Partial
Graph/numerator preparation	FeynKit DOT to Symbolica coefficients 4 model-boundary, 13 preparation, 4 physical numerator and 19 CLI smoke checks pass.	Verified
Physical MadLoop comparison	Graph-isolated top loops and massless box Integrated ggH, ggHH, ggHHH and massless up-box pass their retained MadLoop comparisons. ggHHH Float128/256 also agrees with original PV/hybrid; original IBP still disagrees and all-mode reports remain failed.	Partial

Snapshot boundary. Counts describe the executed component, public-execution and selected physical checks below. They do not certify all kinematics, higher-point ranks or every physical graph. Open discrepancies remain listed with the original tolerance. Four physical triangle and five massive/complex box cases pass six-component mixed derivative checks. Historical timeouts remain visible. The N5 rank-six public rescue verifies all 210 moments / 840 slots at 128/256 bits, including an independent coefficient. Binary64 TIRust and original IBP still exceed the fixed 10^{-10} criterion for that point. The frozen oracle passes 31 cases: 30 at binary64 and one generalized triangle through verified rescue. Broader domains remain pending.

48.1 Known original-reference defect

For shifts $(0,0,0,0)$ and $(0,1,0,0)$, squared masses $(4,9)$, $\mu = 3$ and numerator q_x^4, original PV and hybrid report stability but disagree with an independent Feynman-parameter integral. Their UV residues agree.

Method	Finite coefficient / i
Parameter integral	45.4316300015058951832997…
Original IBP	45.43163000150602
Original PV / hybrid	46.11123944517427

Evidence: `validation/pv-high-rank/evidence.json`. The original PV branch is retained unchanged; the relevant TIRust comparisons use original IBP and the independent parameter-integral result.

The dedicated original-IREGI/COLLIER table covers ordinary, non-dotted N2/N3 rank0/1 inputs, batches1/32. Its numerical and timing results do not qualify broader N4–N7, singular or IR-divergent families.

49 Measurement protocol

Pin release builds to one CPU core and record thread counts, hardware, OS, toolchain, flags, dependency/input hashes, seeds and warm-up. Identify cold or warm caches. Retain raw repeated-block observations, median, dispersion and an output checksum.

Preparation includes graph import, numerator construction and evaluator compilation. Time numerator, reduction, masters and total independently; summed instrumented stages are only diagnostic. Separate library costs from subprocess/file I/O and match physical routing, modes and stage boundaries.

Use identical samples at batch widths 1, 4, 8, 16, 32 and 64, matching numerical precision and including rescue costs. Retain failed lanes. Every speed gate requires correctness: stable binary64 uses the fixed 10^{-10} mixed tolerance on dimensionally normalized outputs. Gate singleton latency and batch-32 throughput per topology/rank/mass family, with IBP, PV and hybrid separate. The target envelope includes seven propagators, rank seven, generalized powers and dimension shifts.

The kernel harness now records every requested configuration before starting subprocesses, preserving interrupted and unrun rows. The latest Python suite passes 80 ledger, provenance, protocol and comparison checks. The optimized joint ggH smoke passes at batches 1/3; its short zero-warm-up timings are diagnostic. The subsequent quiet physical results are reported separately.

Private slim RealBall passes 88 core checks, eight kernel and 15 typed cases, two ggH packets, and 10 separate planar-interaction checks with retained dependencies. Its first four timing rows give two noisy real-lane results, one complex-lane parity and one gain. The longer cohort is reported separately; the candidate remains unadopted.

With the accepted license, 360 batched calls across two workers match sequential A0–D0 values, status and metadata for f64/Float128 and second-order Dual branches. A separate four-worker cold-specialization test passes shared ownership and cleanup checks. These qualify retained SymbolicaBatch snapshots; native kernels remain separate. The earlier rank-one box rejection is retained in `validation/concurrent-workspaces/`.

50 Rebuilding this report

Update `docs/data/results.json` only from retained measurements and rebuild with `bash scripts/build-benchmark-report.sh`. The report uses Libertinus Serif and DejaVu Sans/Mono. Supply their installation directories through Typst's `--font-path` option when needed. Reference checkout revisions are retained in the JSON data. They record the audited source trees and must not be mistaken for a verified dependency build.

50.1 Validation reproduction and remaining checks

Run recorded commands with the pinned environment from `scripts/env.sh`. Exact commands and source hashes accompany each JSON result. The frozen 31-case oracle includes one explicit 128/256-bit rescue; all 30 legacy fixtures pass through 22 primary results and eight verified rescues. Three active rescue, four bubble-Taylor, three scalar API and nineteen CLI smoke checks pass. The new 68-test default-stack acceptance is a separate source scope; earlier failures remain retained.

The final frozen master source passes 27 checks, including extreme-scale B0, six-component scale-only jets in all families, mixed input precision and the full physical D0 derivative batch; see `validation/scalar-masters/qualification.json`. The separate exact factory-parity/default-stack qualification is described above. Generated native kernels remain opt-in; no new timing gate passes.

An independent Symbolica Gamma expansion checks 200 tensor and 40 generalized requests with analytic bubbles disabled/enabled: 480 evaluations and 1920 slots. Nine extracted FeynKit partial-fraction and two planar formula tests pass; the separately qualified explicit planar API is recorded below.

N6 inline 128/256-bit arrays converge over 1320 slots within 6.46×10^{-29}; N6 rank7 and N7 arrays remain unqualified. The later restricted public N6 rank2/3 qualification is recorded separately below. Adopted alias-use collection preserves 29 paired cases, 86 successful lanes and 4520 components. Frozen collector-plus-sum and private amplitude-variable probes reach the 600 s bound (601.232/601.329 s with teardown), after primal amplitudes but before master activations finish; neither produces arrays. The latest private variant passes 48 variable-set, 34 amplitude/error and 73 semantic checks. Earlier timeouts remain in `validation/public-highpoint/` no timing gate follows.

51 Physical fixture qualification

Single graph	Pointwise numerator	Integrated TIRust result
Top triangle: $gg \to H$	Verified	Verified
Top box: $gg \to HH$	Verified	Verified
Top pentagon: $gg \to HHH$	Verified	PV/hybrid + MadLoop
Massless up-quark box	Verified	Verified

51.1 Physical reference contract

Fixtures preserve parameters, actual HELAS polarizations, graph factors and source hashes. The triangle uses two orientations with grouping disabled. MadLoop shifts are $(0, p_1, p_1 + p_2)$ with polynomial variable $x = -Q$; its adapter includes the odd-rank sign. For the FeynKit routing, $N_{\mathrm{FK}(q)} = -\frac{1}{2} P_{\mathrm{ML},1}(p_1 - q)$ with matched couplings/polarizations. Do not apply color or direction factors twice.

Five captures contain 40 graph records, including a second triangle helicity; these are not 40 integrated validations. Thirteen preparation and four selected physical numerator comparisons pass. Actual evaluated momenta are exported because MadLoop may correct the requested point.

Fresh preserved-input Fortran replay matches all 301 numerator coefficients, 903 tensor Laurent entries and 12 contracted slots across four graphs exactly; eight alignment boundary tests pass. Legacy-default ggH is unchanged. This is reference-only qualification; later TIRust checks belong to the separate joint run. Equivalent full-graph timing retains TIRust/MadLoop ranks 2/3 for ggH, 4/4 for both boxes and 4/5 for ggHHH, and their declared routing. It does not qualify tensor-stage ratios. Sources: `validation/benchmark/physical-alignment/`.

The recorded binary64 $gg \to H$ CLI passes all 15 moments / 60 slots against original IREGI within 1.86×10^{-15} normalized error and MadLoop within 9.38×10^{-15}, including poles. Values and binary identity remain in `validation/physical/ggh-cli-comparison.json` this is not a fresh replay. Nineteen CLI smoke checks pass. Binary64 comparison arithmetic certifies no extra digits from higher working precision; short unpinned timings are excluded.

The historical top-box binary64 CLI rejects: PV fails its cancellation guard and IBP fallback is nonfinite. Float128/256 reruns pass all 280 slots against all three reference modes within 1.15×10^{-13}, and MadLoop within 4.77×10^{-15}. Each freshly prepares/evaluates the numerator. Their normalized tensor difference is 6.09×10^{-37} under binary64 comparison arithmetic. See `validation/physical/gghh-cli-precision-convergence.json`. The Symbolica-backed model adapter passes four boundary, thirteen preparation and four numerator tests while preserving other numerical tokens.

The 80949db massless up-quark box passes all 70 moments / 280 slots against each original mode: maximum error 2.85×10^{-14} against IBP and 7.11×10^{-15} against PV/hybrid. MadLoop finite/single/double coefficients agree within 6.74×10^{-15} with fixed-color factor $-\frac{1}{8}$. Its one-run wall time is unqualified; see `validation/physical/gggg-up-cli-comparison.json`.

Complete top-pentagon CLI reruns at Float 128/256 now agree with original PV/hybrid for all 280 slots within 4.63×10^{-14}, and with MadLoop within 9.30×10^{-14}. Between-run tensor difference is 7.79×10^{-17} under binary64 comparison arithmetic. Original IBP still differs by 0.60014, so the all-mode report remains failed. See `validation/physical/gghhh-cli-precision-convergence.json` and linked reports. Earlier PV/hybrid replay of 378 captured MadLoop slots was exact.

51.2 Physical derivative qualification

Four triangle cases at 384 bits cover real subthreshold/absorptive and complex-mass kinematics, each with six first, second and mixed Dual components. Finite differences converge with maximum final scaled error 5.47×10^{-28}. One equal-complex D0 case also passes, with error 1.11×10^{-34}. The complete five-case D0 batch now passes at 384 bits, maximum final scaled error 2.53×10^{-30} under the unchanged 2^{-70} criterion. Its 224.83 s validation duration is not a benchmark; earlier 180 s timeouts remain retained. See `validation/scalar-masters/physical_box_full.json`. Exact commands and scope are retained in `crates/iregi-masters/tests/validation_evidence.json`.

52 Open qualification ledger

Unresolved item	Observed evidence and remaining check
Binary64 baseline	Historical TIRust/IBP errors 2.23×10^{-10} / 3.40×10^{-10} fail. Rank-six rescue passes at the retained point.
Pentagon coverage	Rank-six rescue passes 840 slots; saved rank-seven baseline passes 1320. Later precision/routing edits and broader domains require qualification.
Physical pentagon	Float 128/256 agrees with PV/hybrid and MadLoop; original IBP disagrees. Binary64 and all-mode acceptance remain incomplete.
Shifted scalar	Thirteen standalone checks and public f64 fixtures pass. Production small-routing E6 remains Unstable at requested 128. New private mixed-jet checks cover common masses/zero invariants only; broader domains and production integration remain unqualified.
Derivative domains	Five-case mixed D0 checks pass 2^{-70}. Thresholds and unsupported transverse massless/Gram/Cayley domains retain errors.
Physical top box	Historical binary64 guard rejection remains visible; Float 128/256 and preserved 80949db binary64 checks pass at retained points.
Supplied-product N6	Certified scalar working 128/256, requested 53 passes with zero poles and invalid-lane isolation; binary64 rejects. Independent difference 1.53×10^{-23} is below estimated (nonrigorous) 6.11×10^{-22}. N7/tensors/mixed Dual remain pending.
Legacy and precision	All 30 retained cases pass: 22 primary and 8 rescues. Three false-success cases are repaired; conservative active-master error estimates remain noncertified. Broader domains remain unqualified.
Performance	Historical 4c32 interpreter: 7 regressions/1 gain; native: 5 regressions/1 parity/2 gains. 5c9: 6 regressions/2 noisy; 80949: 5/3. Latest 6b24: 7/1. Physical: six noisy/two failures. Overall acceptance is unmet.

52.1 Optimization and experimental evidence

Candidate	Executed evidence and decision
Adopted allocation / guards	Core allocation: 67 current opt2 checks; quiet paired run preserves 92 outputs. Guard context: 76 checks and 120 snapshots, 14 cleanup checks; six isolated gains/six noisy rows. Combined 6b24 optimized IREGI gate remains seven regressions/one noisy row.
Roundoff cache	29/37 stability checks and 120 snapshots pass. Candidate rejected: two regressions/ten noisy rows. Separate accuracy-guard diagnostic: two regressions/six noisy rows.
CLI coefficients	72 ordinary/12 mixed batches pass. Eight opt2 timing rows: five gains/one parity/two noisy; ggHH32 ratio 1.0367, CI 1.0237–1.0501 retains possible regression; unadopted. CSE proves 892 coefficient/223 homogeneity identities. 1281 raw failures coexist with zero dimension-normalized diagnostic failures; no physical acceptance or new timing claim.
Native candidates	Dense f64 B0/C0: two gains/six singleton parities/four batch32 regressions. D0 allocator: three singleton gains/two batch32 regressions/one noisy. Shared B0/C0 kernels: one gain/two parities/seven regressions/two noisy. All rejected; D0 singleton source is unchanged. Separate actual A0–C0 replay passes 1782 outputs from 297 rows at 10^{-10}; native/interpreter values are not bitwise identical. New replay timings are reported separately.
Certificate candidates	Slim RealBall module: 23 checks, including 8 baseline controls. Separate exact-distance receipts: 49 checks and 120 exact snapshots against the same private RealBall baseline; argument rows unchanged. Fourteen timing cases give eight gains/two parities/four noisy, no qualified regression. Private final integration passes 12 receipt/6 accounting checks, counted once with the automatic-planar followup. The final timing cohort is reported separately.
N5 planar API	Promoted explicit unit-power API: rank7 ordinary128/256 passes 1320 slots/convergence; mixed128 passes 5280 slots per accepted lane versus256 boxes, requested53. Transverse jets reject; 9 focused/49 ordinary checks and 20 Fortran-leaf coefficient cases pass. Earlier timeout remains. Private automatic rank1/2 dispatch passes 69 checks; the final 18-test followup is separate. No automatic rank7, production fallback or timing claim.
Finite D0 certification	Explicit API promoted: three actual public tests and current core Cargo checking pass. Exact represented inputs receive component-budget enclosures in the checked near-vacuum domain; upstream routing and quadrature errors are excluded. Prior typed targets are f64/40, DD/80, Float128/100 and Float256/128. Ordinary evaluation and default E6 remain unchanged; no automatic certified-master dispatch.

53 Latest isolated optimization cohorts

These private candidates were measured in one coordinated quiet window on CPU 254. Each has its own frozen sources, binaries and registered protocol; results do not qualify a combined implementation or replace the TIRust/IREGI performance gate. Candidate effects are never multiplied. Failed controls, inconclusive rows and earlier cohorts remain in the evidence.

Private cohort	Per-family outcomes	Controls, scope and decision
Final proof receipts	14 families: 11 gains, 1 parity, 2 regressions; no noisy intervals.	Ordinary bubble singleton regresses 2.90% (CI 2.43–3.27%); ordinary triangle batch32 regresses 0.69% (CI 0.20–1.26%). No blanket adoption. Same retained masters; N5 path inactive. Earlier cohort remains separate.
Slim RealBall, longer run	4 families: 1 gain, 1 noisy, 2 regressions.	Real singleton improves; real batch32 remains noisy (CI width 2.18%). Complex-mass controls regress 0.74%/0.62% at batches1/32. Real shifts remain; no adoption or pooling with the first run.
Native master replay	8 families: 8 gains; all 168 pairs faster.	TIRust(candidate)/TIRust(baseline) ratios 0.143–0.809 on actual A0–C0 master bundles: native candidate, Symbolica baseline. Raw kernels exclude adapter validation/allocation, recurrence and contraction. No production switch or IREGI parity claim.
CLI coefficient CSE0	8 families: 8 gains.	Complete warm coefficient call only. Raw coefficient criterion still fails in baseline and candidates; separate dimension-normalized diagnostic passes. No adoption or physical acceptance claim.
CLI coefficient pair1	8 families: 6 gains, 2 noisy.	ggHHH singleton and up-quark box batch32 remain inconclusive. Preparation observations are retained separately; repeated fixed-point f64 evaluation is not a full-graph timing gate.

Ratios in these cohorts are **TIRust(candidate) / TIRust(baseline)**, using each cohort's own implementation and baseline. Every measured family is retained in the linked JSON, with raw paired observations, 99% confidence intervals and unchanged qualification thresholds. Project compute was held, but unrelated host activity remained uncontrolled. Earlier measurements use different frozen sources or protocols and remain separate. No new production adoption is implied by this page.

53.1 Exact and source-only research

Additional Symbolica CSE leaves A0/B0/C0 exports unchanged: 43/506/223 instructions and 6/13/8 root slots. Eighteen root identities and 470 control checks pass under the structured-SSA assumption. With no instruction saving, the candidate stops before numerical or performance qualification.

The E6 design notes remain proposals. The initial private direct N2 prototype at Float256/request128 certifies 16 components across four successful lanes with one-axis common-mass jets; five expected rejections hide values. Independent audit: 513 checks and 140 hashes pass. Its later typed followup appears separately below. Neither snapshot qualifies public/default E6, general geometry or performance.

A separate private rank-lowering N6 rank-one candidate passes all 20 complex Laurent slots against original IBP at Float128/request53, maximum normalized error 3.17×10^{-13}. The old public path rejects nonfinite coefficients, so this is not paired old/new numerical agreement. Exact constraint tests accompany the frozen development build. Its rank-two probe rejects nonfinite coefficients in 57.823 s, within the 180 s bound, without arrays. Rank-seven, N7, convergence and performance qualification remain pending.

54 Lazy proof capture: complete paired comparison

TIRust(lazy) / TIRust(eager) measures the entire reduction call, including result construction and disposal. Variants share the eager-final baseline, current master adapter and unchanged optimized numerical dependencies; only receipt-capture allocation changes. Symbolica masters remain selected; N5 retry is inactive.

Table 21: **All fourteen workloads are retained.** Ratios below one favor lazy capture. Each cell shows the paired latency ratio, its 99% bootstrap interval and the unchanged per-family verdict.

Workload	Batch 1	Batch 32
Bubble · mixed Dual	0.9945 99% CI 0.9932–0.9969 Gain	0.9996 99% CI 0.9958–1.0021 Compatible with parity
Bubble · ordinary	0.9913 99% CI 0.9871–0.9976 Gain	0.9957 99% CI 0.9935–0.9968 Gain
Bubble · complex / exact zeros	0.9972 99% CI 0.9953–0.9979 Gain	1.0008 99% CI 0.9987–1.0031 Compatible with parity
Bubble · real / exact zeros	0.9984 99% CI 0.9940–1.0045 Compatible with parity	1.0089 99% CI 1.0038–1.0159 Regression
ggH · mixed complex	0.9979 99% CI 0.9957–1.0019 Compatible with parity	0.9999 99% CI 0.9975–1.0017 Compatible with parity
ggH · real / exact zeros	0.9951 99% CI 0.9890–0.9973 Gain	1.0022 99% CI 1.0002–1.0061 Regression
Triangle · ordinary	1.0077 99% CI 1.0053–1.0108 Regression	1.0018 99% CI 0.9981–1.0034 Compatible with parity

Five gains, six parity results and three qualified regressions; no noisy intervals. No adoption. Regressions occur for the real exact-zero bubble at batch 32, real exact-zero ggH at batch 32, and ordinary triangle singleton. Every 99% interval meets the fixed 2% width gate. The all-family no-regression criterion fails; all families remain retained.

Fourteen optimized preflight pairs preserve 10,560 unique complex output slots and exact status/metadata/context/counters. The earlier 26 debug tests remain separate. CPU 254; five warm calls; 41 paired blocks with 21/20 alternating order; retained calibration ≥200 ms on the faster side; 20,000 paired bootstrap resamples; all 1,148 observations retained. Lane/output checks precede and follow timing; iterations unwrap the top-level result. The host is shared.

Scope: N2/N3 f64 and flattened mixed Dual only; no new TIRust/IREGI or TIRust/COLLIER gate. Earlier eager/core and slim-RealBall cohorts remain separate; their ratios cannot be multiplied. Source: `validation/lazy-receipt-performance/`.

55 Additional bounded correctness checks

These completed checks extend the evidence ledger without changing production or any measured timing row. Each private candidate retains its own source and dependency snapshot; the raw-master replay gains do not establish whole-reduction performance for a different candidate.

Bounded study	Observed result and remaining limit
Selected native A0–C0	Eight ordinary f64 N2/N3 rank0/1 families at batches 1/32 pass all 1584 complete tensor coefficients per backend against original IREGI and the Symbolica baseline. Maximum normalized errors are 3.38×10^{-13} against IREGI and 2.07×10^{-13} between backends. The 297 actual master rows retain all 1782 six-output coefficients. Private opt0 build; whole-call performance is unmeasured.
Lazy proof capture	26 tests pass: 12 receipt, 8 lifecycle and 6 accounting checks. Fourteen paired workloads match exact values, status, metadata and counters through seven calls per implementation. Capture is created only when evidence is needed and discarded after each call. This private debug qualification is separate from the later optimized timing table; no adoption.
COLLIER N4 extension	Eight fixed massive cases cover ranks0–3 at batches1/32. Both scalar rows are accepted, maximum normalized error 2.76×10^{-14}. All six higher-rank COLI rows retain `AccFlag=-1`, despite raw agreement within 2.39×10^{-12}. DD mode returns `ErrFlag=-10` for all eight; those arrays are invalid reference evidence. No inputs or thresholds changed; no timings were run.
Fused core CSE	The N2 rank1 native/OneLoop exports remain byte-identical: 1013/850 instructions and 237/147 slots. All 422 exact roots and 730 control checks pass under the structured-SSA assumption. With no structural saving, the candidate stops before numerical or timing work; other topologies remain untested.

55.1 Scope of the independent checks

The selected-native study also passes 34 adapter cases, ten unchanged master paths covering 33 cases and three unchanged whole-core paths. Repeated whole responses are retained; direct adapter repeats are executable assertions with the common output saved. Initial build and fixture failures remain in the package. Native D0, high-point tensors and a production backend switch are outside this qualification.

The lazy-capture allocation occurs inside the existing argument-stage timer. The earlier eager allocation was outside that stage; the later paired comparison therefore uses total latency. The earlier final receipt cohort and its two qualified regressions remain unchanged.

N4 checks use 238 Symbolica-prepared polynomial arrays for metric traces, pinch translation, denominator-difference signs, UV finiteness and odd-time reflection. Their largest identity discrepancy is 2.89×10^{-12}; these constraints do not independently determine every spatial tensor component or override a rejected library status. The initial eight measured IREGI/COLLIER rows remain unchanged.

A retained **initial unmasked** thin-N6 rank2 probe passes seven exact/sidecar checks, then both certified-planar retries fail among six packed N5 children; that attempt produces no parent arrays. The later private mask followup has separate source, numerical and lifecycle qualification below. Those private capsules establish no production or performance gate; the later explicit public six-point API has its own qualification below.

Historical compile/type/lint checks appear separately from later wheel004/gallery execution. Binding and notebook sources follow the pinned Community guide in `PLAN.md`.

Source bindings, raw outputs, failures and reproduction commands are retained in `docs/data/results.json`. No elapsed correctness-run duration on this page is a benchmark. These historical records bind their named source revisions; the optimized TIRust performance gate remains the measured `6b24c03` snapshot, without the later six-point API changes.

56 Additional execution qualification

56.1 Private N6 rank-two mask followup

Fourteen exact checks qualify the thin child assembly, full-shape zero-mask certificates, cache/context binding and route envelopes. Float128 and Float256, both requested 53, return all 60 native complex Laurent slots for the retained base point. Maximum normalized disagreement with original IREGI is 1.165×10^{-12} against the unchanged 10^{-10} gate; maximum 128/256 difference is 1.474×10^{-34} against 2^{-80}.

Check	Retained count	Scope
Precision comparison	2 × 60 slots	One retained base point at Float128/256; requested 53.
Lifecycle	11 calls / 17 accepted lanes	1020 complex slots; seven explicit rejection controls and two empty resets.
Singleton agreement	Zero differences	Values, precision and signed zeros agree with separate singleton calls.
Independent audit	327 bindings / 1140 slots	Saved-data arithmetic and source-archive audit; no kernel rerun.

The lifecycle corpus includes base/scaled points, mixed valid/invalid batches, reordering, chunks and resets. It shows isolation and absence of partial successful arrays; a rejected status alone does not prove the mathematical failure cause. The ordinary primal sensitivity estimate remains nonrigorous for compounded Taylor errors, and imported quadrature errors are unsupported.

Strict scope: private FromShifts/Preserve, unit powers, no dimension shift, N6 rank 2, native raw components. Numerical Dual, complex masses, N7, rank 7, generalized inputs, production adoption and performance remain unqualified. The earlier unmasked rejection and the 30 s exact-test timeout with zero completed tests remain preserved. The later unchanged 14-test gate passes within its separately declared 180 s bound. Sources: `validation/highpoint-thin-n6-mask/` and its independent checkpoint audit.

56.2 Private generic rank-three extension

A separate generic assembly passes 25 exact structural/integrand/context tests. At the retained N6 rank 3 base point, Float128/256 with requested 53 each return all 140 native complex Laurent slots, including 105 exact zero poles. Maximum original-IREGI discrepancy is 2.426×10^{-12} against 10^{-10}; precision difference is 1.474×10^{-34} against 2^{-80}. The exact tests cover structural identities through rank 7; numerical rank 7 is not qualified. This separate package adds no rank 3 lifecycle, numerical Dual, generalized/SuppliedG, N7, public adoption or performance result. The initial test compile failure and historical rank 7 timeouts remain. Source: `validation/highpoint-thin-n6-generic/`.

56.3 Initialization-order diagnostic

Two fresh processes retain 55 complete responses: seven control requests and 48 case requests. All numerical admissions pass; 37 within-process exact comparisons have zero differences. CASE-first reproduces all 32 saved case responses and all three controls exactly. CONTROL-first reproduces a one-ULP C0 difference and changes 24 finite imaginary tensor values per backend in the batch 32 rank-one triangle, by at most nine ULP (2.50×10^{-16}). The whole-case master captures and input/status/diagnostic records remain exact.

Registration/operand-order source facts suggest a mechanism, but no paired instruction exports establish the cause. This is an untimed, schedule-associated diagnostic, not universal determinism or an accuracy-threshold change. The failed v2 preflight remains unmeasured. A later four-engine cohort uses its separately reviewed CASE-first protocol; no diagnostic duration is a benchmark. Source: `validation/selected-native-initialization-order/`.

56.4 Python interface: historical compile and source typing

The PyO3 standalone/native crate passes Cargo checking with one shared Symbolica/Numerica/Graphica/PyO3/FeynKit owner identity. Its ordinary source consumer passes basedpyright with zero errors and warnings. This checks compilation and source-facing type signatures; installed-wheel typing and execution require the separate runtime evidence below. The initial build timeout and retained mypy/upstream-stub failures are preserved rather than counted as successful checks.

The initial notebook source capsule retains its original strict lint and nine display-helper tests. Later explorer, browser and native notebook checks are separately source-bound; historical results do not qualify a later source revision. Sources: `validation/python-interface/checkpoint-compile/` and `validation/python-notebooks/qualification.json`.

57 Historical Python and notebook runtime

57.1 Inspected native wheel baseline

The exact baseline wheel builds, installs and passes shared-owner identity, packaged font/stub checks and installed-consumer typing. Of 29 focused tests, 28 pass; the real upstream graph renderer fails with `painted coordinates must be numbers`. Numerical, precision, Dual, dotted, batch, contraction and shared tensor/graph interoperability checks pass independently of that rendering failure. Tests use the actual shared upstream types from the installed native host.

Release-wheel performance, general mathematical-domain acceptance and full Community-host integration remain unqualified. Historical `6b24c03` measurements describe an earlier kernel. Source: `validation/python-interface/checkpoint-wheel-003/qualification.json`.

57.2 Actual native notebook execution: partial result

The inspected wheel is installed into the marimo 0.24.0 notebook environment. Four native test methods complete: two pass, two have three rendering-error occurrences. The successful controls check the dotted bubble against its mass Dual derivative, precision distinctions beyond binary64, failed-lane isolation and an edited-topology guard. Finite-difference values are displayed; this test adds no independent finite-difference acceptance gate.

Notebook	Actual checks	HTML export	Qualified scope
01: Graph → integral	Real renderer errors	21 cells / 1 error	Bubble, triangle Float and bubble DoubleFloat stop before their numerical assertions.
02: Dots and derivatives	Numerical controls pass	19 successful cells	Massive bubble identity, exact-input digits and invalid-neighbor masking.
03: Benchmarks	3 tests / 6 cohort choices	11 successful cells	Frozen evidence only; no numerical extension imported.

Both native exports and the original error remain archived. Rendering-dependent numerical assertions were unrun in this baseline. The later wheel004 followup below qualifies the named examples and their static graph/math browser displays. Source: `validation/python-notebooks/native-wheel003/`.

57.3 Benchmark explorer and readable narrow charts

The separate explorer source revision passes strict lint and 15 evidence/display-helper tests, including all forty saved ratios and eighty CI endpoints. Actual marimo execution retains every cohort and its original short/noisy gates. Static Chromium desktop/narrow-dark checks preserve eight ratio cells and render without request, HTTP, console or page failures.

The chart now keeps a 930 px SVG inside controlled horizontal overflow, preserving readable labels within a 348 px narrow viewport; the document stays 390 px wide. Desktop cells and narrow/dark grid/chart crops were visually inspected. Earlier loader failure and small-chart screenshots remain historical. The export requires the official marimo CDN frontend and has no reactive backend; browser snapshots do not qualify numerical controls. Sources: `validation/python-notebooks/chart-readable/` and its separate source/runtime/browser predecessors.

58 Historical wheel004 repair and executed gallery

58.1 Exact JSON and actual upstream graph rendering

GDB inspection of the unchanged baseline wheel finds that exact JSON's `arbitrary_precision` feature turns coordinates serialized through `serde_json::Value` into CBOR private-number marker maps. The actual numeric-coordinate decoder rejects those maps. The pinned upstream owner now uses typed line/cubic command tuples and typed collision-line records at both serialization sites. The exact-JSON feature and actual renderer remain in use.

Two earlier owner-test setup/fixture failures remain archived. The final two consumer regression tests pass, exercising both stroke decoding and collision search under exact JSON. Rebuilt local wheel004 then passes all 29 actual focused tests: 17 numerical and 12 interoperability, including real graph rendering, with no failures or skips. Shared-owner identity, packaged assets/stubs and installed consumer typing pass. This is a local development wheel; build durations are validation diagnostics, not timing comparisons. Sources: `validation/python-interface/checkpoint-wheel-004/qualification.json` and `validation/python-interface/rendering-diagnosis-003/`.

58.2 Unchanged notebooks, fresh installed wheel

The approved protocol freezes eleven notebook/test/graph/environment inputs plus the exact wheel and owner receipts. A separate fresh environment preserves both the failed baseline003 environment and the inspected owner004 environment. All four native test methods pass. They exercise bubble f64, bubble DoubleFloat and triangle Float graph preparation, full tensor output shape, batching and contraction, plus the dotted bubble/Dual mass identity, precision distinctions, invalid-neighbor isolation and the edited-topology guard. Finite-difference values are computed for the display; this test adds no separate finite-difference accuracy threshold.

Notebook	Native / evidence checks	Successful HTML cells	Real static browser display
01: Graph → integral	Graph/numerical controls pass	21 / 21	Actual upstream bubble SVG, formulas and four contracted Laurent cards.
02: Dots and derivatives	Identity/precision/ isolation pass	19 / 19	Formulas and four dotted-integral Laurent cards.
03: Benchmarks	3 tests / 6 cohorts	11 / 11	Eight ratios, retained short gate and readable horizontal chart.

All three strict lint checks and 15 display/evidence tests pass. One Chromium pass inspects each export in desktop light and narrow dark views: six views, 1,918 normal frontend requests, no failed requests, HTTP, console or page errors. The mobile document stays 390 px wide; the 930 px chart pans inside its 348 px viewport. Actual graph, numerical cards and chart crops were visually reviewed. The static export uses marimo's official CDN frontend and has no reactive numerical backend. No numerical mock or replacement renderer is used. Release/platform/Community-host and full mathematical or performance qualification remain open. Source: `validation/python-notebooks/native-wheel004/qualification.json`.

59 Explicit public six-point reduction

The source now exposes `SixPointReductionPlan` and `SixPointWorkspace<T>` as a separate, restricted API. It accepts six unit-power propagators, ranks 0–3, dimension shift zero, preserved Cartesian shifts, PV mode and native IREGI raw components. Unsupported settings fail during preparation. Ordinary reduction, hybrid dispatch and CLI defaults retain their existing behavior.

Isolated qualification	Retained count	What was checked
Selected exact tests	34 passed	Assembly, context, certificates and API controls in the frozen isolated build.
Existing ordinary API	12 paired cases / 2200 slots	42 successful lanes and 18 matching failed controls; values, precision, signed zero, metadata and counters preserved.
N6 rank 3	140 slots × 2 precisions	Float128/256, requested accuracy 53; all 105 pole slots are exactly zero at each precision.
N6 rank 2 lifecycle	11 calls / 17 accepted lanes	1020 complex moments and 68 contractions; seven rejection controls and two empty resets.

The rank-three base point agrees with original IREGI within 2.426×10^{-12} under the unchanged dimensionally normalized 10^{-10} gate. The 128/256-bit difference is at most 1.474×10^{-34} against 2^{-80}. The rank-two lifecycle retains mixed valid/invalid batches, scaling, reordering, chunks and post-error reuse. Its contraction check preserves the requested native convention and tensor metadata; it does not require bitwise equality after reassociation.

The implementation packs all six five-point children across active parent lanes, preserves child sensitivity diagnostics and checks complete Taylor cofactor constraints before dropping zero contributions. Failed lanes expose no partial successful coefficient array. These are implementation properties covered by the named tests, not a rigorous numerical error enclosure.

Separate current-Cargo replay. The actual locked non-Python Cargo library test and example builds pass, followed by all 34 selected tests. At the retained rank-three point, both complete 140-slot arrays pass the unchanged original-IREGI and precision-convergence gates. Independent saved-data arithmetic verifies all 280 original comparisons and 140 convergence checks. The isolated qualification remains unchanged; its 12-case ordinary comparison and rank-two lifecycle are not reported as additional current-Cargo runs.

Provenance binds the actual linked executables and 674 runtime inputs, including recorded Cargo artifacts and Rust source guards. This is not a complete native build-input closure: three native archives have supplemental post-run hashes only, without a retroactive pre/post guard claim.

The accepted high-point numerical scope is the retained ordinary Float rank2/3 corpus at requested accuracy 53. Public N6 Dual, binary64/DoubleFloat, arbitrary complex masses, generalized powers/dimension shifts, rank 7, public N7 and performance remain unqualified. Existing ordinary Dual controls above preserve the old API; they do not qualify high-point Dual evaluation. Earlier failed high-point attempts and original-IREGI rank7 discrepancies remain visible. Sources: `validation/highpoint-six-point-api/qualification.json` and `validation/highpoint-six-point-current-cargo/qualification.json` public usage: `docs/six-point.md`.

60 Historical wheel004 C0/D0 mass-Taylor identities

The unchanged installed local wheel004 now passes first, second and mixed squared-mass Taylor identities for rank-zero C0 and D0. Each family uses one fixed massive spacelike point, working precision 128 and 256 bits, and requested accuracy 53. The ordered six-component Taylor layout is `[[0,0],[1,0],[0,1],[2,0],[1,1],[0,2]]`.

Family	Successful evaluations	Laurent identity checks	Precision checks	Maximum normalized identity error
C0	14 / 14	48 / 48	48 / 48	2.509×10^{-36}
D0	14 / 14	48 / 48	48 / 48	3.712×10^{-36}

All 28 evaluations succeed. The 96 identity checks compare the Taylor coefficients with independently prepared raised-power IBP reductions at the appropriate factorial normalization. Another 96 checks compare the two working precisions. Every check includes native UV single, IR single, IR double and finite coefficients; all three poles are exactly zero at these points. The finite coefficients remain general complex values. The saved-data audit checks 192 evaluated complex Laurent slots through 9238 assertions using exact integer ratios and the original dimensionally normalized 10^{-10} criterion.

Both families use the existing native convention and preserved inputs. No fallback or analytic rescue occurs. There is one numerical attempt per family and no retry. The original C0 saved-data audit incorrectly required a zero real finite part; its failed receipt is retained. The corrected audit checks the full complex value, with unchanged numerical output, input, pole tests and tolerance. No new integral evaluation was used to repair it.

Evidence identity. This is wheel004 evidence, bound to native module `cd4df512a134…` and its saved source archive. It is independent of the later public six-point source/Cargo check. The raised-power comparison and independent saved-data arithmetic are not a new external-library or finite-difference validation.

Working at 128/256 bits does not claim that many achieved accuracy bits: the requested target is 53. These two cases do not add general static-Dual or DoubleFloat coverage, massless or threshold derivatives, collinear/Gram/ Cayley boundaries, arbitrary complex-mass domains, further sample batching, physical/MadLoop comparison or performance qualification. Historical broader C0/D0 finite-difference evidence remains a separately bound source scope. Source: `validation/c0d0-dual-identity/qualification.json`.

61 Private capability and retained high-point failures

61.1 Selective native-f64 masters: correctness and metadata

The private opt-in A0/B0/C0 backend passes 17 focused tests and a retained 38-request capture. Per backend, the whole N2/N3 rank0/1 batch1/32 fixtures contain 1,584 complex tensor outputs and 1,782 complex outputs across all six master coefficients. The largest candidate/original dimensionally normalized error is 3.38×10^{-13} against 10^{-10}. Separate selected, unchanged precision/Dual, cache and failure controls remain visible in the package.

The metadata compatibility matrix now has all 16 entries passing across default, selective native, broad native-batch and combined features. The independent continuation completes only the two unfinished metadata checks; earlier timeouts, cancellation and the false final guard remain historical. Metadata checking does not execute the broad/combined numerical test binaries. No production/default change, performance qualification or adoption follows. Sources: `validation/native-f64-opt-in/capability-v1/qualification.json` and `validation/native-f64-opt-in/compatibility-completion-v1/qualification.json`.

61.2 Uniform-branch scheduler: local fixture qualification

The private Symbolica scheduler candidate passes 15 actual tests: five existing batch, six focused and four nested/shared-function tests. Actual old-batch, candidate and scalar paths are compared, including precision, signed zero and active instruction/lane traces for f64, DoubleFloat, mixed Float precision and static/dynamic mixed-order Dual shapes. Two earlier fixture compile failures remain archived. The patch is not applied to live Symbolica; dependent masters/core, full reduction and performance are not qualified by these local tests. Source: `validation/uniform-branch-scheduler/qualification.json`.

61.3 First private N7 scalar capture: no Rust numerical array

The retained dyadic N7 rank0 fixture passes its exact-domain audit, including seven six-point determinants and 21 child coordinate ranks. Its generic structural plan times out at `preparing_plan`, before workspace or scalar evaluation. There are zero Rust numerical slots. Cleanup completes, but total capture time including guards is 300.928 s, exceeding the frozen 300 s bound; the comparison is not spawned.

Original IBP and hybrid report stable matching finite values; PV is unstable and its zeros remain rejected. An independent scalar oracle finishes all six fixed rules (orders 8/12/16, two mappings), but its empirical normalized uncertainty estimate 7.696×10^{-10} exceeds 10^{-10}. That oracle exits with insufficient convergence. No numerical agreement, precision convergence, general N7, higher rank, Dual or performance acceptance follows. The earlier 17 selected exact tests remain a separate structural result. Source: `validation/highpoint-thin-n7-rank0-attempt/qualification.json`.

A separate marker-only constructor probe also times out: 110 s compute, 116.604 s total within its 150 s bound. Its first six-line chart takes 107.091 s inclusive, containing a 91.136 s determinant, 11.652 s inverse and 4.297 s product; the second determinant is active at termination. These nested observations diagnose that probe only. They are not summed, used as performance comparisons or retrospectively attributed to the earlier capture. The constructor remains incomplete, with no workspace, masters or arrays. Source: `validation/highpoint-thin-n7-phase-attribution/qualification.json`.

A private followup uses Symbolica's `RationalPolynomialField` for the matrix coefficient domain. All 22 selected exact tests pass; the rank-zero constructor completes in 58.442 s (66.660 s including guards, within the 150 s bound). This preparation diagnostic creates no workspace, masters or numerical arrays. It establishes no numerical or performance gate and is not applied to live production. Both earlier timeouts remain unchanged. Source: `validation/highpoint-thin-n7-rational-domain/qualification.json`.

62 Precision and high-point research

62.1 Typed direct E6: bounded common-mass fixtures

The unchanged private N2 direct genuine D6-pentagon engine is checked on common complex masses and zero invariants, with four- and six-component mixed Taylor shapes. Across ten groups, 65 lanes succeed and 65 intentionally reject; all 320 successful components satisfy the requested enclosure budget. Rejected lanes hide values. This extends the earlier one-axis prototype; it does not integrate the engine into the public reduction path.

Base type	Working / requested bits	Accepted / rejected lanes	Accepted components
f64	53 / 40	13 / 13	64
DoubleFloat	106 / 80	13 / 13	64
Float	128 / 80	13 / 13	64
Float	256 / 128	13 / 13	64
Float	384 / 128	13 / 13	64

Independent exact audit: 24,889 checks and 166 hashes, zero failures; the root independently reran that audit. Inputs distinguish 2^{-80} where the type permits it, retain mixed input precisions and verify scale-jet independence. Static Mixed4/Mixed6 and dynamic HyperDual metadata transport use Symbolica's Dualizer with NativeFloat workspaces; this is **not** `Workspace<StaticDual<T>>` qualification. Empty, odd, reordered, chunked, fresh, invalid-neighbor and independent concurrent workspaces pass compiled assertions; complete repeated arrays are not serialized.

Float256⟶384 arithmetic widths shrink for 102 positive coordinates and remain zero for 26. This comparison removes twice the retained truncation radius; it does not establish convergence of the fixed N2 mixed-cubic truncation. Instruction-lane visits are structural estimates. Unequal masses, nonzero invariants in this typed extension, generalized powers/dimensions, N4, contour/geometry certification and performance remain outside its scope. Production E6 failures remain unchanged. Source: `validation/e6-direct-n2-typed/`.

62.2 Exact child localization and mixed-jet counterexample

Two retained regular planar charts locate an ordinary child with Gram rank 2/determinant 0 and Cayley rank 4/determinant −4. Its multiplier vanishes at this exact ordinary input; generic symbolic aliasing still requests the singular PV child. This is preparation/identity evidence only: it does not identify the first nonfinite runtime instruction or evaluate a scalar master.

A separate exact mixed-jet counterexample has zero primal and first components but nonzero mixed cofactors −1/−1 and weights −2/−1 in the two charts. **Primal zero is insufficient for pruning.** Any future mask must certify the complete requested cofactor jet while retaining planar and chart regularity checks. No pruning implementation, successful N6 array, production change or performance result is claimed by these sources: `validation/highpoint-child-localization/`.

63 Independent pentagon adjudication

These checks establish selected mathematical values independently of an original-reference stability flag. They do not qualify every pentagon tensor, physical graph, runtime domain or numerical precision.

63.1 Rank-six Euclidean tensor

For squared masses $(4, 9, 16, 25, 36)$, $\mu = 3$ and numerator $q_0^5 q_3$, direct simplex integration gives native finite coefficient $-0.0024238378935447123i$. Independent coordinate mappings agree within 4.34×10^{-19}; reported quadrature error is at most 2.77×10^{-15}. The bound $F \geq 3.75$, angular integration and power counting exclude poles. This uses no reduction/master library; numerical error estimates are not interval certificates. See `validation/pv-high-rank/pentagon-simplex-oracle.json`. Historical binary64 errors remain 2.23×10^{-10} for TIRust and 3.40×10^{-10} for IREGI. Public rescue flags the primary unstable and verifies all 840 slots at 128/256 bits plus this coefficient below 10^{-13}. `validation/rescue-pentagon-rank6.json` retains the binary and assertions; full arrays were not printed and its exact build-source snapshot was not saved.

The vendored Symbolica baseline passes four scope tests, 57 core tests and all 1320 rank-seven slots at 128/256 bits, including the independent coefficient. Its source predates later precision/routing edits; its workspace compilation preserves 329 source/manifest hashes. The 109.44 s cold run, earlier 240 s timeout and 115.35 s isolated run are historical diagnostics. See `validation/symbolica-production-qualification.json`.

63.2 Physical canonical moment 54

For $gg \to HHH$, rank-four moment $q_0 q_3^3$ has UV degree -2; its UV pole must vanish. Original IBP instead returns approximately $3.33691766067 \times 10^{-6} i$ in native normalization. The 128/256-bit PV evaluation suppresses its OneLoop-normalized UV residual from 7.9×10^{-31} to 2.3×10^{-69}, with both IR slots zero. Its converged OneLoop finite coefficient is approximately $3.3474440772594743330 \times 10^{-13} + 2.9254797223941252086 \times 10^{-13} i$. This supports original PV/hybrid and rejects the original IBP result for this component. It is not a full physical tensor comparison. Raw precision evidence is in `validation/pv-high-rank/canonical-pentagon-precision.json canonical-pentagon-adjudication.json` records fixed-gate comparisons.

63.3 Standalone six-dimensional scalar

Use $J_1 = \frac{I_6}{i\pi^3}$, shifts $(0, e_0, e_1, e_2, e_3)$ and squared masses $(10, 12, 15, 18, 22)$. Here $F \geq \frac{39}{4}$, $\det R = 2600$ and $\det G = -16$.

Method	Finite J_1
Independent simplex	$-0.000177140138264008180607652131 0$
Original scalar/general APIs	-0.00014953025024128455; every mode reports stable

Increasing quadrature order and precision through 70 digits converges to the first value. Independent integration of ordinary D4 values over a common mass shift, with a separately integrated tail, agrees within 4.93×10^{-15}. A box-only half-line implementation also agrees to approximately 8.42×10^{-17} or better in the recorded checks. Ordinary D4 and coalesced D4/D6 normalization checks pass. Evidence and reproduction source are in `validation/shifted-pentagon/evidence.json`. Thirteen consolidated D6 tests cover eight genuine-master checks and five endpoints. Retained public f64 fixtures pass finite values, poles, scaling, rejected endpoints and lane isolation; the earlier quadrature-budget failure remains historical. Pre-vendor public DoubleFloat and Float128 tests now pass three scaled lanes at requested 80-bit accuracy and 10^{-23} normalized oracle tolerance. First common-mass Dual primal/derivative checks pass at 2×10^{-14}, with zero poles. DD/Float validation takes 165.69 s, exceeding its historical 150 s bound; this is not a benchmark. Build scope is in `validation/shifted-pentagon/public-core-qualification.json`. Paired vendored DD E6 passes exact allocation parity. Requested128 fails for the small-routing fixture; higher/mixed Dual, broader endpoints and native E6 remain unqualified. The original complex-box discrepancy is retained in `validation/shifted-pentagon/contours.json`.

A private input-enclosure extension passes five D0 pinches and two contractions at the single exact contour point $x = \frac{2}{5}$, Float256/request128: 40 rectangles and 20 rounded mass slots checked. Exact unshifted geometry remains assumed; no quadrature certificate or production E6 qualification follows.

64 Reference conventions

These conventions were read from the supplied original sources. Their presence in this report is not a claim that every adapter has passed its numerical tests.

64.1 Kinematics and scale

Use Minkowski signature $(+,-,-,-)$ with component order (p^0,p^1,p^2,p^3). Propagators are $D_j = (q+p_j)^2 - m_j^2 + i0$, raised to positive integer powers. Generalized integrals use $d = 4 + \mathrm{idim} - 2\varepsilon$ and the native raw measure; dimension shifts introduce powers of π. Native tensor reduction accepts one shift per propagator and removes the first common shift internally, then translates tensor coefficients back. The MadLoop interface instead receives the nonzero shifts and inserts a zero first shift. A reference comparison must preserve the chosen numerator routing when it changes the propagator direction.

The original `MU` / `MUR` argument is an energy scale, not its square. Its scalar wrapper passes μ^2 to QCDLoop and μ to OneLOop. Complex mass-squared inputs must retain their imaginary part; the reference switches to its OneLOop backend for complex masses. OneLoopMaster accepts real external squared invariants, complex squared masses with $\Im(m^2) \le 0$, and a positive real squared scale. Its continued logarithms and polylogarithms own the Feynman branch prescription; no independent branch implementation belongs in the reduction adapter. Preserve inputs by default; legacy on-shell snapping must be requested explicitly. Dual derivatives follow the selected analytic branch away from nondifferentiable boundaries.

64.2 Native poles and MadLoop conversion

The native result order is

$$(U, I, J, F) = (\varepsilon_{\mathrm{UV}}^{-1}, \varepsilon_{\mathrm{IR}}^{-1}, \varepsilon_{\mathrm{IR}}^{-2}, \mathrm{finite}).$$

These symbols label separate coefficients. In particular, a scaleless integral can have canceling UV and IR contributions even when its combined Laurent series is zero. Three combined coefficients do not determine their separate origin. OneLoopMaster uses $(\mathrm{finite}, \varepsilon^{-1}, \varepsilon^{-2})$ ordering.

The original `IMLOOP` adapter combines the native poles and maps a tensor of homogeneous rank r to MadLoop order (finite, simple pole, double pole). Defining $a = \gamma_E + \ln\pi$, $b = \frac{a^2}{2} + \frac{\pi^2}{12}$ and $c_r = \frac{(-1)^r}{i\pi^2}$, its explicit mapping is

$$(F_{\mathrm{ML}}, S_{\mathrm{ML}}, D_{\mathrm{ML}}) = c_r(F + a(U+I) + bJ, U + I + aJ, J).$$

For a OneLOop-normalized scalar master (L_0, L_1, L_2), the scalar wrappers use

$$J = i\pi^2 L_2, \quad U + I = i\pi^2(L_1 - aL_2),$$

$$F = i\pi^2\left(L_0 - aL_1 + \left(\frac{a^2}{2} - \frac{\pi^2}{12}\right)L_2\right).$$

The UV/IR split requires independent integral classification. In particular, the scaleless bubble is $(U, I, J, F) = (i\pi^2, -i\pi^2, 0, 0)$ in the original wrapper. Applying the MadLoop conversion to an already normalized master result would count normalization twice.

64.3 Polynomial and tensor interfaces

A rank bound R has $\binom{R+4}{4}$ Cartesian coefficients in the original four-component basis. `ti_reduce::calc_pos` traverses total degree in ascending order, then exponents of q^0, q^1, q^2 in descending order; the exponent of q^3 is the remaining degree. Thus the start is $\left(1, q^0, q^1, q^2, q^3, (q^0)^2, q^0q^1, q^0q^2, q^0q^3, (q^1)^2, \ldots\right)$. Numerator preparation and tensor output must share this order. The original native and MadLoop tensor conventions differ in multinomial factors: the native path multiplies symmetric components by the factor returned by `all_integers`, whereas `ML5_CONVENTION=.TRUE.` uses unweighted components. The independent oracle sets this flag **without** applying `IMLOOP`, preserving the native four-pole normalization. Mixed-monomial tests detect multiplicity mismatches.

Reduction modes are 0: IBP, 1: PaVe, 2: PaVe with IBP stability improvement. Direction mode 2 reverses the propagator order and shifts the loop origin. The source labels direction modes 4/5 as unimplemented quadruple precision; their numbers are not evidence of high-precision support.

65 Algebra reuse audit

All symbolic algebra belongs to Symbolica and the FeynKit / Spenso / Idenso ecosystem. Before introducing a missing operation, record three checks: Symbolica APIs and implementation; Symbolica usage/examples/tests; and FeynKit/gammaloop facilities and tests. An adapter or data-layout conversion must not grow an independent polynomial, simplification or matrix engine.

65.1 Existing facilities to reuse

- **Tensor identities and contractions:** Idenso contracts metrics and reduces supported gamma/color identities; Spenso carries tensor structure. Keep Lorentz dimension and any gamma-five prescription explicit.
- **Coefficient extraction:** FeynKit's tensor reducer uses Idenso contraction without global expansion, `to_dots()` and the shared `coefficient_list` collector. Preserve scalar factorization until polynomial extraction requires expansion.
- **Propagator and scalar-product algebra:** FeynKit's `IntegralFamily` uses Symbolica for polynomial conversion, exact rank and linear solves, basis completion and mappings. Its Gaussian-rational conversion avoids statistical expression-field zero tests that may discard small coefficients.
- **Numerical preparation:** Compile reusable Symbolica expressions outside the sample loop. Evaluation and scalar-master types must preserve the selected precision and all Dual components without binary64 round trips.
- **Laurent coefficients:** IBP/PV factors and native normalization use cached Symbolica `series` templates and `coefficient` extraction. The three reuse checks are the pinned Symbolica `src/atom/core.rs` and `src/poly/series.rs` APIs; its `src/derivative.rs` pole-series tests; and existing use in `vakint/src/projection.rs` and OneLoopMaster's `src/sheet_exact.rs`. No separate Laurent convolution engine is needed. Symbolica also evaluates the independent Gamma/series triangle oracle.

65.2 HEPKit tensor-reduction optimizations

The local `feynkit-tensor` reducer stores repeated-vector contractions as symmetry orbits instead of factorially many labeled pairings. Its joint-color dynamic program builds orbit intersections; alternating cycles build orbit Gram matrices; Symbolica solves the smaller orbit system. A rank-20 isotropic projection can have one orbit instead of 19!! labeled pairings.

The implementation caches canonical Weingarten coefficients and has a closed isotropic path. General projectors are validated through rank 20; the special all-equal isotropic route reaches rank 32. These are upstream limits, not a statement of iregi-rust's current supported domain.

For denominators with external momenta, declare an independent external basis and average only the transverse space. Odd total tensor ranks need not vanish. Keep D symbolic while constructing high-rank projectors; early integer specialization can make the universal metric basis singular. External Gram inverses can amplify small-Gram numerical errors, so projector preprocessing requires a conditioning-aware comparison against direct recurrences.

This projector reduces tensor numerators, not complete scalar integral families. Reuse it where applicable and benchmark the resulting plan; no factorial-count argument alone establishes a numerical speedup. Retain explicit output/pairing limits and report their failures honestly.

65.3 References and provenance

The untouched generalized IREGI entry point stops during factorial-table initialization: the configured powers/rank request factorial 16, while the helper rejects arguments above 12. A narrow repair in a generated reference copy replaces that table expression by its exact binomial recurrence. The untouched/repaired twins agree exactly on 24 ordinary cases; all 80 table entries match independent integer binomials; seven generalized fixtures check normalization, unit powers, mass derivatives and a tadpole dimension shift. The original `STOP` returns exit code zero, so the harness also requires an explicit result record. Evidence lives in `validation/reference/fixtures.json` and the report's JSON snapshot.

OneLoopMaster documents corrected analytic-continuation defects in its upstream OneLOop reference, including complex-mass box cases. Preserve each disagreement as a visible case and adjudicate it with independent analytic, high-precision or parameter-integral evidence. Agreement must not mean reproducing a known incorrect branch.

Primary references: Alwall et al., appendix C.1 HEPKit TensorReducer Goode et al., tensor projectors Collins–Matsumoto, Weingarten recurrence NinjaRS report style. Local source audit: FeynKit's `feynkit-tensor/src` projector modules and `feynkit-graph/src/integrals.rs` original IREGI's `IREGI4ML5_interface.f90`, `ti_reduce.f90`, `si_reduce.f90`, `mis_warp.f90` and `funlib.f90`.

66 Final private numerical checks and handoff

The following results extend the saved evidence. They do not apply the private seven-point implementation or scheduler patch to the public default reducer, and they introduce no new kernel timings. Earlier failed attempts, reference instabilities and inconclusive oracle or timing rows remain unchanged.

66.1 Seven-point scalar: one fixed ordinary lane

The private rational-polynomial implementation completes generic symbolic preparation independently at Float128 and Float256, with requested accuracy 53. It packs all 21 five-point children into one batch and returns all four native Laurent slots. Every slot agrees with the retained stable original IREGI IBP and Hybrid results at the unchanged dimensionally normalized 10^{-10} gate. The original PV result remains unstable and is not counted as agreement.

Saved numerical gate	Complete checks	Maximum normalized error
Original stable IBP / Hybrid, both precisions	16 Laurent slots	4.783×10^{-12}
Float128 / Float256 convergence	4 Laurent slots	9.102×10^{-34}

All three poles are exactly zero at both precisions. An independent audit reconstructs acceptance using exact rational squared inequalities, without importing the producer comparator. The full complex norm is compared: the tiny real finite component has tracked precision 2, so these results do not claim 128/256-bit componentwise accuracy. The previous fixed scalar oracle remains inconclusive. This is one private rank-zero case, not a qualification of N7 higher ranks, derivatives, generalized interfaces, numerical types, batches, physical graphs or performance.

Capture totals including final provenance checks are 153.623 s and 164.176 s, both within their predeclared 300 s bounds. They include generic preparation, workspace construction and diagnostics and are not benchmark measurements. Source: `validation/highpoint-thin-n7-rank0-rational-numeric/qualification.json`.

66.2 Uniform-branch scheduler: complete retained dependent corpus

Separate matching Symbolica/OneLoop/masters/core chains pass 60 requests, 30 per process. Exact complete records agree across schedulers and repeated or retained workspaces: eight ordinary N2/N3 rank0/1 batch1/32 families cover 1,584 tensor and 1,782 master complex values. Direct master controls cover 67 cases and 133 rows, including 106 successes and 27 explicit error/null rows. All six master outputs, requested Taylor components, represented values, precision, signed zero, diagnostics, masks and execution counters are checked. Independent saved-data and byte-copy audits pass; earlier compiler failures and the stricter admission-schema revision are retained.

This qualifies the frozen opt0 corpus only. The patch is not applied to production, and neither optimized performance nor full-domain acceptance is established. Fresh matched optimized chains are still needed: the older optimized closure predates a current Symbolica polynomial guard. Source: `validation/uniform-branch-dependent/qualification.json`.

67 Resumed native-master validation

Development resumed on 25 October 2026. The seven previously unrun constructor cases pass in the frozen default/selective builds, with two complete test listings and independent audits. All 646 bound inputs remained unchanged and all nine owned process groups exited. Together with twenty earlier ordinary cases, this completes the 27-case focused inventory across two separate runs. The earlier twenty cases were not rerun.

New constructor gate	Complete cases	Interpretation
Default / selective policy composition	3 + 4 passed	Construction and policy routing; no six-point integral values

The bounded run used 282.090 s including provenance and cleanup, within its unchanged 300 s limit. This is a diagnostic test duration, not a kernel benchmark. Source: `validation/native-f64-sixpoint-constructors/qualification.json`.

67.1 Complete native-policy tensor comparisons

Distinct baseline and selective-policy cores pass the original 38-request corpus: 30 TIRust requests and eight original IREGI requests. The eight ordinary bubble/triangle rank-zero/one families include singleton and batch-32 inputs. All complete tensor and captured master arrays are checked before accepting the policy; repeated workspaces and retained controls preserve full records.

Whole-call comparison	Complex slots / variant	Maximum normalized error
Complete tensor arrays	1,584	3.378×10^{-13} versus IREGI
Captured scalar masters	1,782	1.695×10^{-14}
Baseline versus candidate tensors	1,584	2.061×10^{-13}

The unchanged gate is 10^{-10}. D0, bubble derivatives, Dual and precision controls remain exact by represented values, precision and sign. This is an unoptimized frozen dependency context, not a current-Cargo or performance measurement. Source: `validation/native-f64-sixpoint-whole/qualification.json`.

67.2 Retained six-point behavior under both feature configurations

All ten numerical producers and eight saved gates pass in the default and selective combined source chains. The explicitly supported six-point parent and children use Symbolica. Checks include 24 ordinary paired cases, 4,400 complex Taylor slots, 560 rank-three original-reference comparisons and 280 128/256-bit convergence comparisons. The largest normalized errors are 2.426×10^{-12} against IREGI and 1.474×10^{-34} across precision. All 420 checked pole slots are exactly zero. Twenty-two lifecycle calls cover reordered/repeated/chunked samples, failures, empty calls and contractions; values, precision and sign match retained singleton results exactly.

Independent numerical and provenance audits pass; all 17 owned process groups exited. No f64/DoubleFloat high-point accuracy or public six-point Dual claim follows. Source: `validation/native-f64-sixpoint-regression/qualification.json`.

68 High-point numeric pilot and open qualifications

The private rank-zero six-point pilot evaluates one massive real physical point. Repeated and batched profiles do not increase the number of independent physical points. Missing arrays remain failures.

Numeric domain	Completed Laurent slots	Outcome at unchanged bounds
f64	0 / 40	All ten intended positive lanes rejected by cancellation guard; two invalid controls rejected.
DoubleFloat	40 / 40	Ten successful lanes, two invalid controls and one empty call pass.
Float128 dynamic Dual	24 / 240	One affine singleton passes; later schedule times out. Full domain fails.
Float256 dynamic Dual	0 / 240	Work deadline reached before a complete output. Precision-pair gate has zero comparisons.

DoubleFloat agrees with original IREGI within 8.175×10^{-14} and the retained Float256 scalar target within 6.420×10^{-30}. The available Float128 prefix has 24 complete complex coefficients, 18 exactly zero poles, and maximum dilation-identity error 1.013×10^{-34}. Two formal variables describe one physical dilation direction; the nonlinear mixed profile remains incomplete. Independent checks reconstruct all 64 available outputs and 1,736 exact input components, including seeds below binary64 resolution.

Each domain retained its 180-second aggregate bound and 20-second postwork reserve; no failed attempt was retried or extended in this pilot. Its timings are lifecycle diagnostics. Public high-point Dual accuracy, derivative error transport, general independent derivative directions and full convergence remain unqualified. Source: `validation/highpoint-six-point-numeric-types/qualification.json`.

68.1 Exact routing preparation

A private optimization delegates bounded affine routing matrices to Symbolica's rational-polynomial matrix solver. All 16 selected exact algebra tests pass, including actual generic N6/N7 scalar-compiler routing, generalized power/dimension identities, Gaussian coefficients and explicit fallback for opaque or nonpolynomial entries. The initial missing-environment build failure and interrupted scheduling attempt remain recorded. This is not a full-core numerical or performance qualification and is not applied to production. Source: `validation/routing-rational-preparation/qualification.json`.

The optional native policy passes the current locked-Cargo checks detailed below. The restricted public seven-point proposal and shared exact-master preparation cache remain separate candidates. Matched optimized scheduler products pass fresh numerical admission. Their timing cohort is retained below, with final performance eligibility false because of substantial sibling-core activity. Existing IREGI/COLLIER timings retain their contexts and TIRust-numerator ratios. The full goal remains active: complete N6/N7 rank/power/dimension coverage, high-point precision/Dual qualification, broader physical/exceptional checks and performance parity per family remain required.

69 Optional native policy: current Cargo integration

The Rust working tree integrates the explicit `MasterExecutionPolicy::NativeF64Abc` policy through the `native-f64-abc` feature. Oneloopmaster's generated Rust evaluators handle ordinary f64 A0/B0/C0 requests instruction-major across active lanes. D0, bubble derivatives, DoubleFloat, arbitrary precision and every static or dynamic Dual layout retain Symbolica under the selective policy. Selecting the policy without its feature returns an explicit preparation error.

Actual locked, offline Cargo builds and runtime checks pass for the default and selective configurations, without private capture instrumentation or substituted project archives. Current-source metadata records verify the expected feature forwarding. The executable copies are taken from their exact Cargo artifact records before another configuration can overwrite a shared example path.

Current configuration	Selected exact tests	Public five-lane triangle probe
Default	9 / 9 pass	Default and explicit Symbolica records agree exactly.
Selective feature	17 / 17 pass	Default, Symbolica and native policies pass complete tensor checks.

These 26 selected cases cover policy routing, ordered Dual layouts, independent scratch, precision integrity and family exclusions. They are a focused subset, not a claim that every project test was rerun. The N3/rank-one probe uses three retained physical points, one invalid-scale neighbor and one repeated point. All 400 complex Laurent comparisons against original IREGI pass the unchanged dimensionally normalized 10^{-10} gate, with maximum error 2.346×10^{-13}. The 80 direct selected-versus-Symbolica comparisons have maximum error 1.486×10^{-13}. Invalid lanes contain no successful tensor array; values, precision, signs and execution counters remain checked.

69.1 Retained compatibility failure

Legacy feature configuration	Metadata outcome	Qualification boundary
Broad `native-batch`	Timeout	Reached the fixed 240 s frontend limit; group cleanup and unchanged sources verified.
Broad + selective	Unrun	Later matrix step was not attempted after the timeout.

The failed matrix attempt remains preserved, with no retry or extended bound. A separate reviewed continuation admits the already-passed default/selective metadata rows for their actual builds and numerical checks. Neither the complete legacy matrix nor broad native runtime is qualified by this checkpoint. The large legacy D0 evaluator received no LLVM code-generation request.

All 39 owned process groups exited. Independent source, artifact, numerical and saved-data audits pass for the stated native-only scope. Six-point numerical regressions remain the separate preceding capsule; no new N6/N7 domain or performance claim follows from the Cargo checks. Backend selection is exposed through the Rust API; CLI/Python selection is future work. Default constructors retain their backend behavior. The explicit six-point parent and children use Symbolica. Source and reproduction commands: `validation/native-f64-sixpoint-current-cargo/qualification.json` and `docs/native-masters.md`.

70 Fresh scheduler experiment: retained, ineligible

Final performance eligibility: false. The experiment ran on CPU 34 while its SMT sibling, CPU 290, had substantial unrelated activity (69.6% non-idle over the enclosing observation interval). All mathematical checks pass; timing arithmetic and raw gates are retained. These measurements establish no uncontended parity or production-default improvement.

This private comparison pins `2c1f334` and generic Symbolica masters in the same inherited optimized dependency context for both TIRust variants. It does not measure the later opt-in generated f64 policy. Fourteen matched compiler products, captured and uncaptured complete-array admission, and six-process preflight pass. The earlier linker-map admission failure remains recorded; its five valid products were reused and nine were built subsequently.

The one-shot run retains all eight massive Euclidean bubble/triangle families, 48 balanced quartets per family, 1,768 requests and 1,600 timed replies. Transport lies outside the timed kernel. All six processes exit successfully and final controls pass. No short or noisy family is removed or rerun.

70.1 Absolute kernel times

Medians in **µs per complete batch call**. For batch 32, divide by 32 for the amortized per-sample time; the machine-readable record contains both units. These absolute medians and the paired ratio medians on the next page are different statistics: ratios are not calculated by dividing this table.

N / rank / batch	TIRust baseline	TIRust candidate	IREGI	COLLIER
2 / 0 / 1	2.209	2.278	0.944	0.760
2 / 0 / 32	43.881	40.940	28.438	24.661
2 / 1 / 1	7.099	7.291	3.188	1.199
2 / 1 / 32	264.385	259.023	157.852	65.059
3 / 0 / 1	31.145	31.746	7.818	6.235
3 / 0 / 32	412.617	382.028	164.619	130.904
3 / 1 / 1	43.299	42.683	16.052	9.677
3 / 1 / 32	964.618	906.794	525.433	314.702

The raw candidate/baseline observations include about 6% lower time for N2/rank0/batch32 and N3/rank1/batch32, with 2–3% singleton regressions for N2/rank0 and N3/rank0. This mixed result does not support globally adopting the scheduler candidate, even before considering contention. The production scheduler remains unchanged. Higher-precision and Dual performance are not measured in this cohort.

Reproduce the saved-evidence checks without evaluating new integrals: `python3 validation/scheduler-four-engine-fresh-2c1/verify.py`. Then run `python3 validation/scheduler-four-engine-fresh-2c1-supplement-check.py` to check the two historical nested manifests omitted by the original seal. Machine-readable source: `validation/scheduler-four-engine-fresh-2c1/performance.json`.

71 Fresh scheduler experiment: all paired ratios

Every numerator is TIRust. **B** denotes baseline and **C** the private candidate. Each large cell gives the paired median and 99% bootstrap interval. Color shows the raw ratio only; **every cell is ineligible for final acceptance** because of the shared-core context. Short blocks remain amber.

N / rank / batch	B / IREGI	C / IREGI	B / COLLIER	C / COLLIER	C / B
2 / 0 / 1	2.334 2.315–2.340 Slower	2.412 2.388–2.420 Slower	2.894 2.879–2.919 Slower	2.993 2.973–3.013 Slower	1.033 1.030–1.036 Slower
2 / 0 / 32	1.531 1.497–1.542 Noise	1.435 1.400–1.446 Noise	1.756 1.727–1.786 Noise	1.648 1.622–1.668 Noise	0.935 0.930–0.945 Lower time
2 / 1 / 1	2.313 2.267–2.345 Noise	2.336 2.311–2.350 Slower	5.850 5.800–5.908 Slower	5.936 5.772–6.012 Noise	1.010 0.995–1.022 Noise
2 / 1 / 32	1.671 1.668–1.677 Short block	1.638 1.635–1.644 Short block	4.061 4.051–4.068 Short block	3.979 3.974–3.986 Short block	0.981 0.980–0.984 Short block
3 / 0 / 1	3.982 3.967–3.995 Slower	4.062 4.052–4.072 Slower	4.991 4.966–5.000 Slower	5.085 5.057–5.102 Slower	1.019 1.016–1.021 Slower
3 / 0 / 32	2.433 2.415–2.478 Short block	2.201 2.180–2.294 Short block	3.140 3.112–3.159 Short block	2.830 2.819–2.875 Short block	0.905 0.901–0.923 Short block
3 / 1 / 1	2.719 2.630–2.735 Noise	2.641 2.572–2.678 Noise	4.518 4.435–4.561 Noise	4.391 4.305–4.454 Noise	0.977 0.961–0.983 Noise
3 / 1 / 32	1.818 1.746–1.837 Noise	1.717 1.648–1.725 Noise	3.059 2.818–3.069 Noise	2.876 2.668–2.883 Noise	0.939 0.937–0.941 Lower time

Raw gates: **12 regressions, 16 noise-inconclusive, 10 short-block-inconclusive and 2 improvements**. The short families are N2/rank1/batch32 (one short block) and N3/rank0/batch32 (50 short blocks). The unchanged minimum block duration is 100 ms. All 384 quartets and all 40 confidence intervals remain available, including these failures of timing admission.

Independent saved-data review reproduces the numerical admissions, complete request accounting, quartet orders and all ratio arithmetic. A separate environment review sets final acceptance false without rewriting raw gates. A future cohort needs fresh placement checks and a new measurement protocol; no extrapolation to current native-policy or high-point performance is made.

Source and full scope: `validation/scheduler-four-engine-fresh-2c1/qualification.json`.